

STAT 423 Winter 2026 - Final Project

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Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
  orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
  ifelse(grepl("Country", orig_data$cdr_genre), "Country",
  ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
  ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
  danceability = orig_data$danceability, happiness = orig_data$happiness,
  loudness_db = orig_data$loudness_d_b,
  length_sec = orig_data$length_sec)

cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
  unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- subpop_60s <- cleaned_data[cleaned_data$year >= i - 10 &
    cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
```

```

if(nrow(subpop) < 50) {
  message(paste("Skipping", decade_label, "- Not enough data points."))
  next
}

factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
  function(x) length(unique(na.omit(x))))

if(any(factor_counts < 2)) {
  message(paste("Skipping", decade_label, "due to single-level factor:",
    names(factor_counts)[factor_counts < 2]))
  next
}

first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
  + danceability + happiness + bpm + loudness_db + length_sec)

second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
  + age*energy + age*danceability + age*happiness + age*bpm
  + age*loudness_db + age*length_sec + race*genre + race*energy
  + race*danceability + race*happiness + race*bpm + race*loudness_db
  + race*length_sec + gender*genre + gender*energy + gender*danceability
  + gender*happiness + gender*bpm + gender*loudness_db + gender*length_sec )

# --- Plotting (Wrapped in 'try' so one bad plot doesn't stop the whole loop) ---
try({
  plot(first_model, which = 1:2)
  boxcox(first_model, plotit=TRUE)
  plot(second_model, which = 1:2)
  boxcox(second_model, plotit=TRUE)
})
optimal_model <- step(second_model, direction="backward")

model_results[[decade_label]] <- list(
  "first" = first_model,
  "second" = second_model,
  "optimal" = optimal_model
)

print(decade_label)

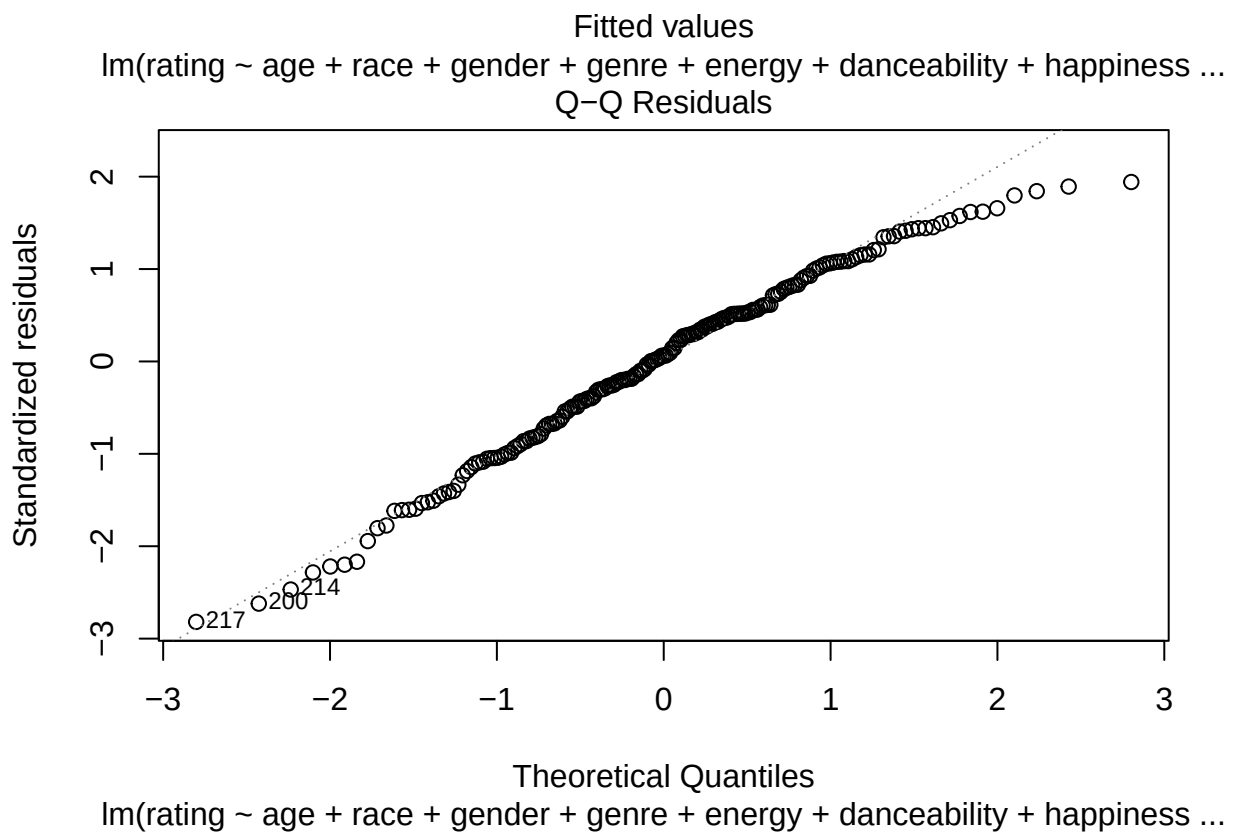
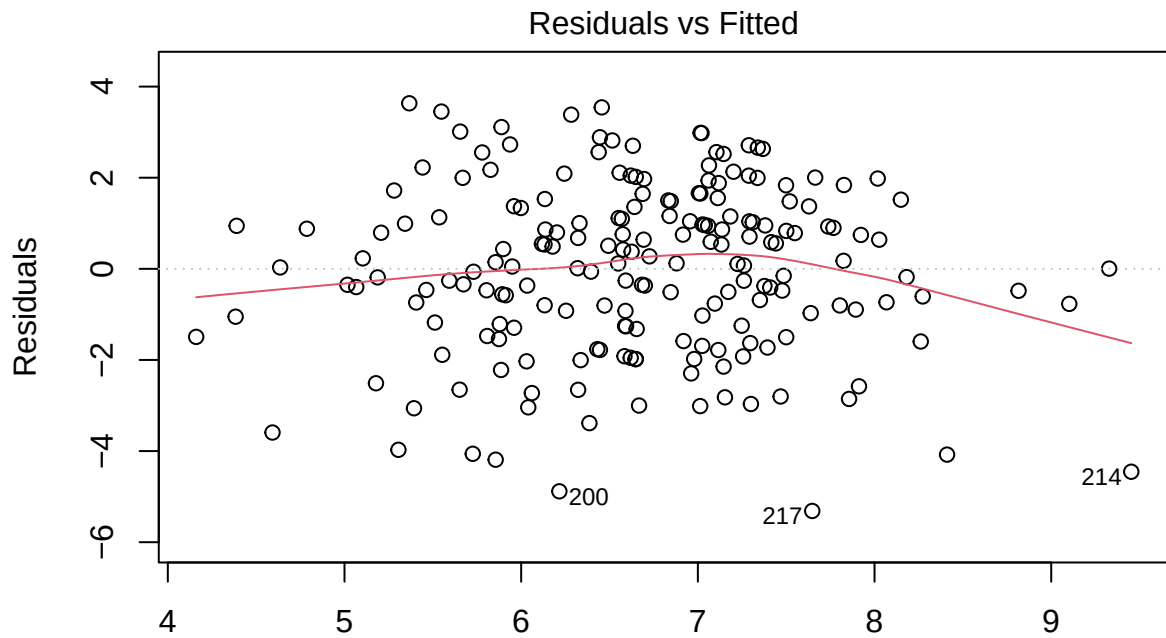
summary(first_model)

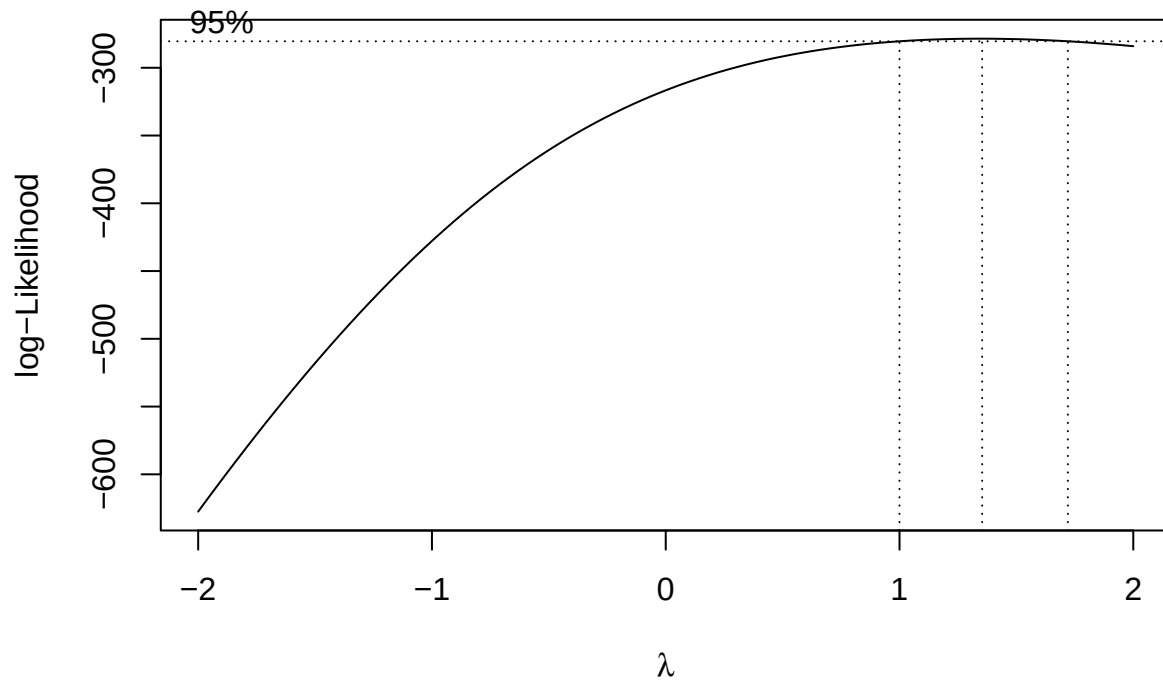
summary(second_model)

summary(optimal_model)
}

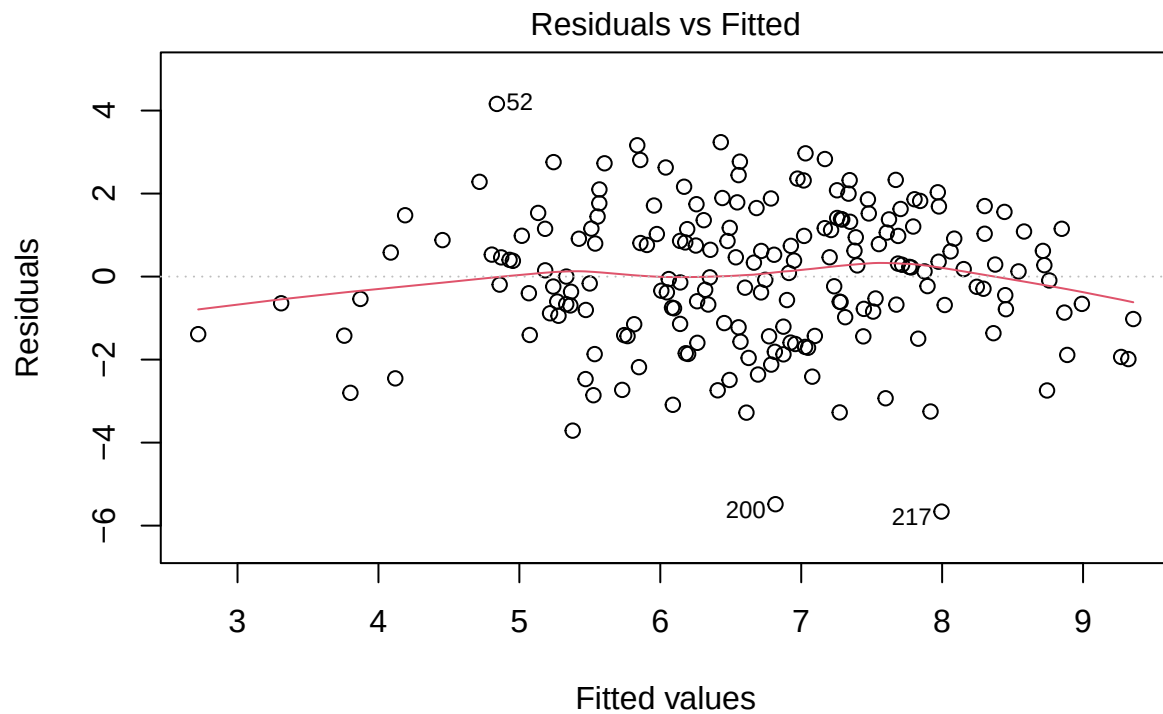
```

```
## Skipping 1950s - Not enough data points.
```

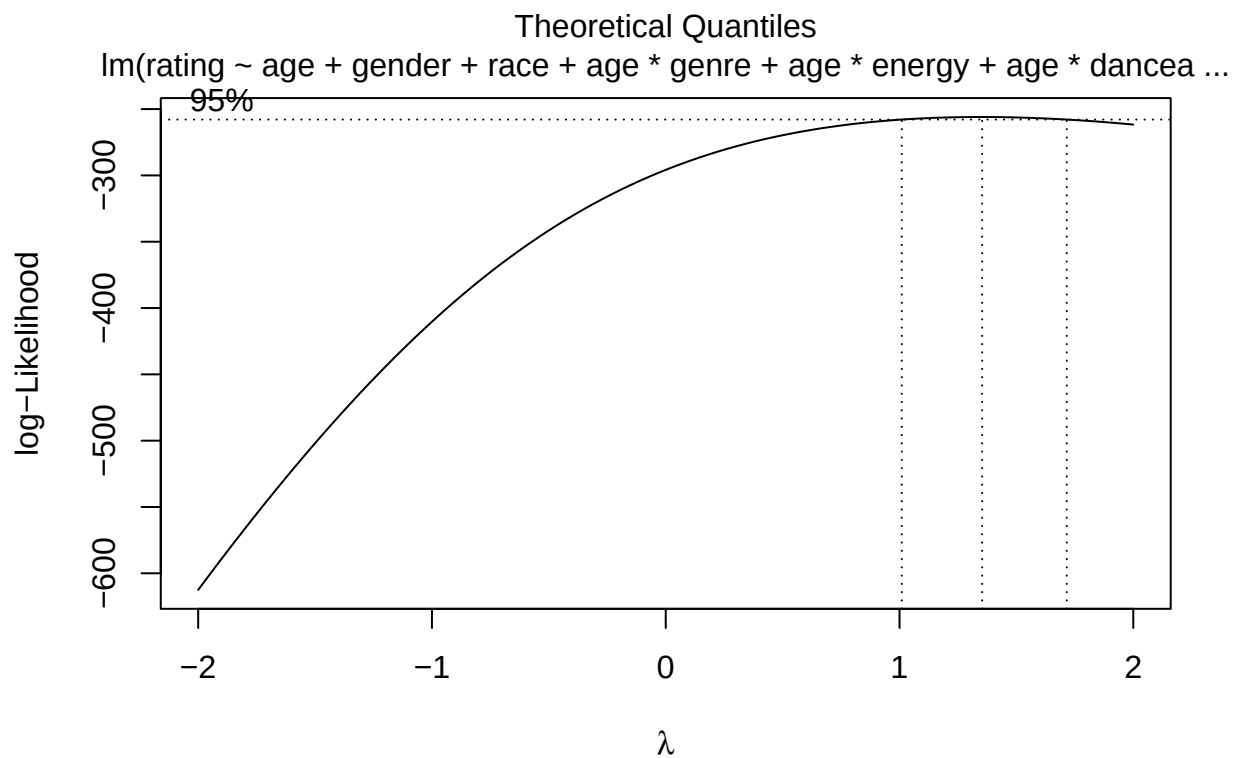
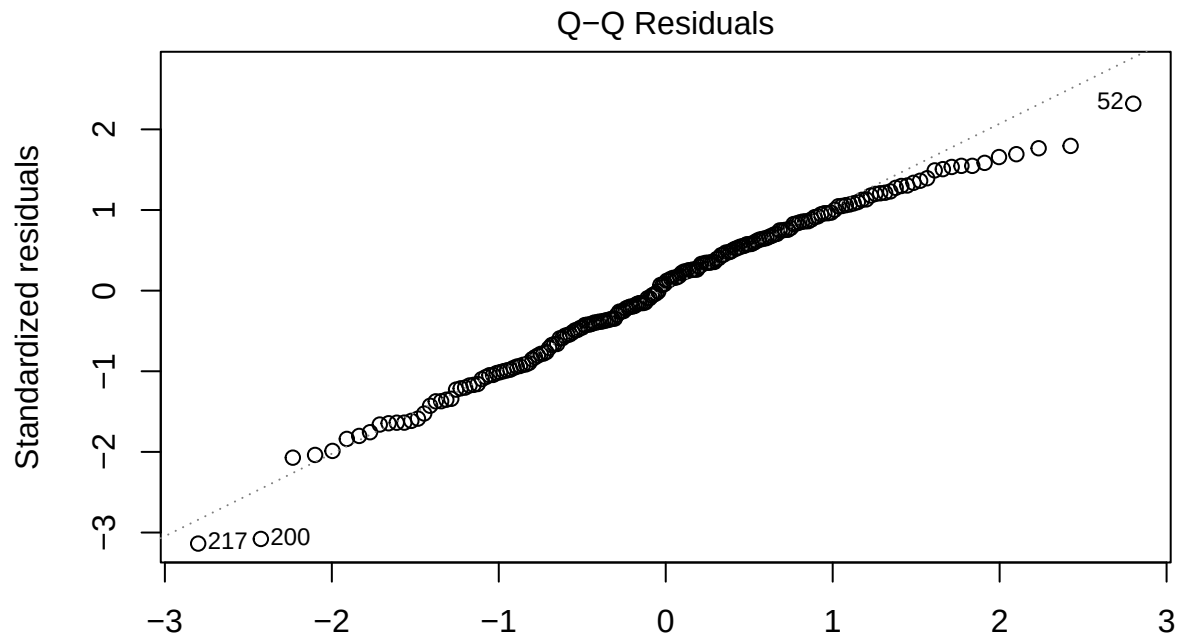




```
## Warning: not plotting observations with leverage one:
## 48
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=281.18
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##          Df Sum of Sq    RSS    AIC
## - age:genre      4      6.3839 542.36 275.51
## - gender:genre    3      0.9873 536.97 275.54
## - age:energy      1      0.1576 536.14 279.23
## - age:length_sec  1      0.2498 536.23 279.27
## - race:bpm        1      0.3150 536.30 279.29
## - age:bpm         1      0.3777 536.36 279.31
## - age:happiness   1      0.4214 536.40 279.33
## - gender:bpm      1      0.8413 536.82 279.49
## - race:happiness  1      0.9928 536.97 279.54
## - race:loudness_db 1      1.1628 537.14 279.60
## - age:danceability 1      1.7691 537.75 279.82
## - gender:length_sec 1      2.3411 538.32 280.04
## - age:loudness_db 1      2.7934 538.77 280.20
## - race:energy     1      4.0209 540.00 280.65
## <none>                                535.98 281.18
## - race:length_sec  1      6.7873 542.77 281.65
## - gender:energy     1      7.2507 543.23 281.82
## - race:danceability 1      8.3951 544.38 282.24
## - race:genre        3     19.8442 555.82 282.34
## - gender:loudness_db 1      8.9645 544.94 282.44
## - gender:happiness  1      9.1272 545.11 282.50
## - gender:danceability 1     30.1379 566.12 289.95
##
## Step:  AIC=275.51
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:energy +
##          age:danceability + age:happiness + age:bpm + age:loudness_db +
##          age:length_sec + race:genre + race:energy + race:danceability +
##          race:happiness + race:bpm + race:loudness_db + race:length_sec +
##          gender:genre + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - gender:genre      3      2.648 545.01 270.47
## - age:energy         1      0.023 542.39 273.52
## - age:length_sec     1      0.040 542.40 273.52
## - age:bpm            1      0.091 542.45 273.54
## - age:happiness      1      0.318 542.68 273.62
## - race:bpm           1      0.667 543.03 273.75
## - race:happiness     1      0.669 543.03 273.75
## - race:loudness_db   1      1.144 543.51 273.92
## - age:loudness_db    1      1.414 543.78 274.02
## - gender:length_sec  1      1.771 544.14 274.15
## - gender:bpm         1      1.934 544.30 274.21
## - age:danceability   1      2.999 545.36 274.60
## - race:energy        1      3.834 546.20 274.90
## <none>                                542.36 275.51
## - race:length_sec    1      6.161 548.53 275.73
## - gender:energy       1      7.395 549.76 276.18
## - race:danceability  1      8.293 550.66 276.50
## - gender:happiness   1      8.431 550.79 276.55
## - gender:loudness_db 1      8.748 551.11 276.66
## - race:genre         3     28.215 570.58 279.50

```

```

## - gender:danceability 1 32.747 575.11 285.06
##
## Step: AIC=270.47
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:bpm 1 0.018 545.03 268.48
## - age:energy 1 0.022 545.03 268.48
## - age:length_sec 1 0.127 545.14 268.51
## - age:happiness 1 0.321 545.33 268.58
## - race:loudness_db 1 0.556 545.57 268.67
## - race:bpm 1 0.661 545.67 268.71
## - race:happiness 1 1.298 546.31 268.94
## - gender:bpm 1 1.387 546.40 268.97
## - age:loudness_db 1 1.579 546.59 269.04
## - gender:length_sec 1 1.847 546.86 269.13
## - age:danceability 1 2.747 547.76 269.46
## - race:energy 1 3.197 548.21 269.62
## <none> 545.01 270.47
## - gender:energy 1 6.580 551.59 270.83
## - race:length_sec 1 6.863 551.88 270.93
## - gender:happiness 1 8.063 553.07 271.36
## - gender:loudness_db 1 8.252 553.26 271.43
## - race:danceability 1 9.685 554.70 271.94
## - race:genre 3 30.197 575.21 275.09
## - gender:danceability 1 32.042 577.05 279.72
##
## Step: AIC=268.47
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:loudness_db + age:length_sec +
## race:genre + race:energy + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:energy 1 0.019 545.05 266.48
## - age:length_sec 1 0.133 545.16 266.52
## - age:happiness 1 0.304 545.33 266.58
## - race:loudness_db 1 0.543 545.57 266.67
## - race:bpm 1 0.683 545.71 266.72
## - race:happiness 1 1.318 546.35 266.95
## - gender:bpm 1 1.375 546.41 266.97
## - age:loudness_db 1 1.616 546.65 267.06
## - gender:length_sec 1 1.829 546.86 267.13
## - race:energy 1 3.191 548.22 267.62
## - age:danceability 1 3.256 548.29 267.65

```

```

## <none> 545.03 268.48
## - gender:energy 1 6.586 551.62 268.84
## - race:length_sec 1 7.035 552.07 269.00
## - gender:happiness 1 8.050 553.08 269.36
## - gender:loudness_db 1 8.236 553.27 269.43
## - race:danceability 1 9.688 554.72 269.95
## - race:genre 3 30.248 575.28 273.12
## - gender:danceability 1 32.457 577.49 277.87
##
## Step: AIC=266.48
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:happiness + age:loudness_db + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - age:length_sec 1 0.122 545.17 264.53
## - age:happiness 1 0.354 545.40 264.61
## - race:loudness_db 1 0.553 545.60 264.68
## - race:bpm 1 0.669 545.72 264.72
## - race:happiness 1 1.320 546.37 264.96
## - gender:bpm 1 1.397 546.45 264.99
## - gender:length_sec 1 1.813 546.86 265.14
## - age:loudness_db 1 2.605 547.65 265.42
## - age:danceability 1 3.281 548.33 265.66
## - race:energy 1 3.431 548.48 265.72
## <none> 545.05 266.48
## - gender:energy 1 6.572 551.62 266.84
## - race:length_sec 1 7.064 552.11 267.02
## - gender:happiness 1 8.032 553.08 267.36
## - gender:loudness_db 1 8.265 553.31 267.45
## - race:danceability 1 9.700 554.75 267.96
## - race:genre 3 30.254 575.30 271.12
## - gender:danceability 1 32.678 577.73 275.95
##
## Step: AIC=264.53
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:happiness + age:loudness_db + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 0.271 545.44 262.62
## - race:loudness_db 1 0.590 545.76 262.74
## - race:bpm 1 0.796 545.97 262.81
## - race:happiness 1 1.332 546.50 263.01
## - gender:bpm 1 1.422 546.59 263.04
## - gender:length_sec 1 1.953 547.12 263.23
## - age:loudness_db 1 2.507 547.68 263.43
## - age:danceability 1 3.162 548.33 263.67

```



```

## - race:energy          1      3.593 548.76 263.82
## <none>                  545.17 264.53
## - gender:energy        1      6.527 551.70 264.87
## - race:length_sec      1      7.135 552.31 265.09
## - gender:happiness     1      7.931 553.10 265.37
## - gender:loudness_db   1      8.277 553.45 265.49
## - race:danceability    1      9.791 554.96 266.03
## - race:genre           3     30.745 575.92 269.33
## - gender:danceability  1     32.799 577.97 274.04
##
## Step: AIC=262.62
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db      1      0.609 546.05 260.84
## - race:bpm              1      0.786 546.23 260.91
## - race:happiness        1      1.233 546.67 261.07
## - gender:bpm            1      1.457 546.90 261.15
## - gender:length_sec     1      2.090 547.53 261.38
## - age:danceability      1      3.012 548.45 261.71
## - age:loudness_db       1      3.366 548.81 261.83
## - race:energy           1      3.726 549.17 261.96
## <none>                   545.44 262.62
## - gender:energy         1      6.614 552.06 263.00
## - race:length_sec       1      7.221 552.66 263.21
## - gender:happiness      1      7.671 553.11 263.38
## - gender:loudness_db    1      8.356 553.80 263.62
## - race:danceability     1      9.685 555.13 264.09
## - race:genre            3     30.717 576.16 267.42
## - gender:danceability   1     32.532 577.97 272.04
##
## Step: AIC=260.84
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm              1      0.758 546.81 259.12
## - race:happiness        1      1.140 547.19 259.25
## - gender:bpm            1      1.556 547.61 259.40
## - gender:length_sec     1      1.836 547.89 259.50
## - age:danceability      1      3.084 549.13 259.95
## - race:energy           1      3.350 549.40 260.05
## - age:loudness_db       1      3.601 549.65 260.14
## <none>                   546.05 260.84
## - gender:energy         1      6.219 552.27 261.07

```

```

## - race:length_sec      1      6.678 552.73 261.24
## - gender:happiness     1      7.681 553.73 261.60
## - gender:loudness_db   1      7.792 553.84 261.63
## - race:danceability    1      9.970 556.02 262.41
## - race:genre           3     30.684 576.73 265.61
## - gender:danceability  1     32.813 578.86 270.34
##
## Step: AIC=259.12
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:length_sec + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.864 547.67 257.43
## - gender:bpm          1      1.059 547.87 257.50
## - gender:length_sec    1      1.782 548.59 257.76
## - race:energy          1      2.937 549.75 258.17
## - age:loudness_db      1      3.349 550.16 258.32
## - age:danceability     1      3.376 550.18 258.33
## <none>                  546.81 259.12
## - gender:energy        1      6.306 553.11 259.38
## - race:length_sec      1      6.687 553.50 259.51
## - gender:loudness_db   1      8.286 555.09 260.08
## - gender:happiness     1      8.709 555.52 260.23
## - race:danceability    1      9.911 556.72 260.65
## - race:genre           3     31.011 577.82 263.98
## - gender:danceability  1     33.009 579.82 268.66
##
## Step: AIC=257.43
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm          1      0.865 548.54 255.74
## - gender:length_sec    1      1.573 549.25 255.99
## - age:danceability     1      3.165 550.84 256.56
## - age:loudness_db      1      3.329 551.00 256.62
## <none>                  547.67 257.43
## - race:energy          1      5.919 553.59 257.55
## - race:length_sec      1      6.071 553.74 257.60
## - gender:energy        1      6.920 554.59 257.90
## - gender:loudness_db   1      8.084 555.76 258.31
## - race:danceability    1      9.447 557.12 258.80
## - gender:happiness     1     11.330 559.00 259.46
## - race:genre           3     32.597 580.27 262.82
## - gender:danceability  1     34.647 582.32 267.51
##
## Step: AIC=255.74
## rating ~ age + gender + race + genre + energy + danceability +

```

```

##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##      Df Sum of Sq    RSS    AIC
## - gender:length_sec    1      1.365 549.90 254.23
## - bpm                  1      1.607 550.14 254.31
## - age:danceability     1      3.185 551.72 254.88
## - age:loudness_db      1      3.620 552.16 255.03
## <none>                  548.54 255.74
## - race:energy          1      5.613 554.15 255.74
## - race:length_sec      1      6.189 554.73 255.95
## - gender:energy        1      6.192 554.73 255.95
## - gender:loudness_db   1      7.441 555.98 256.39
## - race:danceability    1      9.572 558.11 257.15
## - gender:happiness     1     12.437 560.97 258.15
## - race:genre           3     33.928 582.47 261.56
## - gender:danceability  1     34.121 582.66 265.63
##
## Step: AIC=254.23
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##      Df Sum of Sq    RSS    AIC
## - bpm                  1      1.740 551.64 252.85
## - age:danceability     1      3.052 552.95 253.32
## - age:loudness_db      1      3.350 553.25 253.42
## - race:length_sec      1      4.840 554.74 253.95
## - race:energy          1      4.922 554.82 253.98
## <none>                  549.90 254.23
## - gender:energy        1      5.917 555.82 254.34
## - gender:loudness_db   1      7.215 557.12 254.80
## - race:danceability    1      8.651 558.55 255.30
## - gender:happiness     1     15.057 564.96 257.55
## - race:genre           3     33.710 583.61 259.95
## - gender:danceability  1     35.155 585.06 264.44
##
## Step: AIC=252.85
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##      Df Sum of Sq    RSS    AIC
## - age:loudness_db      1      3.113 554.76 251.96
## - age:danceability     1      3.743 555.39 252.18
## - race:energy          1      4.950 556.59 252.61
## - race:length_sec      1      4.981 556.62 252.62
## <none>                  551.64 252.85

```

```

## - gender:energy      1      6.148 557.79 253.03
## - gender:loudness_db 1      8.025 559.67 253.70
## - race:danceability  1      8.473 560.12 253.85
## - gender:happiness   1     14.433 566.08 255.94
## - race:genre         3     34.698 586.34 258.87
## - gender:danceability 1     34.474 586.12 262.79
##
## Step: AIC=251.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:danceability +
##      race:genre + race:energy + race:danceability + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - age:danceability  1      4.140 558.90 251.42
## - race:energy      1      4.411 559.17 251.52
## <none>                                554.76 251.96
## - gender:energy    1      5.774 560.53 252.00
## - race:length_sec  1      5.824 560.58 252.02
## - gender:loudness_db 1      7.492 562.25 252.60
## - race:danceability 1      8.520 563.28 252.96
## - gender:happiness 1     13.672 568.43 254.75
## - race:genre       3     35.753 590.51 258.26
## - gender:danceability 1     34.021 588.78 261.68
##
## Step: AIC=251.42
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:energy + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:energy    1      5.246 564.14 251.26
## - race:energy      1      5.291 564.19 251.28
## <none>                                558.90 251.42
## - gender:loudness_db 1      6.576 565.47 251.73
## - race:length_sec  1      6.646 565.54 251.75
## - race:danceability 1      8.657 567.55 252.45
## - age              1     11.424 570.32 253.41
## - gender:happiness  1     14.543 573.44 254.49
## - race:genre       3     35.352 594.25 257.51
## - gender:danceability 1     31.578 590.47 260.25
##
## Step: AIC=251.26
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db 1      1.810 565.95 249.90
## - race:energy        1      3.897 568.04 250.62
## - race:length_sec    1      4.870 569.01 250.96

```

```

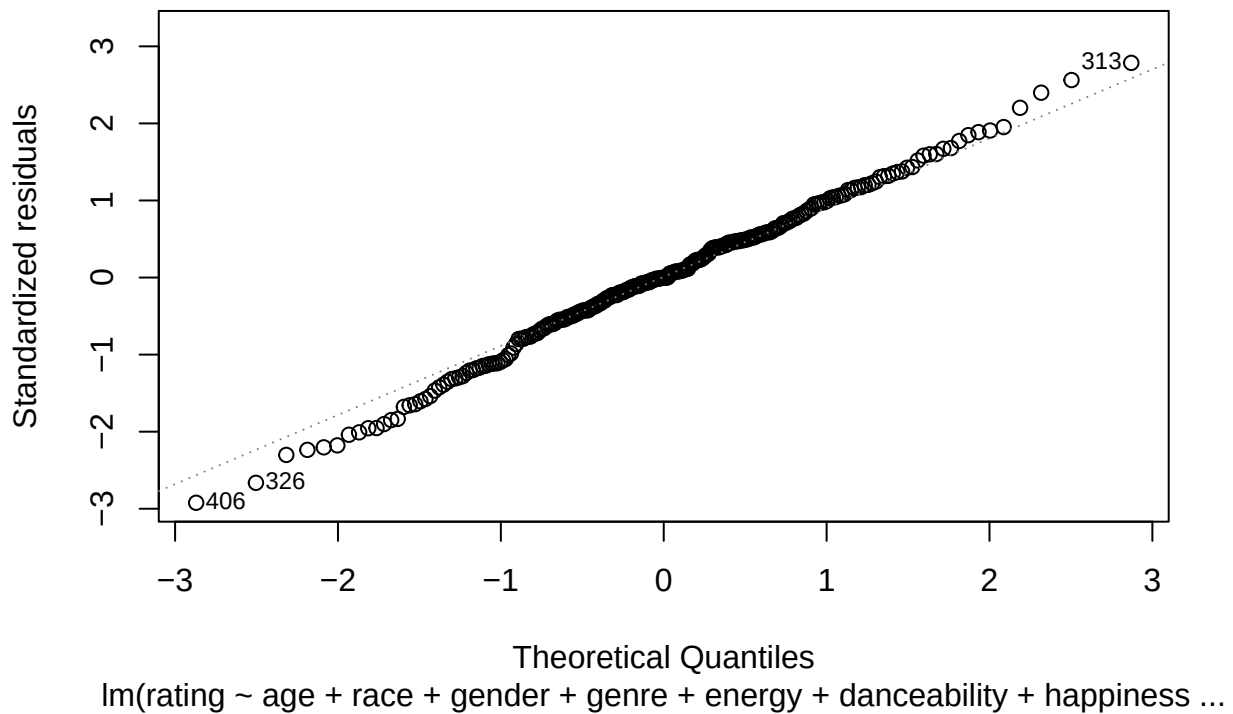
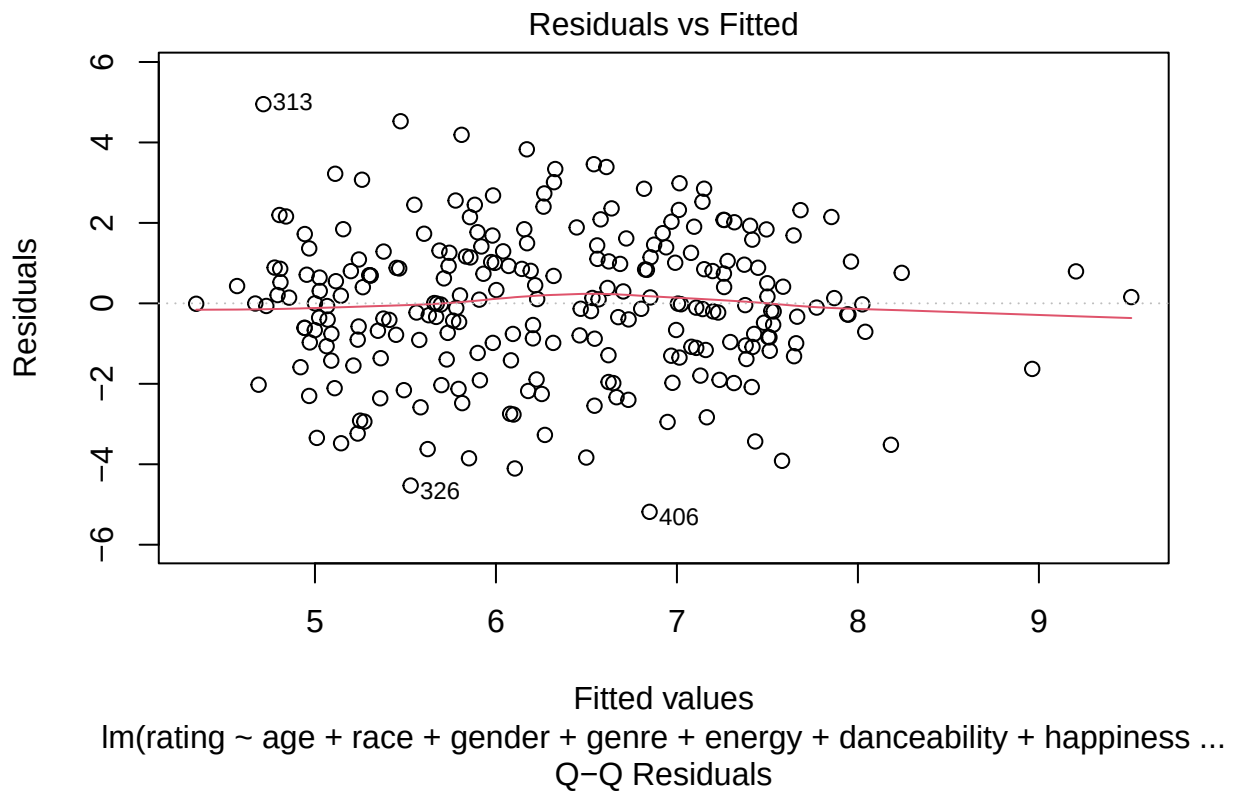
## <none>                                564.14 251.26
## - race:danceability    1      8.107 572.25 252.07
## - age                  1      9.432 573.57 252.53
## - gender:happiness     1      9.738 573.88 252.64
## - race:genre           3     35.039 599.18 257.13
## - gender:danceability  1     30.692 594.83 259.70
##
## Step: AIC=249.9
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:danceability +
##      gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## - race:length_sec    1      4.800 570.75 249.56
## - loudness_db        1      4.839 570.79 249.57
## <none>                                565.95 249.90
## - race:energy        1      6.645 572.60 250.19
## - race:danceability  1      9.038 574.99 251.02
## - age                1     10.136 576.09 251.39
## - gender:happiness   1     12.208 578.16 252.10
## - race:genre         3     33.271 599.22 255.15
## - gender:danceability 1     30.208 596.16 258.14
##
## Step: AIC=249.56
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + gender:danceability + gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## - loudness_db        1      4.255 575.01 249.02
## <none>                                570.75 249.56
## - race:energy        1      6.452 577.20 249.77
## - age                1     10.174 580.93 251.04
## - gender:happiness   1     11.277 582.03 251.41
## - race:danceability  1     12.736 583.49 251.91
## - race:genre         3     28.783 599.54 253.25
## - length_sec         1     17.833 588.58 253.62
## - gender:danceability 1     33.114 603.87 258.67
##
## Step: AIC=249.02
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + length_sec + race:genre + race:energy + race:danceability +
##      gender:danceability + gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## <none>                                575.01 249.02
## - race:energy        1      6.391 581.40 249.20
## - age                1      9.812 584.82 250.36
## - gender:happiness   1     11.732 586.74 251.00
## - race:danceability  1     14.130 589.14 251.81
## - race:genre         3     28.484 603.49 252.55
## - length_sec         1     18.065 593.07 253.12
## - gender:danceability 1     34.156 609.16 258.39

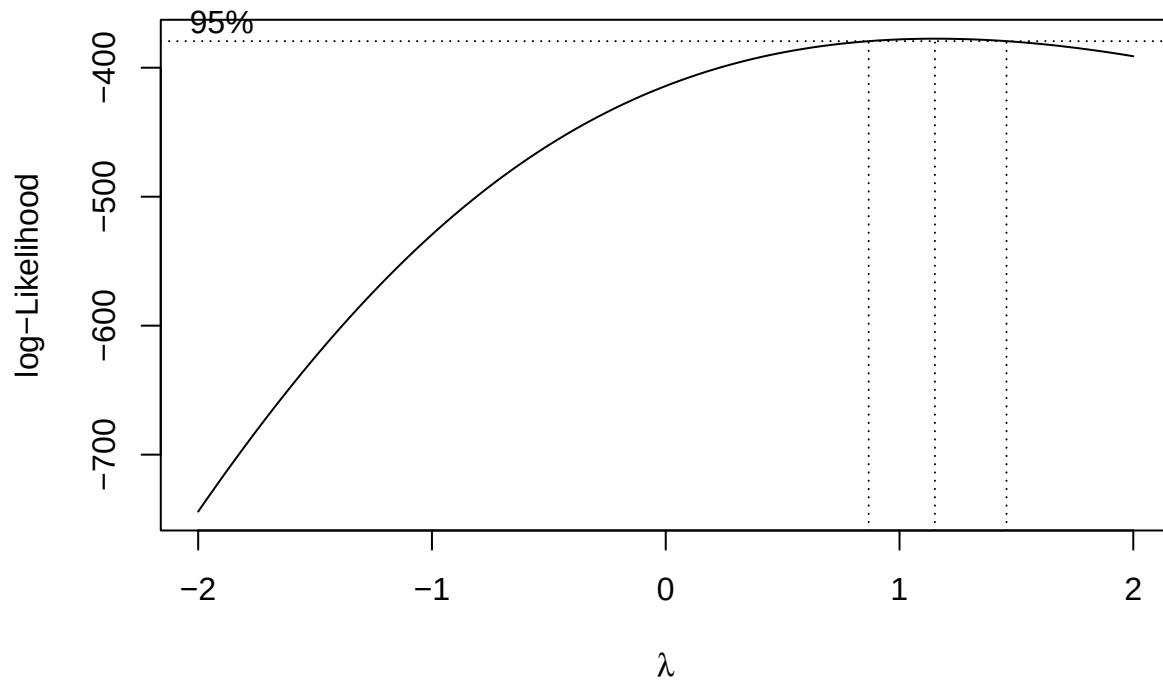
```

```
## [1] "1960s"
```

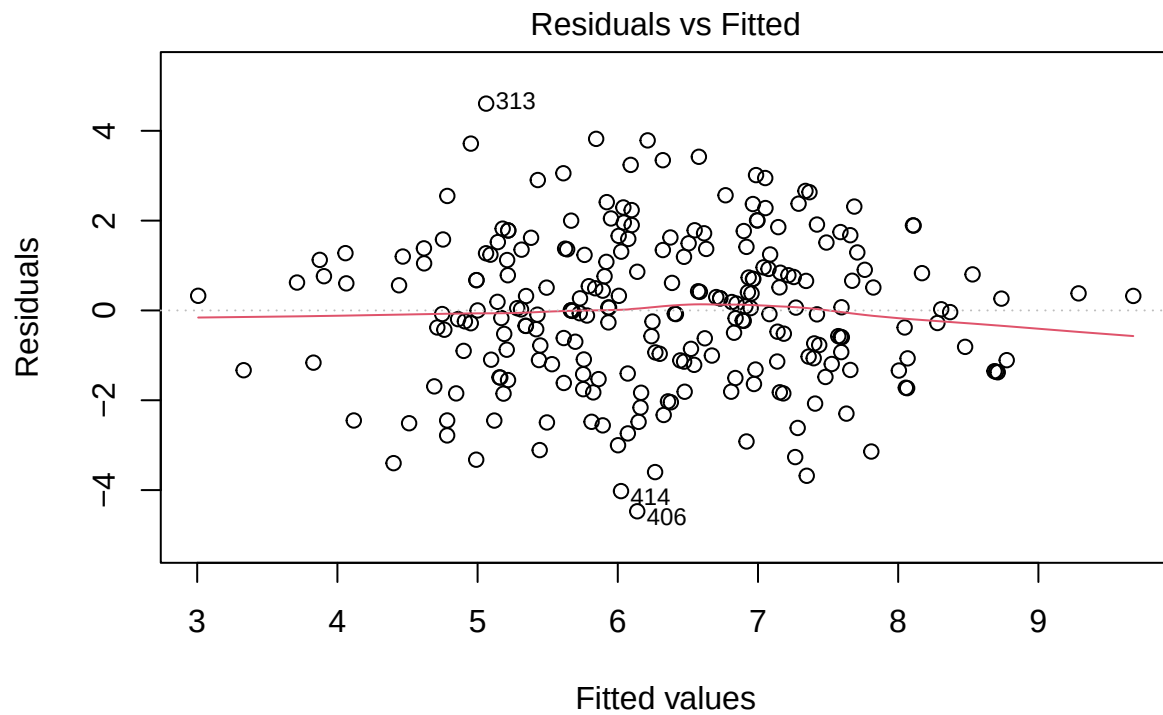
```
## Warning: not plotting observations with leverage one:
```

```
## 239
```

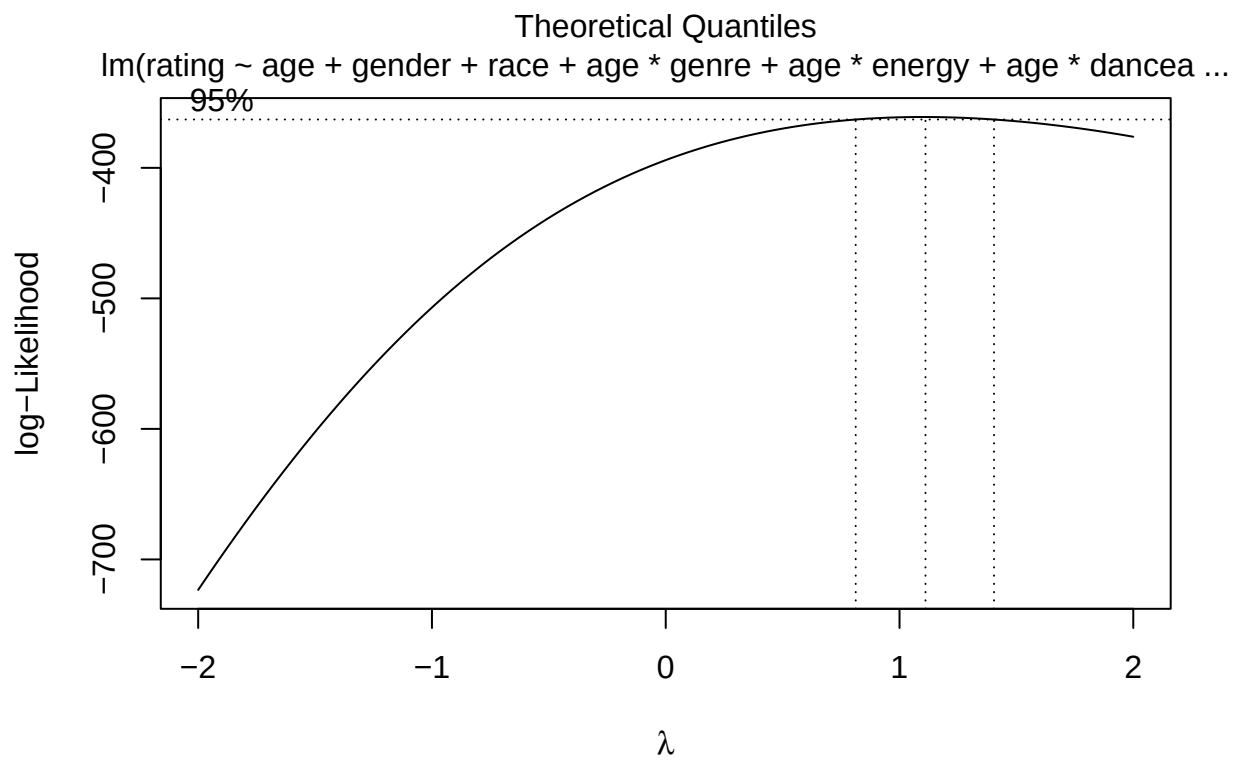
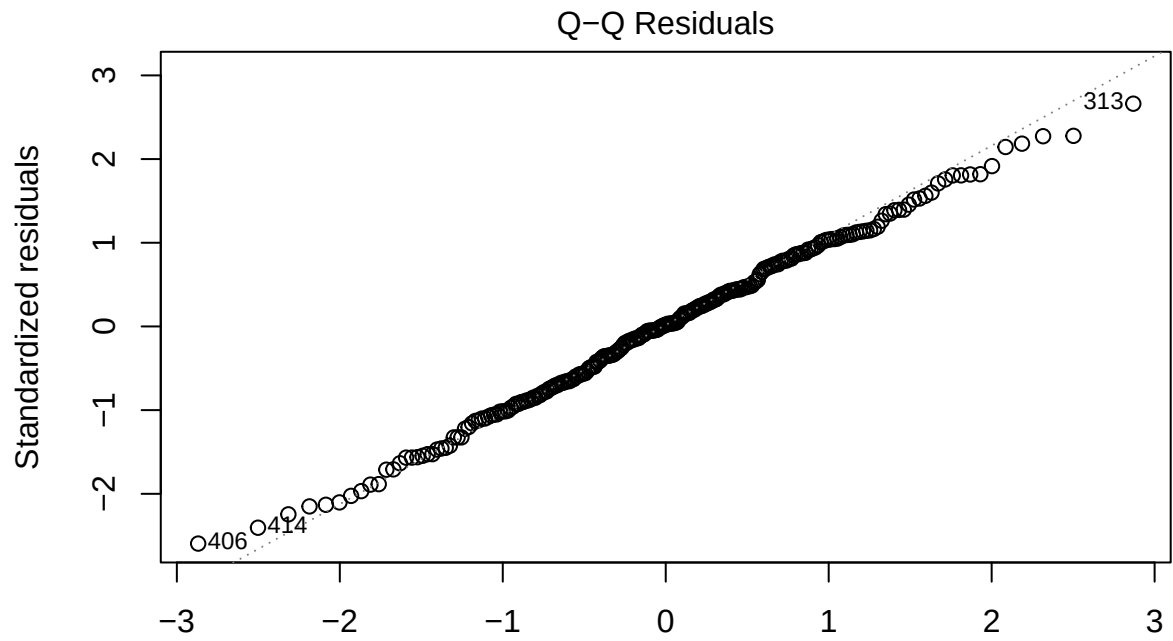




```
## Warning: not plotting observations with leverage one:
## 38, 239
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=327.31
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```



```

##              Df Sum of Sq    RSS    AIC
## - gender:genre      3      4.5148 671.33 322.96
## - race:happiness     1      0.0509 666.87 325.33
## - gender:bpm         1      0.0575 666.87 325.33
## - race:loudness_db   1      0.0637 666.88 325.33
## - gender:happiness   1      0.0693 666.88 325.33
## - gender:danceability 1      0.0693 666.88 325.33
## - race:danceability  1      0.5466 667.36 325.51
## - age:loudness_db    1      0.6594 667.47 325.55
## - age:danceability   1      1.2483 668.06 325.77
## - gender:loudness_db 1      1.7006 668.52 325.93
## - gender:energy      1      1.9573 668.77 326.03
## - race:length_sec    1      2.1587 668.97 326.10
## - race:genre         3     14.9204 681.74 326.73
## - age:energy         1      3.9706 670.79 326.76
## <none>                666.81 327.31
## - age:happiness      1      5.9102 672.73 327.47
## - race:bpm          1      6.7182 673.53 327.76
## - race:energy        1      7.1011 673.92 327.90
## - age:bpm           1     10.1186 676.93 329.00
## - gender:length_sec  1     12.1319 678.95 329.72
## - age:genre          3     23.7323 690.55 329.88
## - age:length_sec     1     14.2379 681.05 330.48
##
## Step:  AIC=322.96
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:genre +
##          age:energy + age:danceability + age:happiness + age:bpm +
##          age:loudness_db + age:length_sec + race:genre + race:energy +
##          race:danceability + race:happiness + race:bpm + race:loudness_db +
##          race:length_sec + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db   1      0.0027 671.33 320.96
## - gender:danceability 1      0.0064 671.34 320.96
## - gender:happiness    1      0.0151 671.34 320.97
## - race:happiness      1      0.0240 671.35 320.97
## - gender:bpm          1      0.3552 671.69 321.09
## - race:danceability   1      0.4579 671.79 321.13
## - age:loudness_db     1      0.4797 671.81 321.14
## - gender:loudness_db  1      0.6231 671.95 321.19
## - gender:energy       1      0.7602 672.09 321.24
## - age:danceability    1      1.0366 672.37 321.34
## - race:length_sec     1      2.5747 673.90 321.90
## - race:genre          3     14.1348 685.46 322.07
## - age:energy         1      3.4393 674.77 322.21
## <none>                671.33 322.96
## - age:happiness      1      6.3026 677.63 323.25
## - race:energy        1      6.7154 678.05 323.40
## - race:bpm          1      7.2589 678.59 323.60
## - age:bpm           1     11.3985 682.73 325.09
## - age:genre          3     24.5416 695.87 325.76
## - age:length_sec     1     14.3055 685.64 326.13

```

```

## - gender:length_sec      1    16.1867 687.52 326.80
##
## Step: AIC=320.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - gender:danceability  1      0.0071 671.34 318.96
## - gender:happiness     1      0.0159 671.35 318.97
## - race:happiness       1      0.0312 671.36 318.97
## - gender:bpm           1      0.3731 671.71 319.10
## - race:danceability    1      0.4623 671.79 319.13
## - age:loudness_db      1      0.4824 671.81 319.14
## - gender:loudness_db   1      0.6332 671.97 319.19
## - gender:energy        1      0.7591 672.09 319.24
## - age:danceability     1      1.0353 672.37 319.34
## - race:length_sec      1      2.5770 673.91 319.90
## - race:genre           3     14.3567 685.69 320.15
## - age:energy           1      3.4369 674.77 320.21
## <none>                  671.33 320.96
## - age:happiness        1      6.3995 677.73 321.29
## - race:bpm             1      7.4374 678.77 321.66
## - age:bpm              1     11.4625 682.79 323.11
## - race:energy          1     11.7473 683.08 323.21
## - age:genre            3     25.2477 696.58 324.01
## - age:length_sec       1     14.4572 685.79 324.18
## - gender:length_sec    1     16.1849 687.52 324.80
##
## Step: AIC=318.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - gender:happiness     1      0.0100 671.35 316.97
## - race:happiness       1      0.0299 671.37 316.98
## - gender:bpm           1      0.3868 671.73 317.11
## - race:danceability    1      0.4569 671.80 317.13
## - age:loudness_db      1      0.4817 671.82 317.14
## - gender:loudness_db   1      0.6269 671.97 317.19
## - gender:energy        1      0.7663 672.11 317.24
## - age:danceability     1      1.0643 672.40 317.35
## - race:length_sec      1      2.5720 673.91 317.90
## - race:genre           3     14.3666 685.71 318.15
## - age:energy           1      3.4407 674.78 318.22

```

```

## <none>                                671.34 318.96
## - age:happiness          1      6.4179 677.76 319.30
## - race:bpm               1      7.4310 678.77 319.66
## - age:bpm                1     11.4599 682.80 321.11
## - race:energy            1     11.7443 683.08 321.21
## - age:genre              3     25.2747 696.61 322.02
## - age:length_sec         1     14.5147 685.85 322.20
## - gender:length_sec      1     16.4894 687.83 322.91
##
## Step:  AIC=316.97
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.0225 671.37 314.98
## - gender:bpm          1      0.3845 671.73 315.11
## - race:danceability    1      0.4543 671.80 315.13
## - age:loudness_db      1      0.4860 671.84 315.15
## - gender:loudness_db   1      0.7785 672.13 315.25
## - age:danceability     1      1.0593 672.41 315.35
## - gender:energy        1      1.4978 672.85 315.51
## - race:length_sec      1      2.6242 673.97 315.92
## - race:genre           3     14.4707 685.82 316.19
## - age:energy           1      3.4312 674.78 316.22
## <none>                                671.35 316.97
## - age:happiness        1      6.4119 677.76 317.30
## - race:bpm             1      7.4527 678.80 317.67
## - age:bpm              1     11.4618 682.81 319.12
## - race:energy          1     12.8095 684.16 319.60
## - age:genre            3     25.3765 696.73 320.06
## - age:length_sec       1     14.8398 686.19 320.32
## - gender:length_sec    1     16.7887 688.14 321.02
##
## Step:  AIC=314.98
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:bpm + race:length_sec + gender:energy +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm          1      0.3886 671.76 313.12
## - age:loudness_db      1      0.4659 671.84 313.15
## - race:danceability    1      0.4709 671.84 313.15
## - gender:loudness_db   1      0.7622 672.13 313.25
## - age:danceability     1      1.1477 672.52 313.39
## - gender:energy        1      1.5000 672.87 313.52
## - race:length_sec      1      2.6029 673.98 313.92
## - race:genre           3     14.5057 685.88 314.21

```

```

## - age:energy      1      3.4088 674.78 314.22
## <none>              671.37 314.98
## - age:happiness   1      6.6747 678.05 315.40
## - race:bpm        1      7.5442 678.92 315.71
## - age:bpm         1     11.4466 682.82 317.12
## - age:genre       3     25.7596 697.13 318.20
## - age:length_sec  1     14.8393 686.21 318.33
## - race:energy     1     15.8498 687.22 318.69
## - gender:length_sec 1     17.0128 688.38 319.11
##
## Step:  AIC=313.12
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:bpm + race:length_sec + gender:energy +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      0.4025 672.16 311.26
## - age:loudness_db    1      0.5193 672.28 311.31
## - gender:loudness_db 1      0.8559 672.62 311.43
## - age:danceability   1      1.1403 672.90 311.53
## - gender:energy      1      1.8979 673.66 311.81
## - race:length_sec    1      2.5654 674.33 312.05
## - race:genre         3     14.4815 686.24 312.34
## - age:energy         1      3.5062 675.27 312.39
## <none>                671.76 313.12
## - age:happiness      1      6.6135 678.37 313.52
## - race:bpm           1      7.8870 679.65 313.98
## - age:bpm            1     11.5684 683.33 315.30
## - age:genre          3     25.4697 697.23 316.24
## - age:length_sec     1     14.8834 686.64 316.49
## - race:energy        1     15.7902 687.55 316.81
## - gender:length_sec  1     17.0058 688.77 317.24
##
## Step:  AIC=311.26
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:bpm + race:length_sec + gender:energy + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db    1      0.5933 672.76 309.48
## - age:danceability   1      1.1692 673.33 309.69
## - gender:loudness_db 1      1.1994 673.36 309.70
## - gender:energy      1      2.2366 674.40 310.08
## - race:length_sec    1      2.4276 674.59 310.15
## - age:energy         1      3.5750 675.74 310.56
## - race:genre         3     16.4217 688.59 311.18
## <none>                672.16 311.26
## - age:happiness      1      6.8066 678.97 311.73

```

```

## - race:bpm          1    7.6990 679.86 312.06
## - age:bpm           1   11.6400 683.80 313.47
## - age:genre         3   25.2709 697.43 314.31
## - age:length_sec    1   15.1451 687.31 314.72
## - race:energy        1   16.3393 688.50 315.15
## - gender:length_sec  1   16.7323 688.90 315.29
##
## Step: AIC=309.48
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:length_sec + race:genre + race:energy + race:bpm + race:length_sec +
## gender:energy + gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - gender:loudness_db  1    0.9620 673.72 307.83
## - age:danceability    1    1.1381 673.89 307.89
## - gender:energy        1    2.0913 674.85 308.24
## - race:length_sec     1    2.3781 675.13 308.35
## - age:energy          1    3.5224 676.28 308.76
## - race:genre          3   16.1447 688.90 309.29
## <none>                672.76 309.48
## - race:bpm           1    7.5503 680.31 310.21
## - age:happiness       1   10.0098 682.77 311.10
## - age:bpm            1   11.5872 684.34 311.67
## - age:genre           3   25.0042 697.76 312.42
## - race:energy         1   15.7540 688.51 313.15
## - age:length_sec      1   16.6943 689.45 313.49
## - gender:length_sec   1   17.5417 690.30 313.79
##
## Step: AIC=307.83
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:length_sec + race:genre + race:energy + race:bpm + race:length_sec +
## gender:energy + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - age:danceability    1    1.0841 674.80 306.23
## - gender:energy        1    1.1310 674.85 306.24
## - race:length_sec     1    2.5843 676.30 306.77
## - age:energy          1    3.8324 677.55 307.22
## <none>                673.72 307.83
## - race:genre          3   17.8821 691.60 308.25
## - race:bpm            1    7.2729 680.99 308.46
## - loudness_db         1    7.5347 681.25 308.56
## - age:happiness       1    9.8237 683.54 309.38
## - age:bpm            1   11.4953 685.21 309.98
## - age:genre           3   25.1786 698.90 310.82
## - race:energy         1   16.9276 690.65 311.91
## - age:length_sec      1   17.0886 690.81 311.97
## - gender:length_sec   1   18.2172 691.94 312.37
##
## Step: AIC=306.22

```

```

## rating ~ age + gender + race + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##     race:energy + race:bpm + race:length_sec + gender:energy +
##     gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - danceability      1    0.6482 675.45 304.46
## - gender:energy      1    1.0456 675.85 304.60
## - race:length_sec    1    2.6561 677.46 305.19
## - age:energy         1    3.3828 678.19 305.45
## <none>                674.80 306.23
## - race:genre        3   18.0765 692.88 306.70
## - race:bpm          1    7.0787 681.88 306.78
## - loudness_db       1    7.4797 682.28 306.93
## - age:happiness     1    9.8679 684.67 307.78
## - age:bpm           1   12.2391 687.04 308.63
## - age:genre         3   25.6748 700.48 309.37
## - age:length_sec    1   16.0048 690.81 309.97
## - race:energy       1   16.9075 691.71 310.29
## - gender:length_sec  1   17.8339 692.64 310.62
##
## Step:  AIC=304.46
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##     loudness_db + length_sec + age:genre + age:energy + age:happiness +
##     age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##     race:length_sec + gender:energy + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - gender:energy      1    1.0840 676.53 302.85
## - race:length_sec    1    2.4834 677.93 303.36
## - age:energy         1    3.3649 678.82 303.68
## <none>                675.45 304.46
## - race:genre        3   17.6009 693.05 304.76
## - race:bpm          1    7.2391 682.69 305.07
## - loudness_db       1    7.3376 682.79 305.11
## - age:happiness     1    9.7081 685.16 305.96
## - age:bpm           1   12.3161 687.77 306.89
## - age:genre         3   25.2517 700.70 307.45
## - age:length_sec    1   15.9393 691.39 308.17
## - race:energy       1   16.7780 692.23 308.47
## - gender:length_sec  1   18.1069 693.56 308.94
##
## Step:  AIC=302.85
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##     loudness_db + length_sec + age:genre + age:energy + age:happiness +
##     age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##     race:length_sec + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - race:length_sec    1    2.5400 679.07 301.77
## - age:energy         1    3.2509 679.79 302.03
## <none>                676.53 302.85
## - race:genre        3   18.5325 695.07 303.47

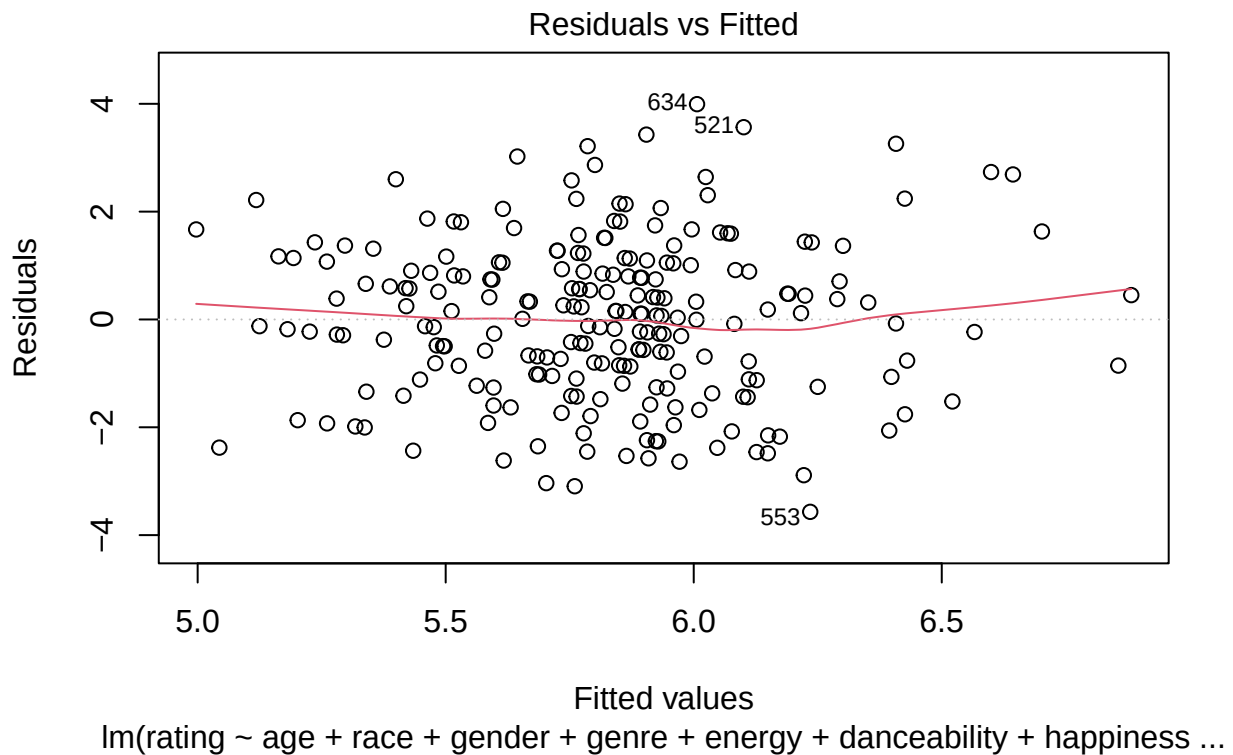
```

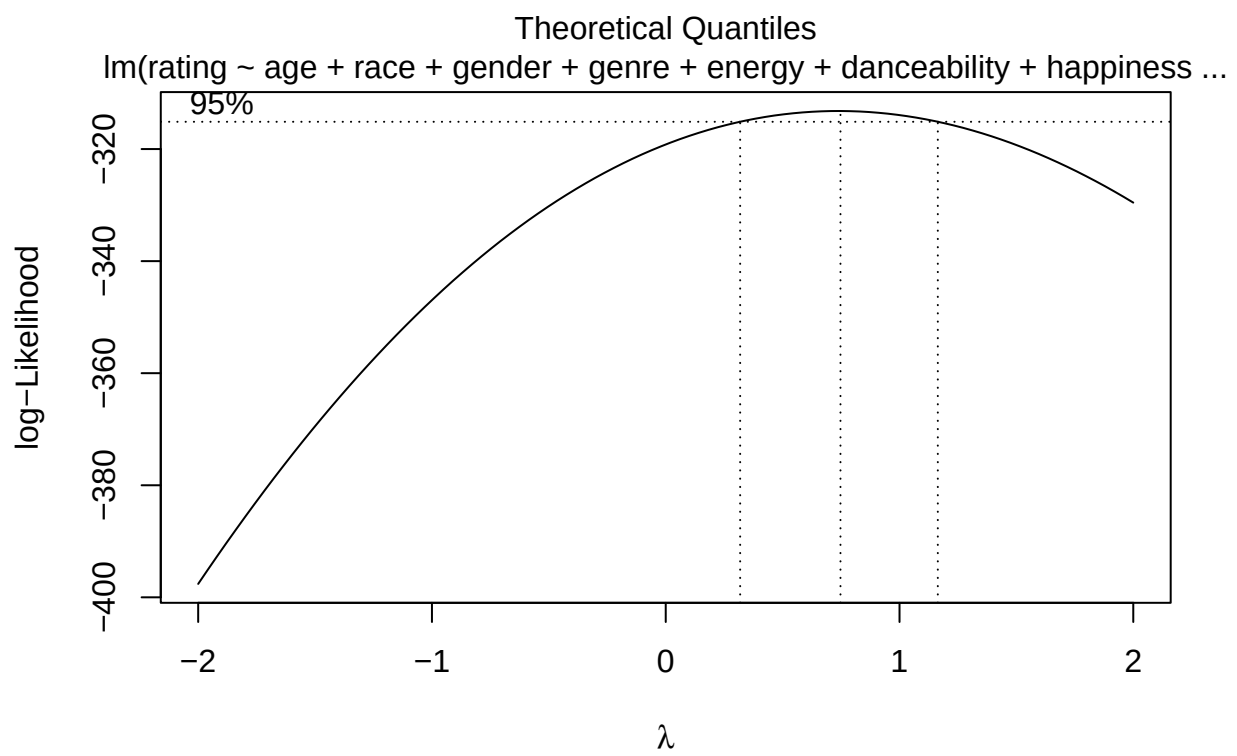
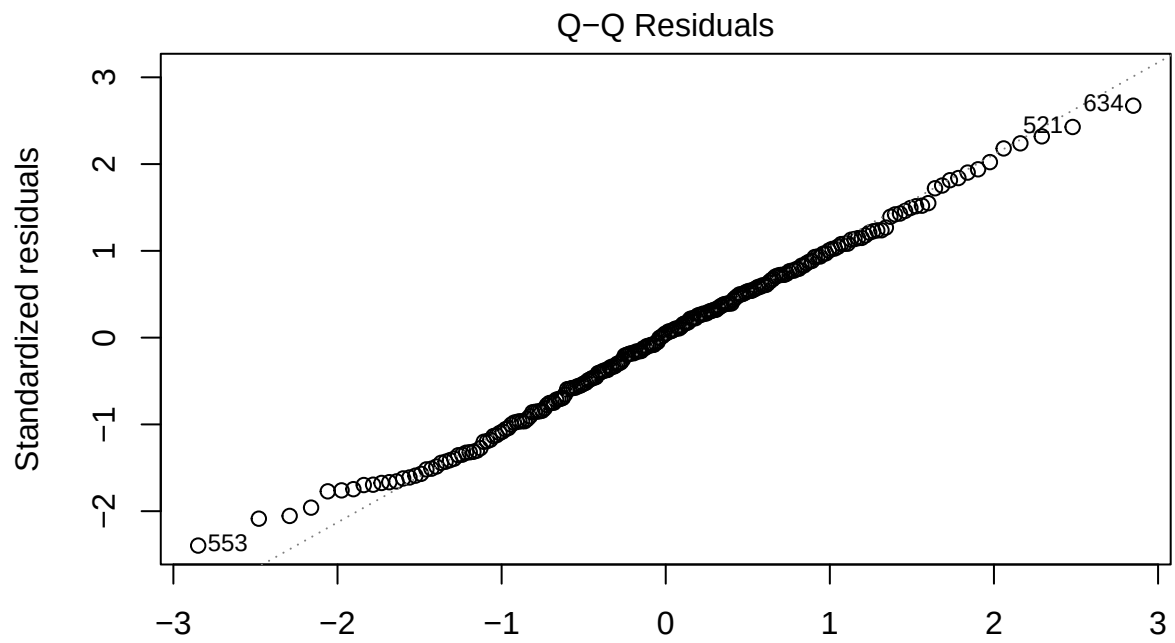
```

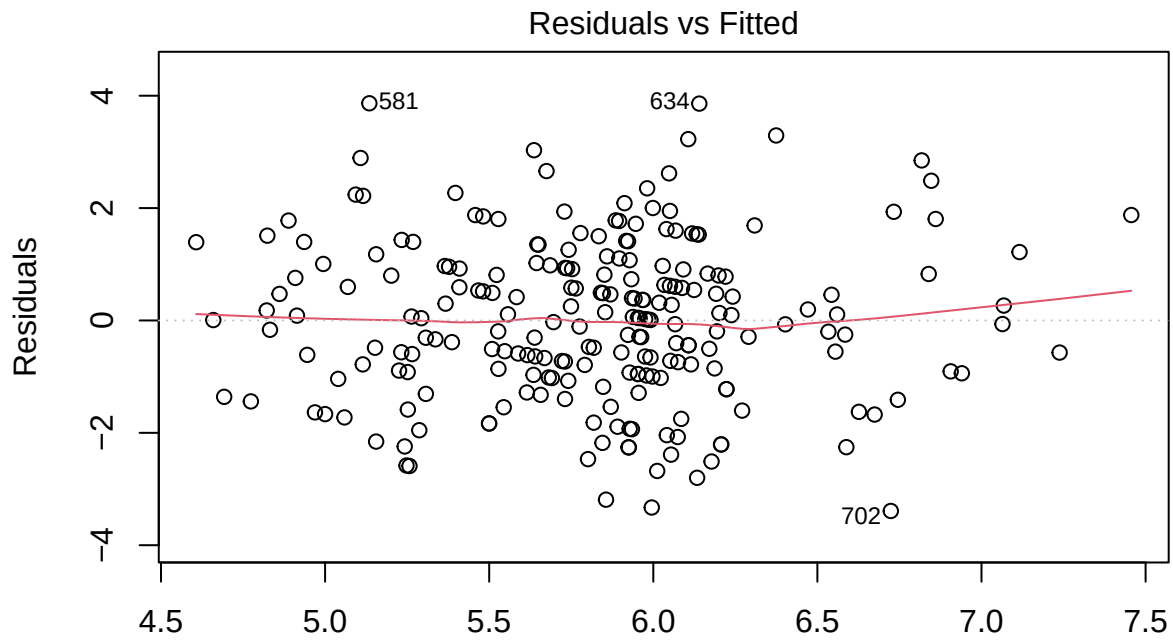
## - race:bpm          1    7.7735 684.31 303.65
## - loudness_db       1    8.4575 684.99 303.90
## - age:happiness     1    9.4000 685.93 304.23
## - age:bpm          1   12.2406 688.78 305.25
## - age:genre        3   24.6657 701.20 305.63
## - age:length_sec   1   15.6285 692.16 306.45
## - race:energy       1   16.3123 692.85 306.69
## - gender:length_sec 1   18.0731 694.61 307.31
##
## Step:  AIC=301.77
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##         loudness_db + length_sec + age:genre + age:energy + age:happiness +
##         age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##         gender:length_sec
##
##           Df Sum of Sq    RSS    AIC
## - age:energy      1    3.6200 682.69 301.07
## - race:genre      3   16.7212 695.80 301.73
## <none>              679.07 301.77
## - race:bpm        1    7.1994 686.27 302.36
## - age:happiness    1    8.3054 687.38 302.75
## - loudness_db      1    8.6956 687.77 302.89
## - age:bpm          1   12.2490 691.32 304.15
## - age:genre        3   24.0867 703.16 304.31
## - race:energy       1   14.9064 693.98 305.09
## - age:length_sec   1   16.2469 695.32 305.56
## - gender:length_sec 1   16.4467 695.52 305.63
##
## Step:  AIC=301.07
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##         loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##         age:length_sec + race:genre + race:energy + race:bpm + gender:length_sec
##
##           Df Sum of Sq    RSS    AIC
## - race:genre      3   16.5383 699.23 300.94
## <none>              682.69 301.07
## - race:bpm        1    6.1217 688.82 301.26
## - loudness_db      1    7.8042 690.50 301.86
## - age:genre        3   21.4854 704.18 302.67
## - age:bpm          1   11.7041 694.40 303.24
## - race:energy       1   14.0155 696.71 304.05
## - gender:length_sec 1   16.6269 699.32 304.97
## - age:length_sec    1   17.3916 700.09 305.24
## - age:happiness     1   23.6421 706.34 307.41
##
## Step:  AIC=300.94
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##         loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##         age:length_sec + race:energy + race:bpm + gender:length_sec
##
##           Df Sum of Sq    RSS    AIC
## - race:bpm        1    3.8475 703.08 300.28
## <none>              699.23 300.94
## - age:bpm          1    6.7730 706.01 301.30

```

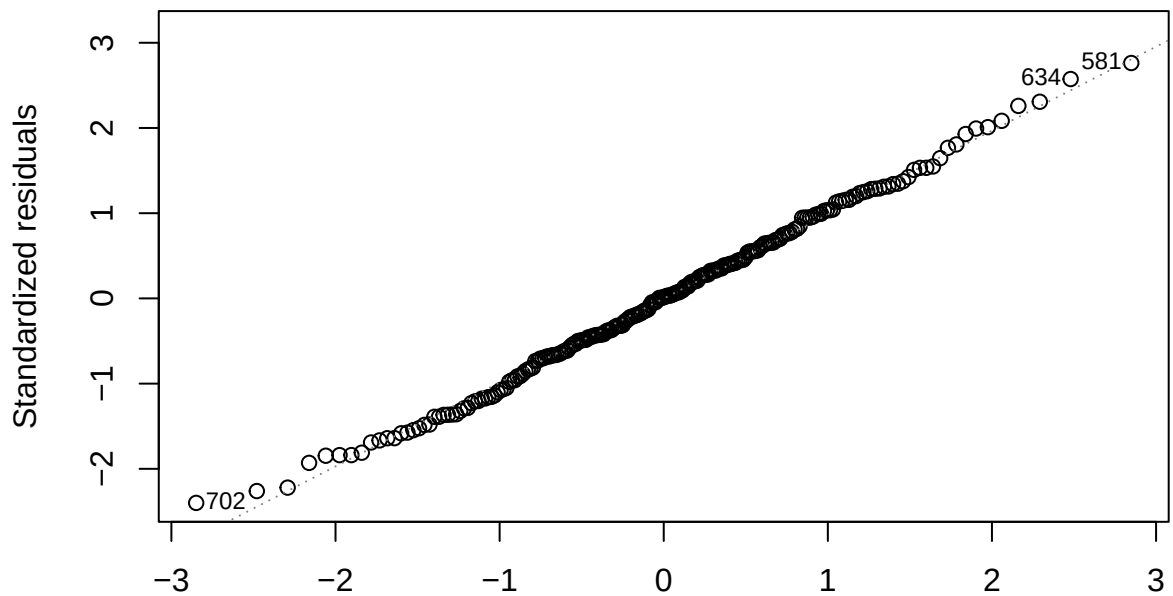
```
## - loudness_db      1      7.1708 706.40 301.44
## - race:energy      1      8.1624 707.40 301.78
## - age:genre        3     20.3701 719.60 301.97
## - age:happiness    1     19.1919 718.43 305.57
## - gender:length_sec 1     19.3127 718.55 305.61
## - age:length_sec   1     21.0050 720.24 306.19
##
## Step: AIC=300.28
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##      age:length_sec + race:energy + gender:length_sec
##
##              Df Sum of Sq  RSS   AIC
## <none>                703.08 300.28
## - loudness_db        1    5.8449 708.93 300.31
## - age:bpm            1    7.7319 710.81 300.96
## - race:energy        1    7.8422 710.92 301.00
## - age:genre          3   19.6097 722.69 301.02
## - age:happiness      1   17.9609 721.04 304.46
## - gender:length_sec  1   19.5122 722.59 304.99
## - age:length_sec     1   21.1947 724.28 305.56
## [1] "1970s"
```



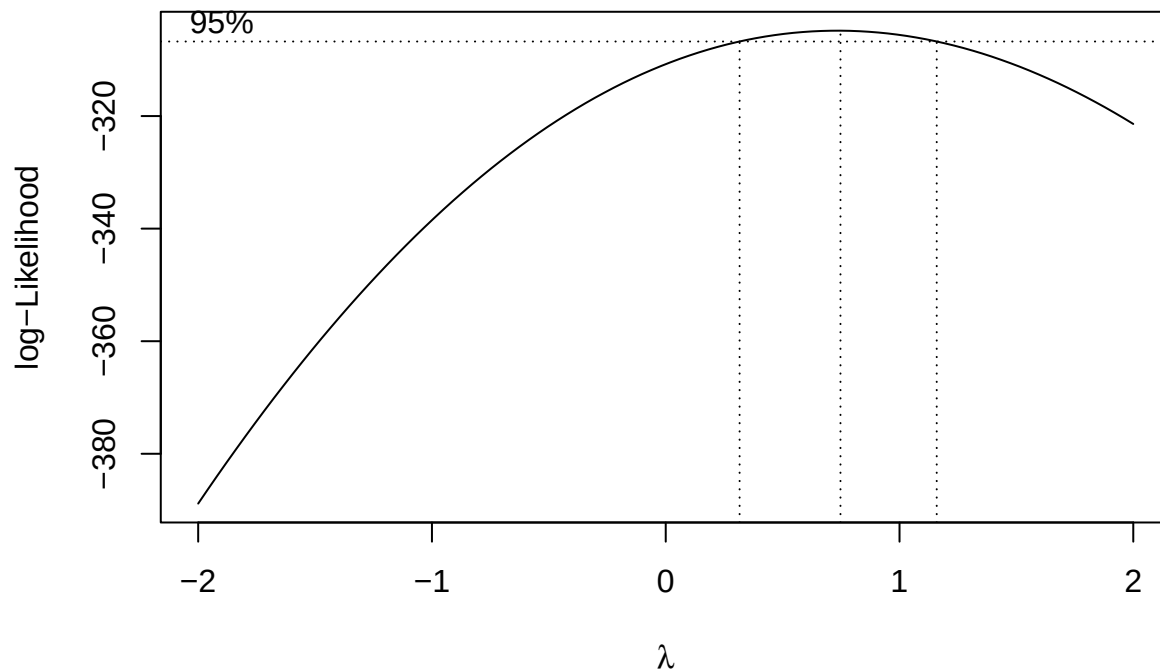




Fitted values
 $\text{lm}(\text{rating} \sim \text{age} + \text{gender} + \text{race} + \text{age} * \text{genre} + \text{age} * \text{energy} + \text{age} * \text{dancea} \dots)$
 Q-Q Residuals



Theoretical Quantiles
 $\text{lm}(\text{rating} \sim \text{age} + \text{gender} + \text{race} + \text{age} * \text{genre} + \text{age} * \text{energy} + \text{age} * \text{dancea} \dots)$



```
## Start: AIC=231.98
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##
##           Df Sum of Sq  RSS   AIC
## - age:genre      2    1.9499 461.86 228.95
## - age:energy      1     0.0012 459.91 229.98
## - age:bpm         1     0.0300 459.94 230.00
## - gender:length_sec 1     0.0874 460.00 230.03
## - age:length_sec   1     0.1084 460.02 230.04
## - age:danceability 1     0.1513 460.06 230.06
## - gender:bpm       1     0.1817 460.09 230.07
## - race:energy       1     0.2867 460.19 230.12
## - race:loudness_db  1     0.2893 460.20 230.13
## - gender:danceability 1     0.3624 460.27 230.16
## - gender:happiness  1     0.3944 460.30 230.18
## - age:loudness_db   1     0.4693 460.38 230.22
## - race:happiness    1     0.5862 460.49 230.27
## - race:danceability 1     0.6606 460.57 230.31
## - gender:loudness_db 1     0.9713 460.88 230.46
## - gender:genre      2     5.2307 465.14 230.56
## - gender:energy      1     1.5799 461.49 230.76
## - age:happiness      1     2.2112 462.12 231.08
## - race:length_sec    1     2.6993 462.61 231.32
## <none>                                459.91 231.98
## - race:bpm          1     4.0558 463.96 231.99
## - race:genre         2     9.6085 469.52 232.70
```

```

##
## Step: AIC=228.95
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:bpm 1 0.0045 461.86 226.95
## - age:energy 1 0.0295 461.89 226.96
## - gender:length_sec 1 0.0540 461.91 226.97
## - age:danceability 1 0.1190 461.98 227.01
## - age:length_sec 1 0.1455 462.00 227.02
## - race:loudness_db 1 0.2041 462.06 227.05
## - race:energy 1 0.2552 462.11 227.07
## - gender:bpm 1 0.2679 462.13 227.08
## - gender:danceability 1 0.4433 462.30 227.17
## - race:happiness 1 0.5201 462.38 227.20
## - race:danceability 1 0.5212 462.38 227.21
## - gender:genre 2 4.6735 466.53 227.24
## - gender:happiness 1 0.6296 462.49 227.26
## - age:loudness_db 1 0.6516 462.51 227.27
## - gender:loudness_db 1 1.2440 463.10 227.56
## - gender:energy 1 1.6035 463.46 227.74
## - age:happiness 1 2.3741 464.23 228.12
## - race:length_sec 1 2.5741 464.43 228.22
## <none> 461.86 228.95
## - race:bpm 1 4.3388 466.20 229.08
## - race:genre 2 10.2311 472.09 229.94
##
## Step: AIC=226.95
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:loudness_db + age:length_sec +
## race:genre + race:energy + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:genre +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:energy 1 0.0261 461.89 224.96
## - gender:length_sec 1 0.0523 461.91 224.98
## - age:danceability 1 0.1262 461.99 225.01
## - age:length_sec 1 0.1489 462.01 225.02
## - race:loudness_db 1 0.2017 462.06 225.05
## - race:energy 1 0.2510 462.11 225.07
## - gender:bpm 1 0.2874 462.15 225.09
## - gender:danceability 1 0.4557 462.32 225.18
## - race:danceability 1 0.5182 462.38 225.21
## - race:happiness 1 0.5209 462.38 225.21
## - gender:genre 2 4.6699 466.53 225.24

```

```

## - gender:happiness      1      0.6337 462.50 225.26
## - age:loudness_db       1      0.6645 462.53 225.28
## - gender:loudness_db    1      1.2642 463.13 225.57
## - gender:energy         1      1.5993 463.46 225.74
## - age:happiness         1      2.3730 464.24 226.12
## - race:length_sec       1      2.5699 464.43 226.22
## <none>                  461.86 226.95
## - race:bpm              1      4.4004 466.26 227.11
## - race:genre            2     10.3181 472.18 227.99
##
## Step:  AIC=224.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:happiness + age:loudness_db + age:length_sec + race:genre +
##      race:energy + race:danceability + race:happiness + race:bpm +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:length_sec      1    0.0538 461.94 222.99
## - age:danceability       1    0.1239 462.01 223.02
## - age:length_sec         1    0.1623 462.05 223.04
## - race:loudness_db       1    0.1880 462.08 223.06
## - race:energy            1    0.2289 462.12 223.08
## - gender:bpm             1    0.2800 462.17 223.10
## - gender:danceability    1    0.4581 462.35 223.19
## - race:happiness         1    0.4974 462.39 223.21
## - race:danceability      1    0.5180 462.41 223.22
## - gender:genre           2    4.6438 466.53 223.24
## - gender:happiness       1    0.6203 462.51 223.27
## - gender:loudness_db     1    1.2404 463.13 223.57
## - gender:energy         1    1.5733 463.46 223.74
## - age:loudness_db        1    2.1843 464.07 224.04
## - race:length_sec        1    2.5787 464.47 224.23
## - age:happiness          1    2.8850 464.77 224.38
## <none>                   461.89 224.96
## - race:bpm              1    4.4130 466.30 225.13
## - race:genre            2   10.3089 472.20 226.00
##
## Step:  AIC=222.99
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:happiness + age:loudness_db + age:length_sec + race:genre +
##      race:energy + race:danceability + race:happiness + race:bpm +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq   RSS   AIC
## - age:danceability       1    0.1331 462.08 221.06
## - age:length_sec         1    0.1717 462.11 221.07
## - race:loudness_db       1    0.1770 462.12 221.08
## - race:energy            1    0.2288 462.17 221.10
## - gender:bpm             1    0.3042 462.25 221.14

```

```

## - gender:danceability 1 0.4512 462.39 221.21
## - race:happiness 1 0.5282 462.47 221.25
## - race:danceability 1 0.5683 462.51 221.27
## - gender:genre 2 4.6989 466.64 221.30
## - gender:happiness 1 0.7816 462.72 221.38
## - gender:loudness_db 1 1.3202 463.26 221.64
## - gender:energy 1 1.7167 463.66 221.84
## - age:loudness_db 1 2.1634 464.11 222.06
## - race:length_sec 1 2.5613 464.50 222.25
## - age:happiness 1 2.8505 464.79 222.39
## <none> 461.94 222.99
## - race:bpm 1 4.3683 466.31 223.13
## - race:genre 2 10.2581 472.20 224.00
##
## Step: AIC=221.06
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:happiness +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:genre + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db
##
## Df Sum of Sq RSS AIC
## - race:loudness_db 1 0.1780 462.25 219.14
## - age:length_sec 1 0.1984 462.27 219.15
## - race:energy 1 0.2120 462.29 219.16
## - gender:bpm 1 0.2824 462.36 219.19
## - gender:danceability 1 0.4577 462.53 219.28
## - race:happiness 1 0.5060 462.58 219.31
## - race:danceability 1 0.5069 462.58 219.31
## - gender:happiness 1 0.7686 462.84 219.43
## - gender:genre 2 4.9016 466.98 219.46
## - gender:loudness_db 1 1.2314 463.31 219.66
## - gender:energy 1 1.6534 463.73 219.87
## - age:loudness_db 1 2.0440 464.12 220.06
## - race:length_sec 1 2.6033 464.68 220.34
## - age:happiness 1 3.9750 466.05 221.01
## <none> 462.08 221.06
## - race:bpm 1 4.2954 466.37 221.16
## - race:genre 2 10.2976 472.37 222.08
##
## Step: AIC=219.14
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:happiness +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db
##
## Df Sum of Sq RSS AIC
## - race:energy 1 0.0542 462.31 217.17
## - age:length_sec 1 0.2032 462.46 217.24
## - gender:bpm 1 0.2909 462.54 217.29
## - race:happiness 1 0.3706 462.62 217.33

```

```

## - race:danceability      1      0.4256 462.68 217.35
## - gender:danceability    1      0.5276 462.78 217.40
## - gender:genre           2      4.7716 467.02 217.48
## - gender:happiness       1      0.7445 463.00 217.51
## - gender:loudness_db     1      1.3677 463.62 217.82
## - gender:energy          1      1.7275 463.98 217.99
## - age:loudness_db        1      2.1081 464.36 218.18
## - race:length_sec        1      2.4861 464.74 218.37
## - age:happiness          1      3.9373 466.19 219.08
## <none>                   462.25 219.14
## - race:bpm               1      4.2157 466.47 219.21
## - race:genre             2     10.1449 472.40 220.09
##
## Step:  AIC=217.17
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:happiness +
##         age:loudness_db + age:length_sec + race:genre + race:danceability +
##         race:happiness + race:bpm + race:length_sec + gender:genre +
##         gender:energy + gender:danceability + gender:happiness +
##         gender:bpm + gender:loudness_db
##
##              Df Sum of Sq  RSS    AIC
## - age:length_sec      1      0.1868 462.49 215.26
## - gender:bpm           1      0.2846 462.59 215.31
## - race:happiness       1      0.3169 462.62 215.33
## - race:danceability    1      0.3881 462.70 215.36
## - gender:danceability  1      0.5096 462.82 215.42
## - gender:genre         2      4.9490 467.26 215.60
## - gender:happiness     1      0.8685 463.18 215.60
## - gender:loudness_db   1      1.3731 463.68 215.85
## - gender:energy        1      1.8062 464.11 216.06
## - age:loudness_db      1      2.0596 464.37 216.18
## - race:length_sec      1      2.4341 464.74 216.37
## - age:happiness        1      3.8898 466.20 217.08
## <none>                 462.31 217.17
## - race:bpm            1      4.3531 466.66 217.31
## - race:genre           2     10.5639 472.87 218.32
##
## Step:  AIC=215.26
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:happiness +
##         age:loudness_db + race:genre + race:danceability + race:happiness +
##         race:bpm + race:length_sec + gender:genre + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq  RSS    AIC
## - gender:bpm           1      0.2664 462.76 213.39
## - race:happiness       1      0.3265 462.82 213.42
## - race:danceability    1      0.4058 462.90 213.46
## - gender:danceability  1      0.4601 462.95 213.49
## - gender:happiness     1      0.8225 463.32 213.67
## - gender:genre         2      4.9349 467.43 213.68
## - gender:loudness_db   1      1.3852 463.88 213.94
## - gender:energy        1      1.8147 464.31 214.16

```

```

## - age:loudness_db      1      2.0156 464.51 214.25
## - race:length_sec      1      2.7952 465.29 214.64
## <none>                  462.49 215.26
## - race:bpm             1      4.4195 466.91 215.43
## - age:happiness         1      4.4987 466.99 215.47
## - race:genre            2     10.4449 472.94 216.35
##
## Step:  AIC=213.39
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:danceability + race:happiness +
##      race:bpm + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.3541 463.11 211.57
## - gender:danceability  1      0.3876 463.15 211.58
## - race:danceability    1      0.4207 463.18 211.60
## - gender:happiness     1      0.8365 463.60 211.81
## - gender:genre         2      5.1465 467.91 211.91
## - gender:loudness_db   1      1.2417 464.00 212.00
## - gender:energy        1      1.6599 464.42 212.21
## - age:loudness_db      1      1.8410 464.60 212.30
## - race:length_sec      1      3.0825 465.84 212.91
## <none>                  462.76 213.39
## - race:bpm             1      4.5066 467.27 213.60
## - age:happiness         1      4.5776 467.34 213.64
## - race:genre            2     10.4590 473.22 214.49
##
## Step:  AIC=211.57
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:danceability + race:bpm +
##      race:length_sec + gender:genre + gender:energy + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability    1      0.1513 463.27 209.64
## - gender:danceability  1      0.3694 463.48 209.75
## - gender:happiness     1      0.9129 464.03 210.02
## - gender:genre         2      5.1128 468.23 210.07
## - gender:loudness_db   1      1.1987 464.31 210.16
## - age:loudness_db      1      1.7438 464.86 210.42
## - gender:energy        1      1.7740 464.89 210.44
## - race:length_sec      1      2.8635 465.98 210.97
## <none>                  463.11 211.57
## - age:happiness        1      4.3610 467.48 211.70
## - race:bpm             1      4.9196 468.03 211.98
## - race:genre            2     10.7089 473.82 212.78
##
## Step:  AIC=209.64
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:bpm + race:length_sec +

```



```

##      gender:genre + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:danceability  1      0.3893 463.66 207.83
## - gender:happiness     1      0.9126 464.18 208.09
## - gender:loudness_db   1      1.1555 464.42 208.21
## - gender:genre         2      5.4341 468.70 208.30
## - gender:energy        1      1.7125 464.98 208.48
## - age:loudness_db      1      1.7312 465.00 208.49
## - race:length_sec      1      3.1413 466.41 209.18
## <none>                  463.27 209.64
## - age:happiness        1      4.5633 467.83 209.88
## - race:bpm             1      4.7694 468.04 209.98
## - race:genre           2     10.9350 474.20 210.96
##
## Step:  AIC=207.83
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:bpm + race:length_sec +
##      gender:genre + gender:energy + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - danceability         1      0.7050 464.36 206.18
## - gender:genre          2      5.0799 468.74 206.32
## - gender:loudness_db    1      1.0823 464.74 206.37
## - age:loudness_db       1      1.6314 465.29 206.63
## - gender:energy         1      1.7430 465.40 206.69
## - gender:happiness      1      2.2347 465.89 206.93
## - race:length_sec       1      3.4994 467.15 207.55
## <none>                  463.66 207.83
## - age:happiness         1      4.5415 468.20 208.06
## - race:bpm              1      4.6745 468.33 208.12
## - race:genre            2     11.4150 475.07 209.38
##
## Step:  AIC=206.18
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:happiness + age:loudness_db +
##      race:genre + race:bpm + race:length_sec + gender:genre +
##      gender:energy + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db    1      1.3663 465.73 204.85
## - age:loudness_db       1      1.3747 465.74 204.85
## - gender:genre          2      5.5955 469.96 204.91
## - gender:energy         1      2.2257 466.59 205.27
## - gender:happiness      1      2.7064 467.07 205.50
## - race:length_sec       1      3.5065 467.87 205.90
## <none>                  464.36 206.18
## - age:happiness         1      4.4755 468.84 206.37
## - race:bpm              1      5.5505 469.91 206.89
## - race:genre            2     12.6578 477.02 208.31
##
## Step:  AIC=204.85

```

```

## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:happiness + age:loudness_db +
##   race:genre + race:bpm + race:length_sec + gender:genre +
##   gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - gender:energy      1    0.9021 466.63 203.29
## - age:loudness_db     1    1.2394 466.97 203.46
## - gender:happiness    1    2.0811 467.81 203.87
## - gender:genre        2    6.5221 472.25 204.02
## - race:length_sec     1    3.9711 469.70 204.79
## <none>                465.73 204.85
## - age:happiness       1    4.1688 469.90 204.88
## - race:bpm            1    5.5953 471.32 205.57
## - race:genre          2   13.7119 479.44 207.47
##
## Step:  AIC=203.29
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:happiness + age:loudness_db +
##   race:genre + race:bpm + race:length_sec + gender:genre +
##   gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - energy              1    0.6261 467.25 201.60
## - gender:happiness    1    1.2457 467.87 201.90
## - age:loudness_db     1    1.4788 468.11 202.01
## - gender:genre        2    5.9962 472.62 202.20
## - race:length_sec     1    3.4423 470.07 202.97
## - age:happiness       1    3.9814 470.61 203.23
## <none>                466.63 203.29
## - race:bpm            1    5.2795 471.91 203.86
## - race:genre          2   14.2021 480.83 206.13
##
## Step:  AIC=201.6
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
##   length_sec + age:happiness + age:loudness_db + race:genre +
##   race:bpm + race:length_sec + gender:genre + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - gender:happiness    1    1.3568 468.61 200.26
## - age:loudness_db     1    1.3626 468.62 200.26
## - gender:genre        2    6.2055 473.46 200.60
## - age:happiness       1    3.8434 471.10 201.46
## - race:length_sec     1    3.8805 471.14 201.48
## <none>                467.25 201.60
## - race:bpm            1    5.4241 472.68 202.23
## - race:genre          2   13.7123 480.97 204.19
##
## Step:  AIC=200.26
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
##   length_sec + age:happiness + age:loudness_db + race:genre +
##   race:bpm + race:length_sec + gender:genre
##
##           Df Sum of Sq   RSS   AIC

```

```

## - age:loudness_db 1 1.4500 470.06 198.96
## - gender:genre 2 5.6214 474.23 198.98
## - race:length_sec 1 3.8041 472.42 200.10
## <none> 468.61 200.26
## - age:happiness 1 5.2600 473.87 200.80
## - race:bpm 1 5.4847 474.10 200.91
## - race:genre 2 13.9050 482.52 202.93
##
## Step: AIC=198.96
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
## length_sec + age:happiness + race:genre + race:bpm + race:length_sec +
## gender:genre
##
## Df Sum of Sq RSS AIC
## - loudness_db 1 1.8844 471.95 197.87
## - gender:genre 2 6.1122 476.17 197.91
## <none> 470.06 198.96
## - age:happiness 1 4.1987 474.26 198.99
## - race:length_sec 1 4.4861 474.55 199.13
## - race:bpm 1 5.7202 475.78 199.72
## - race:genre 2 13.6185 483.68 201.47
##
## Step: AIC=197.87
## rating ~ age + gender + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec +
## gender:genre
##
## Df Sum of Sq RSS AIC
## - gender:genre 2 6.0005 477.95 196.75
## - age:happiness 1 3.9888 475.93 197.79
## <none> 471.95 197.87
## - race:length_sec 1 5.4211 477.37 198.48
## - race:bpm 1 7.4563 479.40 199.45
## - race:genre 2 13.6851 485.63 200.39
##
## Step: AIC=196.75
## rating ~ age + gender + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - gender 1 3.2575 481.20 196.30
## - age:happiness 1 3.3913 481.34 196.37
## <none> 477.95 196.75
## - race:length_sec 1 5.9140 483.86 197.56
## - race:bpm 1 6.8757 484.82 198.01
## - race:genre 2 12.0192 489.97 198.42
##
## Step: AIC=196.3
## rating ~ age + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 3.3045 484.51 195.86
## <none> 481.20 196.30

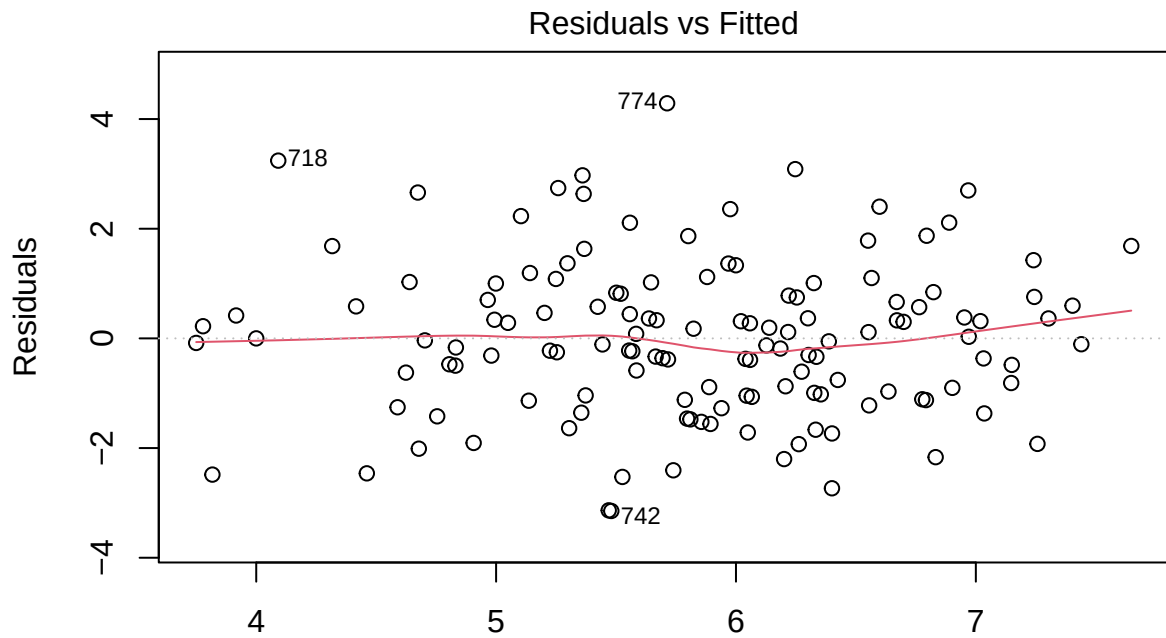
```

```

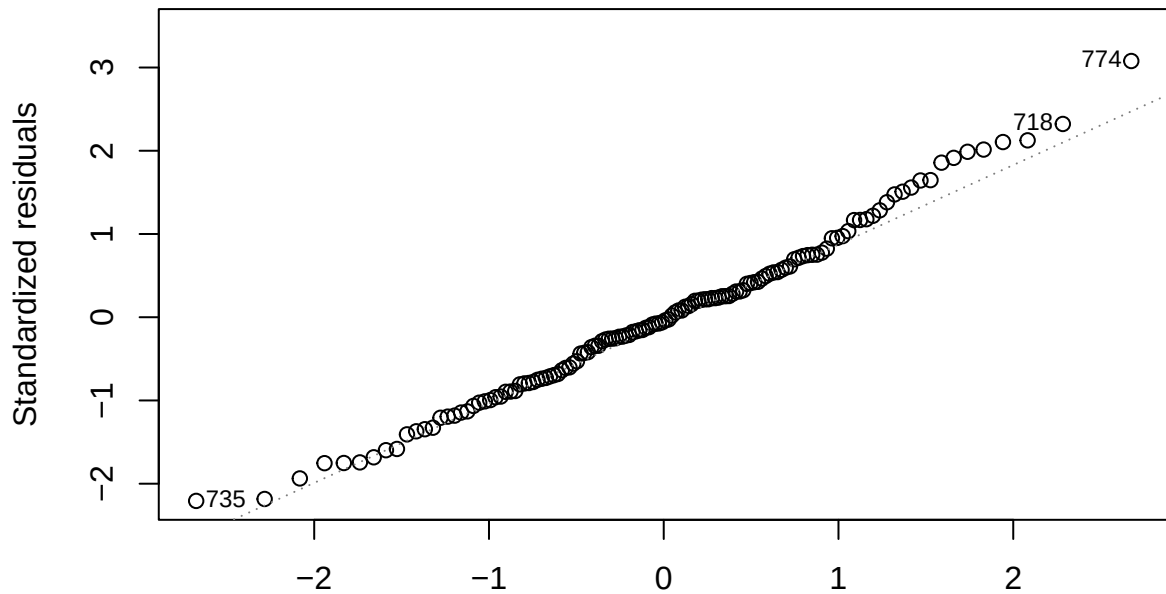
## - race:length_sec 1 5.1720 486.38 196.74
## - race:bpm 1 6.5740 487.78 197.40
## - race:genre 2 13.6543 494.86 198.68
##
## Step: AIC=195.86
## rating ~ age + race + genre + happiness + bpm + length_sec +
## race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - age 1 0.1370 484.65 193.93
## - happiness 1 0.9283 485.44 194.30
## <none> 484.51 195.86
## - race:length_sec 1 5.1063 489.61 196.25
## - race:bpm 1 7.0455 491.55 197.16
## - race:genre 2 13.8550 498.36 198.29
##
## Step: AIC=193.93
## rating ~ race + genre + happiness + bpm + length_sec + race:genre +
## race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - happiness 1 0.9290 485.57 192.37
## <none> 484.65 193.93
## - race:length_sec 1 5.1852 489.83 194.35
## - race:bpm 1 7.3218 491.97 195.35
## - race:genre 2 13.7380 498.38 196.30
##
## Step: AIC=192.37
## rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
## race:length_sec
##
## Df Sum of Sq RSS AIC
## <none> 485.57 192.37
## - race:length_sec 1 5.0687 490.64 192.73
## - race:bpm 1 6.8395 492.41 193.55
## - race:genre 2 13.6538 499.23 194.69
## [1] "1980s"

## Warning: not plotting observations with leverage one:
## 14

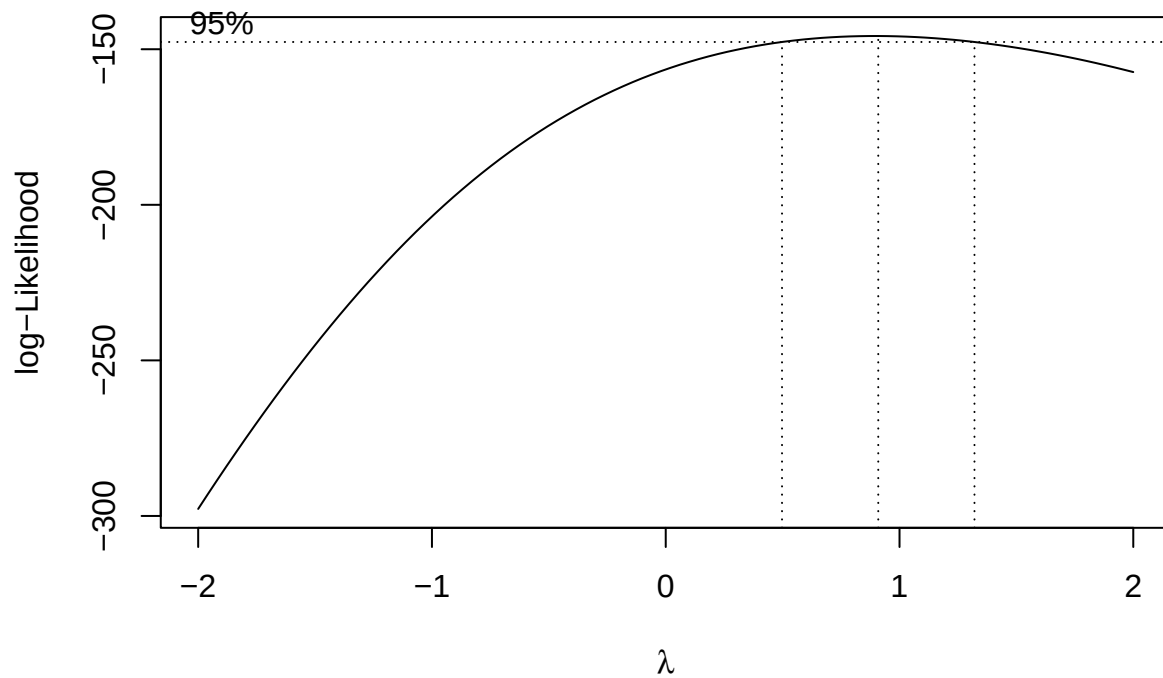
```



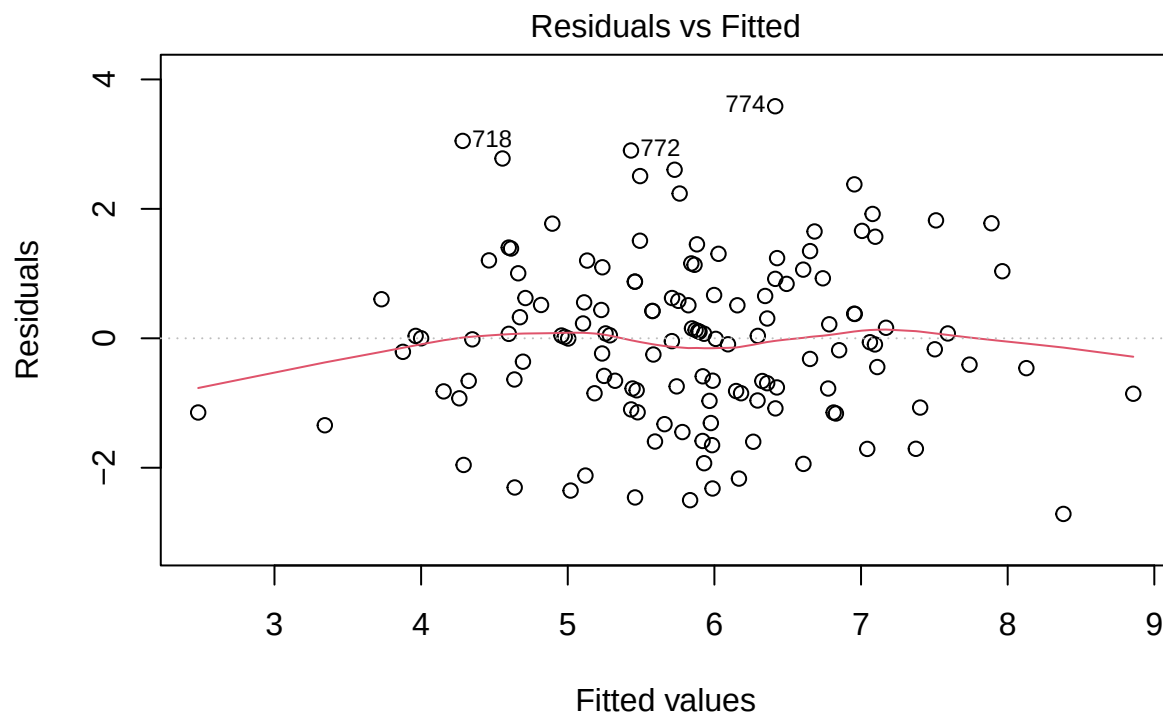
Fitted values
 $\text{lm}(\text{rating} \sim \text{age} + \text{race} + \text{gender} + \text{genre} + \text{energy} + \text{danceability} + \text{happiness} \dots)$
 Q-Q Residuals



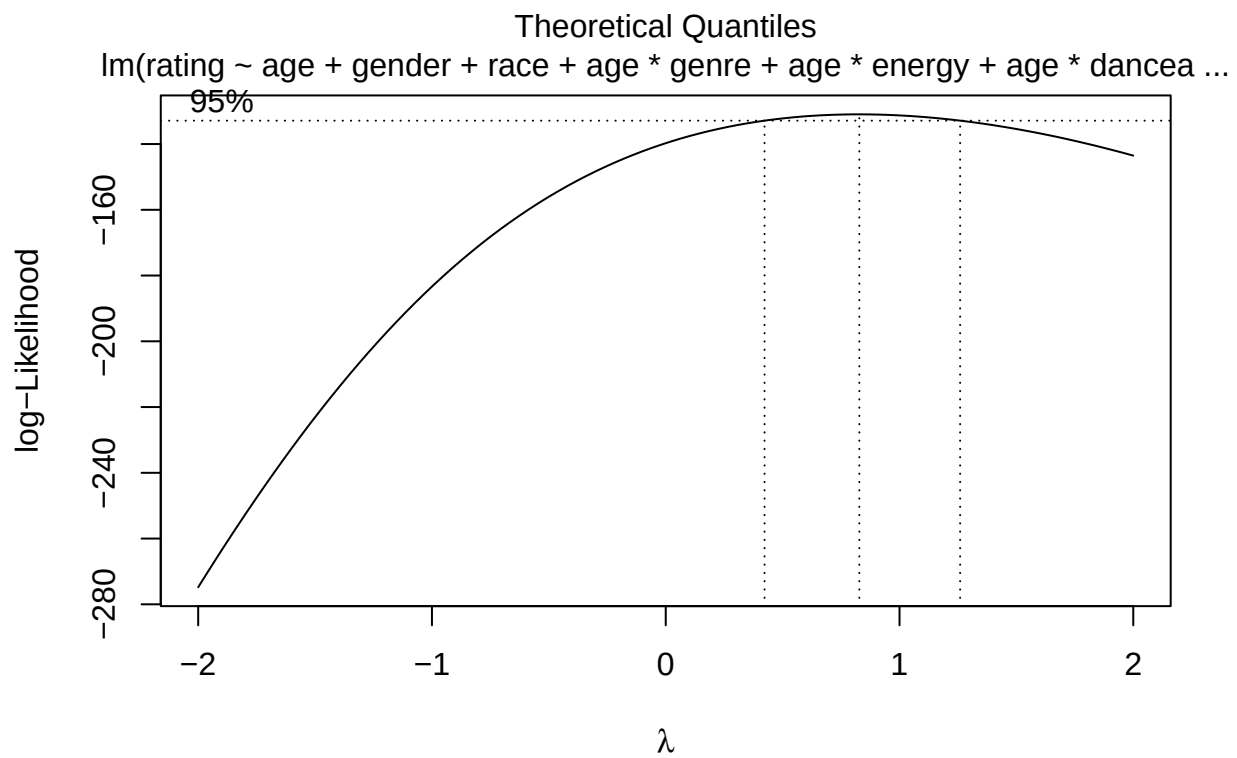
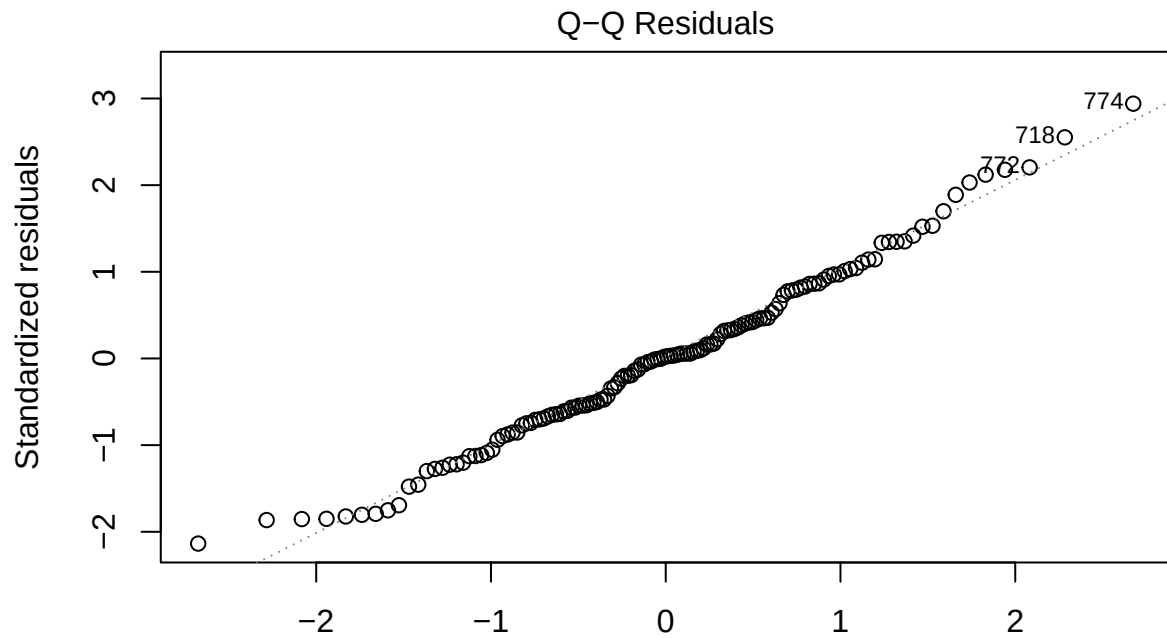
Theoretical Quantiles
 $\text{lm}(\text{rating} \sim \text{age} + \text{race} + \text{gender} + \text{genre} + \text{energy} + \text{danceability} + \text{happiness} \dots)$



```
## Warning: not plotting observations with leverage one:
## 14
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=138.39
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##              Df Sum of Sq    RSS    AIC
## - age:genre      2      1.4073 218.91 135.26
## - race:genre      2      1.9738 219.48 135.61
## - age:happiness    1      0.0000 217.50 136.39
## - race:loudness_db 1      0.0001 217.50 136.39
## - race:happiness    1      0.0116 217.52 136.39
## - race:energy       1      0.0527 217.56 136.42
## - gender:energy     1      0.2695 217.77 136.55
## - race:danceability 1      0.3419 217.85 136.60
## - age:bpm          1      0.4545 217.96 136.67
## - gender:bpm        1      0.6115 218.12 136.77
## - gender:genre      2      4.0892 221.59 136.90
## - gender:happiness   1      1.1061 218.61 137.07
## - gender:danceability 1      1.6610 219.16 137.41
## - age:length_sec     1      1.8941 219.40 137.56
## - gender:loudness_db 1      2.6071 220.11 138.00
## - gender:length_sec   1      2.7471 220.25 138.08
## - age:danceability    1      2.9717 220.48 138.22
## <none>                217.50 138.39
## - race:length_sec     1      4.0141 221.52 138.86
## - race:bpm           1      4.1272 221.63 138.93
## - age:energy          1      5.8923 223.40 140.00
## - age:loudness_db     1     13.3515 230.86 144.43
##
## Step:  AIC=135.26
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:energy +
##          age:danceability + age:happiness + age:bpm + age:loudness_db +
##          age:length_sec + race:genre + race:energy + race:danceability +
##          race:happiness + race:bpm + race:loudness_db + race:length_sec +
##          gender:genre + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:genre      2      2.5187 221.43 132.80
## - race:loudness_db 1      0.0090 218.92 133.26
## - race:happiness    1      0.0153 218.93 133.27
## - age:happiness     1      0.0256 218.94 133.27
## - race:energy       1      0.0468 218.96 133.29
## - gender:energy     1      0.1961 219.11 133.38
## - age:bpm          1      0.2465 219.16 133.41
## - race:danceability 1      0.3715 219.28 133.49
## - gender:bpm        1      0.6364 219.55 133.65
## - gender:happiness   1      1.0560 219.97 133.91
## - gender:danceability 1      1.6690 220.58 134.28
## - gender:loudness_db 1      2.3501 221.26 134.70
## - age:length_sec     1      2.7297 221.64 134.93
## <none>                218.91 135.26
## - gender:genre      2      6.6194 225.53 135.28
## - gender:length_sec   1      3.4204 222.33 135.35
## - race:bpm          1      3.8490 222.76 135.61
## - race:length_sec     1      3.8789 222.79 135.63
## - age:energy         1      7.4455 226.36 137.77
## - age:danceability    1      7.4825 226.39 137.79

```



```

## - age:loudness_db      1    15.2561 234.17 142.35
##
## Step: AIC=132.8
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:energy + race:danceability + race:happiness +
##      race:bpm + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - race:loudness_db      1      0.0193 221.45 130.81
## - race:happiness        1      0.0473 221.48 130.83
## - race:energy           1      0.0674 221.50 130.84
## - age:bpm              1      0.2249 221.66 130.94
## - gender:energy         1      0.4478 221.88 131.07
## - age:happiness         1      0.4952 221.93 131.10
## - gender:bpm           1      0.6623 222.09 131.21
## - gender:happiness      1      0.8000 222.23 131.29
## - race:danceability     1      1.2493 222.68 131.56
## - gender:danceability   1      2.5539 223.98 132.35
## - age:length_sec        1      3.0754 224.50 132.66
## - race:bpm             1      3.2492 224.68 132.77
## <none>                  221.43 132.80
## - gender:genre          2      6.8660 228.30 132.93
## - gender:loudness_db    1      3.9121 225.34 133.17
## - gender:length_sec     1      4.2846 225.72 133.39
## - race:length_sec       1      5.4507 226.88 134.09
## - age:energy            1      6.2558 227.69 134.56
## - age:danceability      1      7.3094 228.74 135.19
## - age:loudness_db      1     14.3725 235.80 139.29
##
## Step: AIC=130.81
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:energy + race:danceability + race:happiness +
##      race:bpm + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - race:energy           1      0.0484 221.50 128.84
## - race:happiness        1      0.0485 221.50 128.84
## - age:bpm              1      0.2510 221.70 128.97
## - gender:energy         1      0.4963 221.95 129.12
## - age:happiness         1      0.5329 221.98 129.14
## - gender:bpm           1      0.6513 222.10 129.21
## - gender:happiness      1      0.8008 222.25 129.30
## - race:danceability     1      1.5682 223.02 129.77
## - gender:danceability   1      2.6802 224.13 130.44
## - age:length_sec        1      3.1452 224.59 130.72
## <none>                  221.45 130.81

```

```

## - race:bpm          1    3.3796 224.83 130.86
## - gender:genre      2    6.9326 228.38 130.97
## - gender:loudness_db 1    3.9778 225.43 131.22
## - gender:length_sec 1    4.3779 225.83 131.46
## - race:length_sec   1    5.4732 226.92 132.11
## - age:energy        1    6.3724 227.82 132.64
## - age:danceability  1    7.3109 228.76 133.20
## - age:loudness_db   1   14.8871 236.34 137.60
##
## Step: AIC=128.84
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:danceability + race:happiness + race:bpm +
## race:length_sec + gender:genre + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq  RSS    AIC
## - race:happiness    1    0.1872 221.69 126.96
## - age:bpm           1    0.2447 221.74 126.99
## - gender:energy      1    0.5987 222.10 127.21
## - gender:bpm         1    0.6284 222.13 127.23
## - age:happiness      1    0.7101 222.21 127.28
## - gender:happiness   1    0.8352 222.33 127.35
## - race:danceability  1    1.5605 223.06 127.79
## - gender:danceability 1    2.6462 224.14 128.45
## - age:length_sec     1    3.1174 224.62 128.73
## <none>                221.50 128.84
## - race:bpm          1    3.3338 224.83 128.86
## - gender:genre       2    6.8872 228.38 128.98
## - gender:loudness_db 1    4.1535 225.65 129.35
## - gender:length_sec  1    4.3350 225.83 129.46
## - race:length_sec    1    5.5009 227.00 130.16
## - age:energy         1    7.2624 228.76 131.20
## - age:danceability   1    7.4297 228.93 131.30
## - age:loudness_db    1   14.8550 236.35 135.61
##
## Step: AIC=126.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:danceability + race:bpm + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq  RSS    AIC
## - age:bpm           1    0.3004 221.99 125.14
## - gender:energy      1    0.6378 222.32 125.34
## - gender:bpm         1    0.6701 222.35 125.36
## - age:happiness      1    0.6978 222.38 125.38
## - gender:happiness   1    0.8244 222.51 125.46
## - gender:danceability 1    2.7098 224.40 126.60
## - age:length_sec     1    2.9973 224.68 126.77
## - race:bpm          1    3.1516 224.84 126.86

```

```

## <none>                221.69 126.96
## - race:danceability    1    3.3507 225.04 126.98
## - gender:genre         2    7.1361 228.82 127.23
## - gender:length_sec    1    4.2697 225.95 127.53
## - gender:loudness_db   1    4.4528 226.14 127.64
## - race:length_sec      1    5.4301 227.12 128.22
## - age:danceability     1    7.3944 229.08 129.39
## - age:energy           1    7.5176 229.20 129.46
## - age:loudness_db      1   15.1325 236.82 133.87
##
## Step:  AIC=125.14
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:loudness_db + age:length_sec +
##      race:danceability + race:bpm + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:bpm      1    0.4598 222.44 123.42
## - age:happiness    1    0.6830 222.67 123.56
## - gender:energy     1    0.6873 222.67 123.56
## - gender:happiness  1    0.9378 222.92 123.71
## - gender:danceability 1    2.8667 224.85 124.87
## - age:length_sec   1    2.8738 224.86 124.88
## - race:danceability 1    3.2933 225.28 125.13
## <none>                221.99 125.14
## - race:bpm         1    3.8355 225.82 125.45
## - gender:genre      2    7.9091 229.89 125.87
## - gender:loudness_db 1    4.6268 226.61 125.92
## - gender:length_sec 1    4.6550 226.64 125.94
## - race:length_sec   1    5.9976 227.98 126.74
## - age:energy        1    7.7882 229.77 127.80
## - age:danceability  1    8.4810 230.47 128.20
## - age:loudness_db   1   15.6726 237.66 132.35
##
## Step:  AIC=123.42
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:loudness_db + age:length_sec +
##      race:danceability + race:bpm + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - age:happiness    1    0.5789 223.02 121.77
## - gender:energy     1    0.8700 223.31 121.95
## - gender:happiness  1    1.0454 223.49 122.05
## - age:length_sec    1    2.8525 225.30 123.14
## - gender:danceability 1    3.1054 225.55 123.29
## - race:danceability 1    3.1871 225.63 123.34
## <none>                222.44 123.42
## - race:bpm         1    3.6837 226.13 123.64
## - gender:length_sec 1    4.8237 227.27 124.32

```

```

## - gender:loudness_db 1 5.1469 227.59 124.51
## - gender:genre 2 8.5771 231.02 124.53
## - race:length_sec 1 6.6767 229.12 125.41
## - age:energy 1 7.9051 230.35 126.13
## - age:danceability 1 8.4988 230.94 126.48
## - age:loudness_db 1 15.6733 238.12 130.61
##
## Step: AIC=121.77
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:loudness_db + age:length_sec + race:danceability +
## race:bpm + race:length_sec + gender:genre + gender:energy +
## gender:danceability + gender:happiness + gender:loudness_db +
## gender:length_sec
##
## Df Sum of Sq RSS AIC
## - gender:energy 1 0.6848 223.71 120.18
## - gender:happiness 1 1.0515 224.07 120.41
## - age:length_sec 1 2.4335 225.46 121.24
## - gender:danceability 1 2.8685 225.89 121.50
## - race:danceability 1 3.1073 226.13 121.64
## <none> 223.02 121.77
## - race:bpm 1 3.4976 226.52 121.87
## - gender:length_sec 1 4.4054 227.43 122.41
## - gender:loudness_db 1 4.8195 227.84 122.66
## - gender:genre 2 8.3224 231.35 122.72
## - race:length_sec 1 6.5107 229.53 123.66
## - age:danceability 1 8.1387 231.16 124.61
## - age:energy 1 12.4540 235.48 127.11
## - age:loudness_db 1 16.2506 239.28 129.26
##
## Step: AIC=120.18
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:loudness_db + age:length_sec + race:danceability +
## race:bpm + race:length_sec + gender:genre + gender:danceability +
## gender:happiness + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - gender:happiness 1 0.4453 224.15 118.45
## - age:length_sec 1 2.1938 225.90 119.50
## - gender:danceability 1 2.9711 226.68 119.97
## - race:danceability 1 3.2804 226.99 120.15
## <none> 223.71 120.18
## - race:bpm 1 3.6180 227.33 120.35
## - gender:loudness_db 1 4.1797 227.89 120.68
## - gender:length_sec 1 4.3795 228.09 120.80
## - gender:genre 2 8.2571 231.97 121.08
## - race:length_sec 1 6.6801 230.39 122.16
## - age:danceability 1 8.1040 231.81 122.99
## - age:energy 1 12.3900 236.10 125.46
## - age:loudness_db 1 15.9089 239.62 127.46
##
## Step: AIC=118.45

```

```

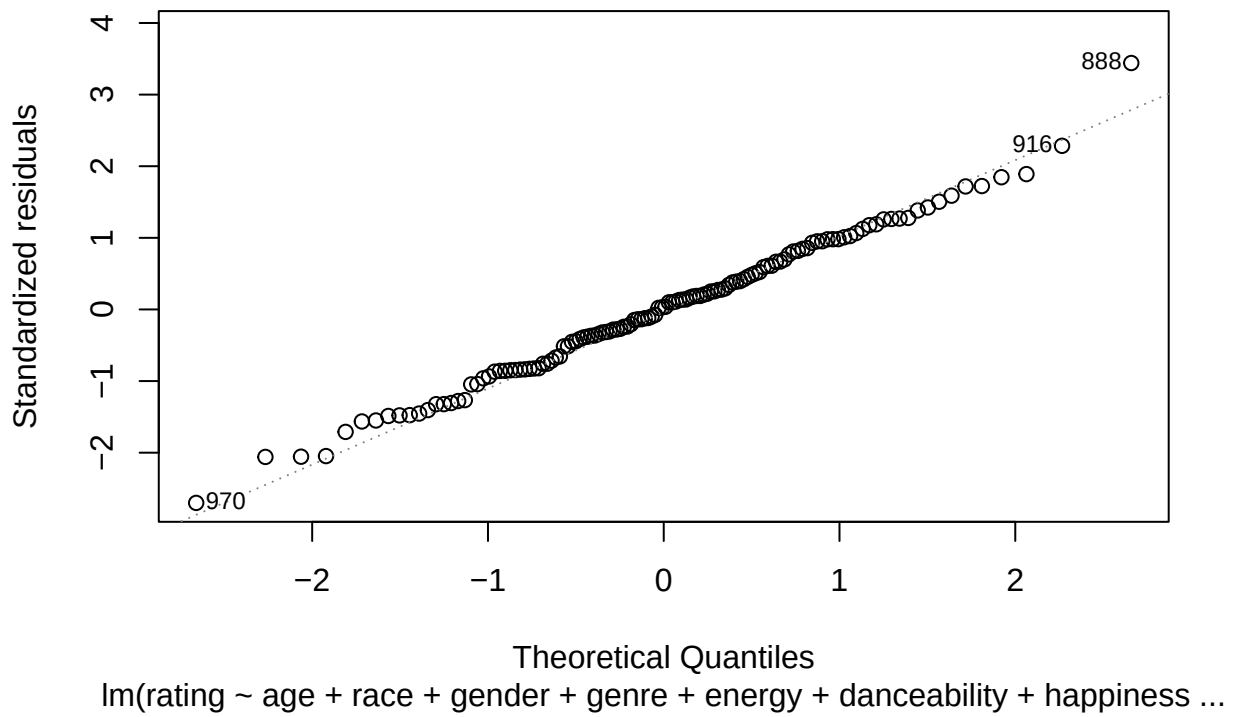
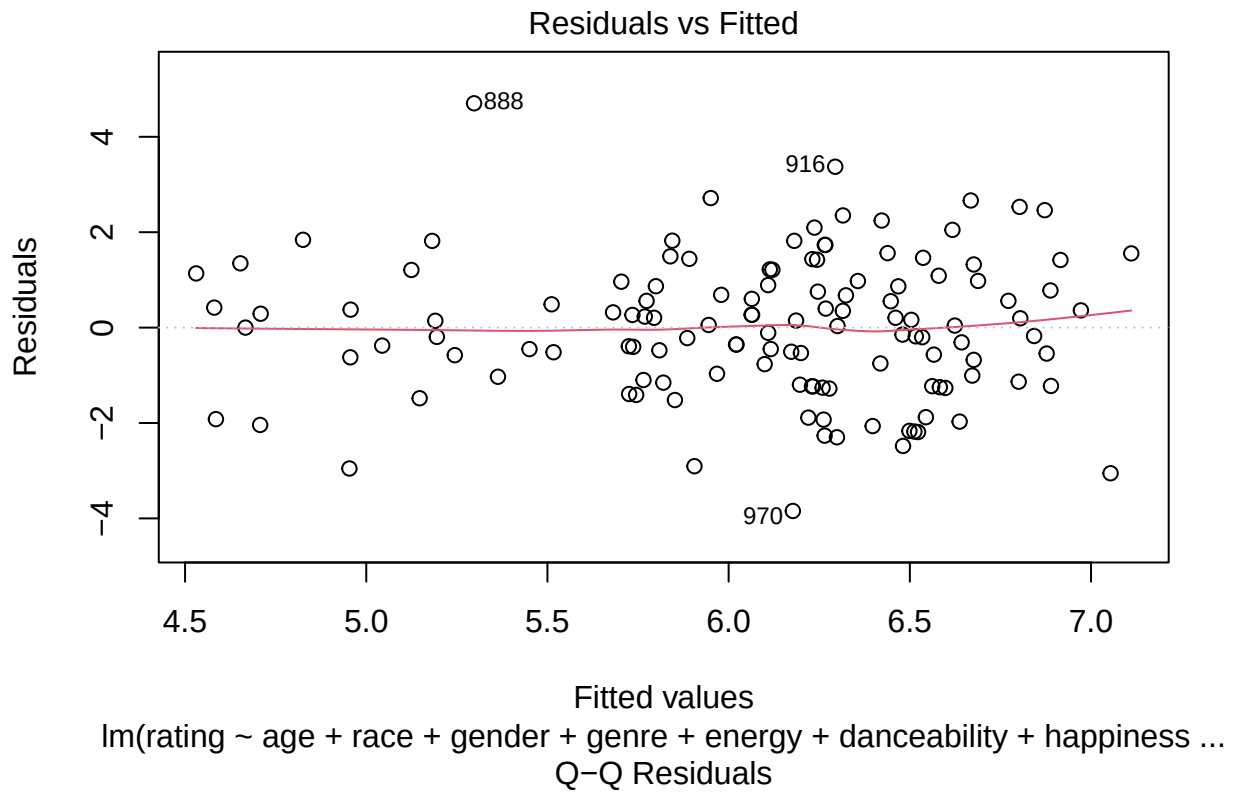
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + age:length_sec + race:danceability +
##   race:bpm + race:length_sec + gender:genre + gender:danceability +
##   gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - age:length_sec      1    1.8649 226.02 117.57
## - race:danceability    1    3.2937 227.45 118.42
## <none>                  224.15 118.45
## - race:bpm            1    3.5714 227.72 118.59
## - gender:length_sec    1    3.9799 228.13 118.83
## - gender:genre         2    7.9761 232.13 119.17
## - gender:danceability  1    4.8451 229.00 119.34
## - gender:loudness_db   1    5.0982 229.25 119.49
## - race:length_sec      1    6.3611 230.51 120.23
## - age:danceability     1    8.3086 232.46 121.37
## - happiness            1    9.3752 233.53 121.98
## - age:energy           1   13.2956 237.45 124.23
## - age:loudness_db      1   16.6391 240.79 126.12
##
## Step:  AIC=117.57
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + race:danceability +
##   race:bpm + race:length_sec + gender:genre + gender:danceability +
##   gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - race:bpm            1    2.9798 229.00 117.34
## <none>                  226.02 117.57
## - race:danceability    1    3.8262 229.84 117.84
## - gender:genre         2    7.8278 233.85 118.17
## - gender:danceability  1    4.4934 230.51 118.23
## - gender:loudness_db   1    5.0710 231.09 118.57
## - race:length_sec      1    5.1283 231.15 118.60
## - gender:length_sec    1    5.5585 231.58 118.85
## - happiness            1    8.8569 234.88 120.76
## - age:danceability     1    9.5638 235.58 121.17
## - age:energy           1   11.6161 237.63 122.34
## - age:loudness_db      1   15.2086 241.23 124.36
##
## Step:  AIC=117.34
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + race:danceability +
##   race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##   gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - bpm                  1    1.7671 230.77 116.38
## <none>                  229.00 117.34
## - gender:danceability  1    4.2421 233.24 117.82
## - gender:genre         2    7.8097 236.81 117.87

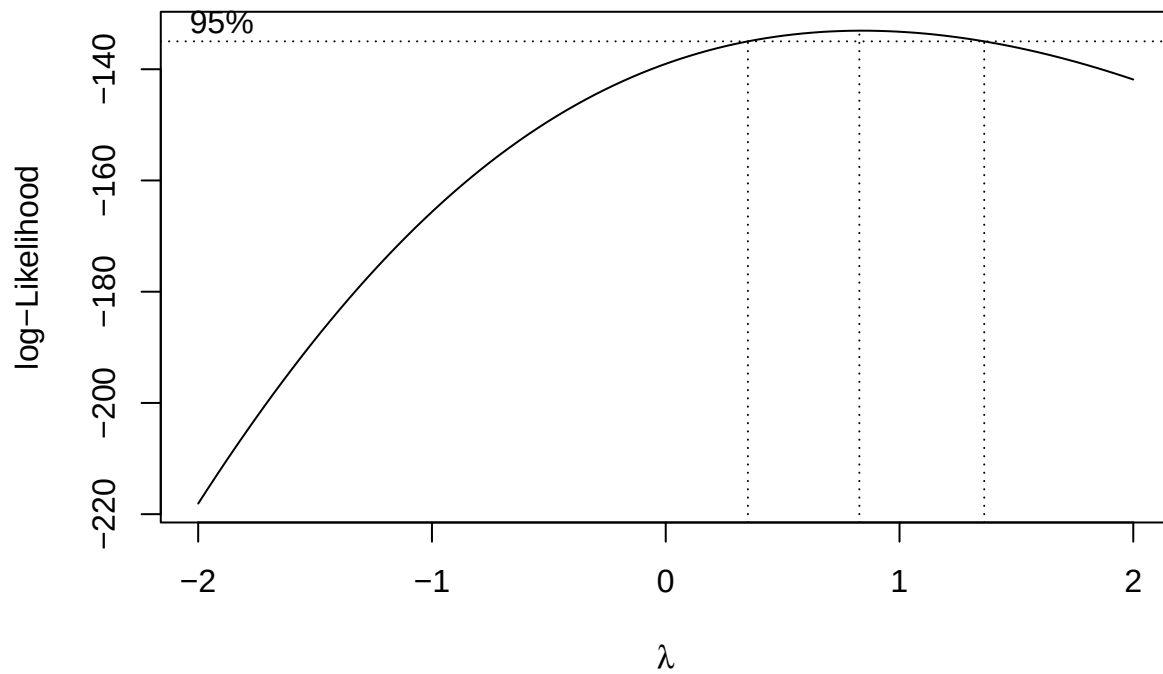
```

```

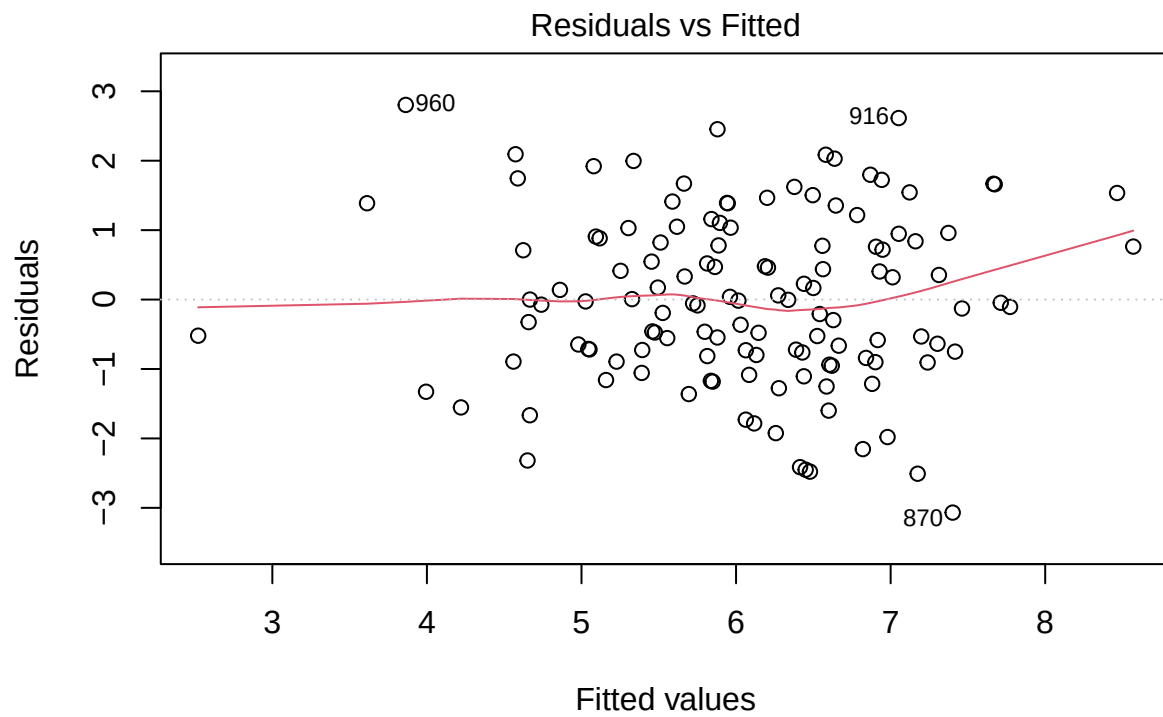
## - race:danceability      1      5.0409 234.04 118.28
## - gender:loudness_db     1      5.3476 234.35 118.46
## - race:length_sec        1      5.7553 234.75 118.69
## - gender:length_sec      1      7.4133 236.41 119.64
## - age:danceability       1      8.5272 237.53 120.28
## - happiness              1     10.1922 239.19 121.22
## - age:energy             1     11.0432 240.04 121.70
## - age:loudness_db        1     14.6957 243.69 123.74
##
## Step:  AIC=116.38
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + loudness_db + length_sec + age:energy + age:danceability +
##         age:loudness_db + race:danceability + race:length_sec + gender:genre +
##         gender:danceability + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## <none>                230.77 116.38
## - gender:genre        2      7.5708 238.34 116.73
## - race:danceability    1      4.3094 235.07 116.88
## - gender:danceability  1      5.4605 236.23 117.53
## - race:length_sec      1      5.6093 236.38 117.62
## - gender:loudness_db   1      5.6884 236.45 117.67
## - gender:length_sec    1      7.4700 238.24 118.68
## - age:danceability     1      8.7287 239.50 119.39
## - age:energy           1     10.4462 241.21 120.35
## - happiness            1     10.7471 241.51 120.52
## - age:loudness_db      1     14.6406 245.41 122.68
## [1] "1990s"
##
## Warning: not plotting observations with leverage one:
##      4

```

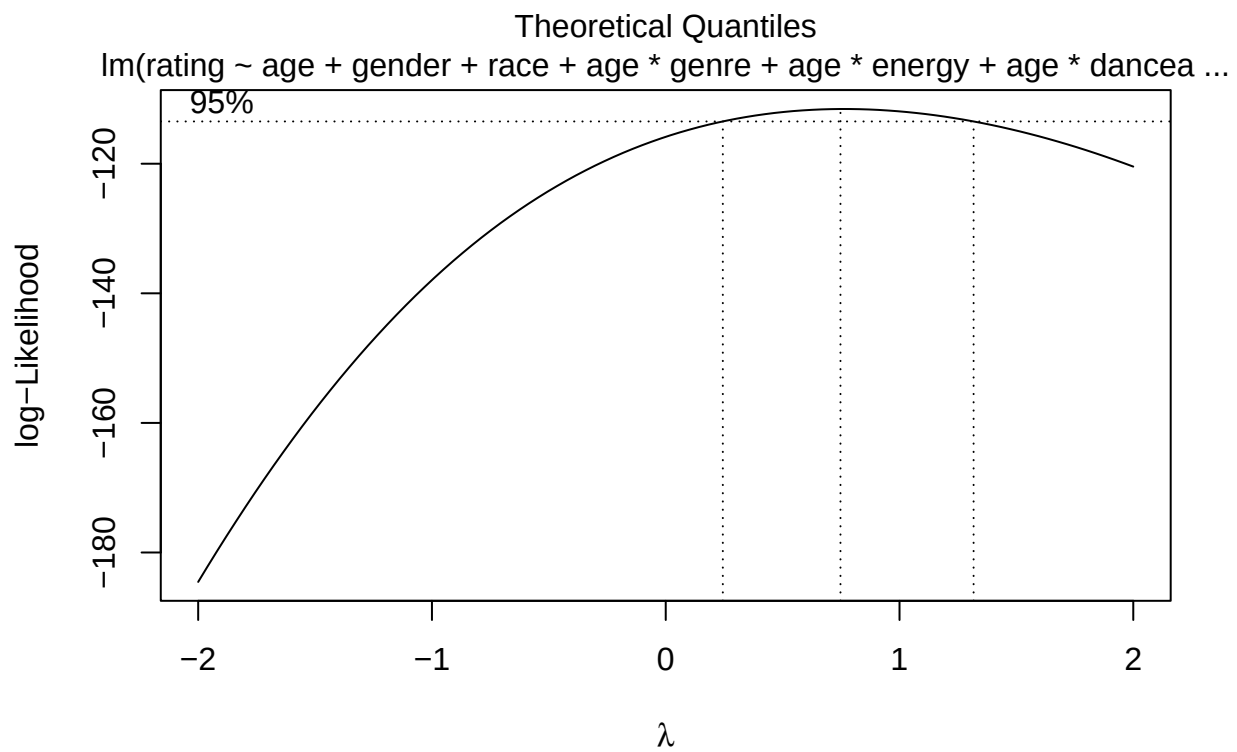
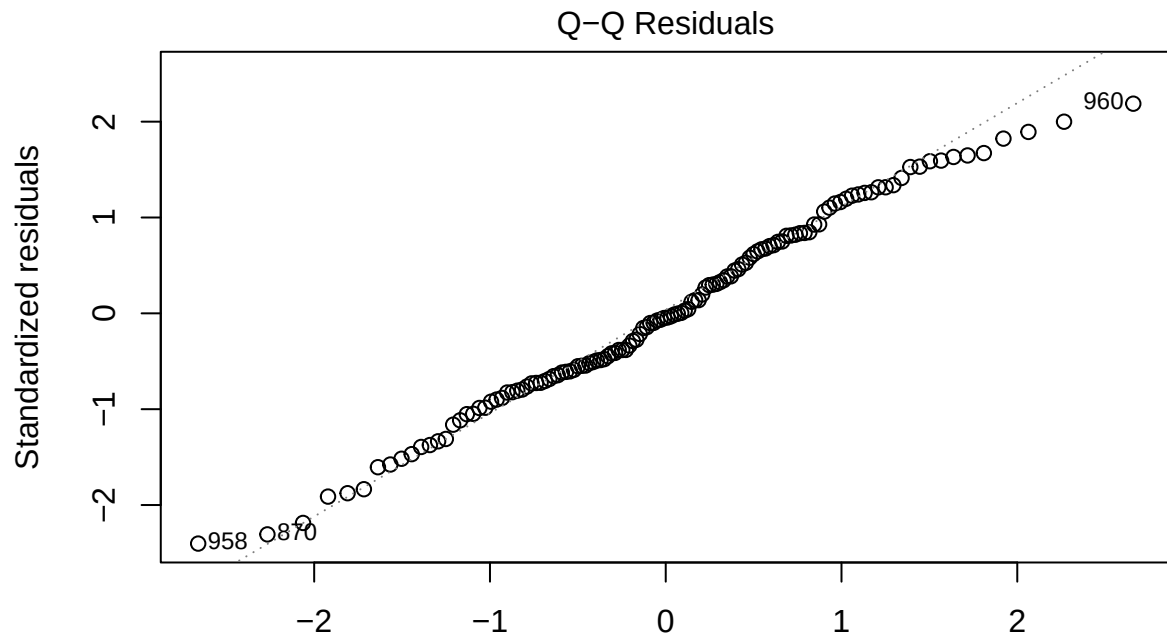




```
## Warning: not plotting observations with leverage one:
## 4
```



Fitted values
lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=124.77
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##              Df Sum of Sq    RSS    AIC
## - gender:genre      2      0.2909 194.49 120.96
## - gender:length_sec  1      0.0570 194.26 122.81
## - age:energy         1      0.1216 194.32 122.85
## - age:bpm           1      0.1778 194.38 122.89
## - age:happiness     1      0.4584 194.66 123.07
## - gender:loudness_db 1      0.4794 194.68 123.09
## - age:length_sec    1      0.9614 195.16 123.41
## - race:energy       1      1.2819 195.48 123.62
## - age:genre         2      4.3972 198.60 123.66
## - age:loudness_db   1      1.3852 195.58 123.69
## - gender:energy     1      1.4248 195.62 123.71
## - race:loudness_db  1      1.7680 195.97 123.94
## - age:danceability  1      2.1475 196.35 124.19
## - race:genre        1      2.4169 196.62 124.36
## <none>              194.20 124.77
## - race:length_sec   1      4.0842 198.28 125.45
## - gender:happiness  1      4.4298 198.63 125.68
## - race:happiness    1      5.1869 199.38 126.17
## - gender:bpm        1      5.6035 199.80 126.44
## - gender:danceability 1      8.3414 202.54 128.19
## - race:danceability 1     12.6344 206.83 130.90
## - race:bpm         1     13.9941 208.19 131.75
##
## Step:  AIC=120.96
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:genre +
##          age:energy + age:danceability + age:happiness + age:bpm +
##          age:loudness_db + age:length_sec + race:genre + race:energy +
##          race:danceability + race:happiness + race:bpm + race:loudness_db +
##          race:length_sec + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:length_sec  1      0.1410 194.63 119.06
## - age:energy         1      0.1432 194.63 119.06
## - age:bpm           1      0.2038 194.69 119.10
## - age:happiness     1      0.5093 195.00 119.30
## - gender:loudness_db 1      0.5341 195.02 119.32
## - age:length_sec    1      0.8897 195.38 119.55
## - race:energy       1      1.0952 195.59 119.69
## - age:loudness_db   1      1.3811 195.87 119.88
## - age:genre         2      4.5885 199.08 119.97
## - gender:energy     1      1.6752 196.16 120.07
## - race:loudness_db  1      1.6796 196.17 120.07
## - age:danceability  1      2.1850 196.68 120.40
## - race:genre        1      2.4594 196.95 120.58
## <none>              194.49 120.96
## - race:happiness    1      4.9200 199.41 122.19
## - race:length_sec   1      4.9339 199.42 122.19
## - gender:happiness  1      4.9712 199.46 122.22
## - gender:bpm        1      5.4597 199.95 122.53
## - gender:danceability 1      8.2237 202.71 124.31
## - race:danceability 1     12.9157 207.41 127.26

```

```

## - race:bpm          1    14.2374 208.73 128.08
##
## Step:  AIC=119.06
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db
##
##
##      Df Sum of Sq    RSS    AIC
## - age:energy      1      0.1503 194.78 117.16
## - age:bpm          1      0.1755 194.81 117.17
## - gender:loudness_db 1      0.5106 195.14 117.39
## - age:happiness    1      0.5595 195.19 117.43
## - age:length_sec   1      1.0576 195.69 117.76
## - race:energy      1      1.1071 195.74 117.79
## - age:loudness_db  1      1.5460 196.18 118.08
## - gender:energy    1      1.6133 196.24 118.12
## - age:genre        2      4.7363 199.37 118.16
## - race:loudness_db 1      2.0322 196.66 118.40
## - age:danceability 1      2.1949 196.83 118.50
## - race:genre       1      2.3311 196.96 118.59
## <none>                                194.63 119.06
## - race:length_sec  1      4.7938 199.42 120.19
## - race:happiness   1      4.7999 199.43 120.20
## - gender:happiness 1      4.8522 199.48 120.23
## - gender:bpm       1      5.7543 200.38 120.81
## - gender:danceability 1      8.1192 202.75 122.33
## - race:danceability 1     12.7794 207.41 125.26
## - race:bpm         1     14.6052 209.24 126.39
##
##
## Step:  AIC=117.16
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db
##
##
##      Df Sum of Sq    RSS    AIC
## - age:bpm          1      0.2614 195.04 115.33
## - age:happiness     1      0.4189 195.20 115.43
## - gender:loudness_db 1      0.4382 195.22 115.45
## - race:energy       1      1.0180 195.80 115.83
## - age:length_sec    1      1.0802 195.86 115.87
## - gender:energy     1      1.5427 196.32 116.17
## - age:genre         2      4.6494 199.43 116.20
## - race:loudness_db  1      2.0250 196.81 116.49
## - age:danceability  1      2.2189 197.00 116.62
## - race:genre        1      2.2472 197.03 116.64
## - age:loudness_db   1      2.7256 197.51 116.95

```

```

## <none>                                194.78 117.16
## - race:length_sec      1      4.6696 199.45 118.21
## - race:happiness       1      4.7170 199.50 118.24
## - gender:happiness     1      5.1363 199.92 118.51
## - gender:bpm           1      5.6174 200.40 118.82
## - gender:danceability  1      8.6958 203.48 120.79
## - race:danceability    1     12.7055 207.49 123.31
## - race:bpm             1     14.5050 209.29 124.42
##
## Step:  AIC=115.33
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:genre + race:energy + race:danceability + race:happiness +
##         race:bpm + race:loudness_db + race:length_sec + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db  1      0.4518 195.49 113.63
## - age:happiness       1      0.5097 195.55 113.67
## - race:energy         1      1.0076 196.05 113.99
## - age:length_sec      1      1.1602 196.20 114.09
## - age:genre           2      4.5286 199.57 114.29
## - gender:energy       1      1.6739 196.72 114.43
## - age:danceability    1      1.9702 197.01 114.63
## - race:loudness_db    1      2.0320 197.07 114.67
## - race:genre          1      2.2829 197.32 114.83
## <none>                                195.04 115.33
## - age:loudness_db     1      3.3223 198.37 115.51
## - race:length_sec     1      4.7071 199.75 116.41
## - race:happiness      1      4.7576 199.80 116.44
## - gender:bpm          1      5.4326 200.47 116.87
## - gender:happiness    1      5.7397 200.78 117.07
## - gender:danceability 1      9.1047 204.15 119.22
## - race:danceability    1     12.6221 207.66 121.42
## - race:bpm            1     14.2503 209.29 122.43
##
## Step:  AIC=113.63
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:genre + race:energy + race:danceability + race:happiness +
##         race:bpm + race:loudness_db + race:length_sec + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:happiness       1      0.6697 196.16 112.07
## - race:energy         1      0.8472 196.34 112.19
## - age:length_sec      1      1.0331 196.53 112.31
## - gender:energy       1      1.2479 196.74 112.45
## - age:genre           2      4.5650 200.06 112.61
## - race:loudness_db    1      1.9267 197.42 112.89
## - race:genre          1      1.9962 197.49 112.94
## - age:danceability    1      2.2628 197.76 113.11

```

```

## <none>                                195.49 113.63
## - age:loudness_db      1      3.6882 199.18 114.04
## - race:length_sec      1      4.3490 199.84 114.47
## - race:happiness       1      4.8955 200.39 114.82
## - gender:bpm           1      5.7848 201.28 115.39
## - gender:happiness     1      5.8293 201.32 115.42
## - gender:danceability  1      8.6722 204.17 117.23
## - race:danceability    1     12.5298 208.02 119.64
## - race:bpm             1     14.1900 209.68 120.67
##
## Step:  AIC=112.07
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:loudness_db + age:length_sec + race:genre +
##         race:energy + race:danceability + race:happiness + race:bpm +
##         race:loudness_db + race:length_sec + gender:energy + gender:danceability +
##         gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      0.6126 196.78 110.47
## - race:energy          1      0.9178 197.08 110.67
## - age:genre            2      4.3793 200.54 110.92
## - gender:energy        1      1.3660 197.53 110.96
## - age:danceability     1      1.9279 198.09 111.33
## - race:loudness_db     1      2.0138 198.18 111.39
## - race:genre           1      2.2968 198.46 111.57
## <none>                  196.16 112.07
## - age:loudness_db      1      3.2745 199.44 112.20
## - race:length_sec      1      4.5901 200.75 113.05
## - race:happiness       1      4.8985 201.06 113.25
## - gender:bpm           1      5.6194 201.78 113.71
## - gender:happiness     1      5.8449 202.01 113.86
## - gender:danceability  1      8.2718 204.44 115.40
## - race:danceability    1     13.9120 210.08 118.91
## - race:bpm             1     14.6554 210.82 119.36
##
## Step:  AIC=110.47
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:loudness_db + race:genre + race:energy +
##         race:danceability + race:happiness + race:bpm + race:loudness_db +
##         race:length_sec + gender:energy + gender:danceability + gender:happiness +
##         gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:genre            2      3.8284 200.60 108.96
## - race:energy          1      1.0157 197.79 109.14
## - gender:energy        1      1.4213 198.20 109.40
## - race:genre           1      1.9124 198.69 109.72
## - race:loudness_db     1      1.9883 198.76 109.77
## - age:danceability     1      2.4006 199.18 110.03
## <none>                  196.78 110.47
## - age:loudness_db      1      3.1558 199.93 110.52
## - race:length_sec      1      4.6725 201.45 111.50

```

```

## - race:happiness      1      4.8863 201.66 111.64
## - gender:happiness    1      5.7000 202.48 112.16
## - gender:bpm          1      6.4126 203.19 112.61
## - gender:danceability 1      7.9788 204.75 113.60
## - race:danceability   1     13.8881 210.66 117.27
## - race:bpm            1     14.4573 211.23 117.62
##
## Step: AIC=108.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:energy      1      1.1880 201.79 107.72
## - gender:energy     1      1.6288 202.23 108.00
## - race:loudness_db  1      2.8090 203.41 108.75
## - race:genre        1      3.0688 203.67 108.92
## <none>              200.60 108.96
## - age:loudness_db   1      3.7711 204.38 109.36
## - race:length_sec   1      3.8413 204.45 109.40
## - race:happiness    1      4.6947 205.30 109.94
## - gender:happiness  1      5.1011 205.71 110.20
## - age:danceability  1      5.1258 205.73 110.21
## - gender:bpm        1      7.4709 208.08 111.67
## - gender:danceability 1      8.4997 209.10 112.31
## - race:danceability 1     15.2674 215.87 116.42
## - race:bpm          1     16.8572 217.46 117.36
##
## Step: AIC=107.72
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db  1      1.6555 203.45 106.77
## - gender:energy     1      2.0790 203.87 107.04
## - race:genre        1      2.9970 204.79 107.62
## <none>              201.79 107.72
## - age:loudness_db   1      3.3881 205.18 107.87
## - race:happiness    1      3.6076 205.40 108.00
## - race:length_sec   1      3.7446 205.54 108.09
## - age:danceability  1      5.2854 207.08 109.05
## - gender:happiness  1      5.9235 207.72 109.45
## - gender:bpm        1      7.7831 209.58 110.60
## - gender:danceability 1      8.3421 210.13 110.94
## - race:danceability 1     14.8282 216.62 114.86
## - race:bpm          1     18.2821 220.07 116.91
##
## Step: AIC=106.77

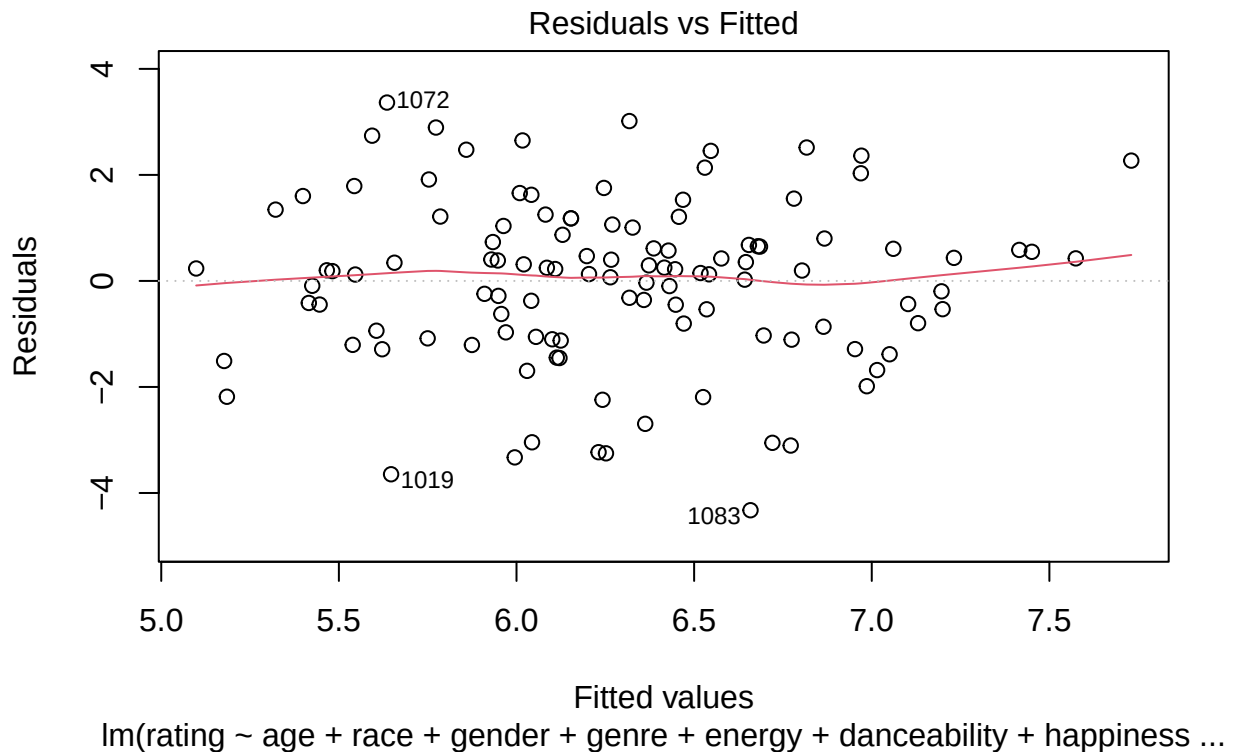
```

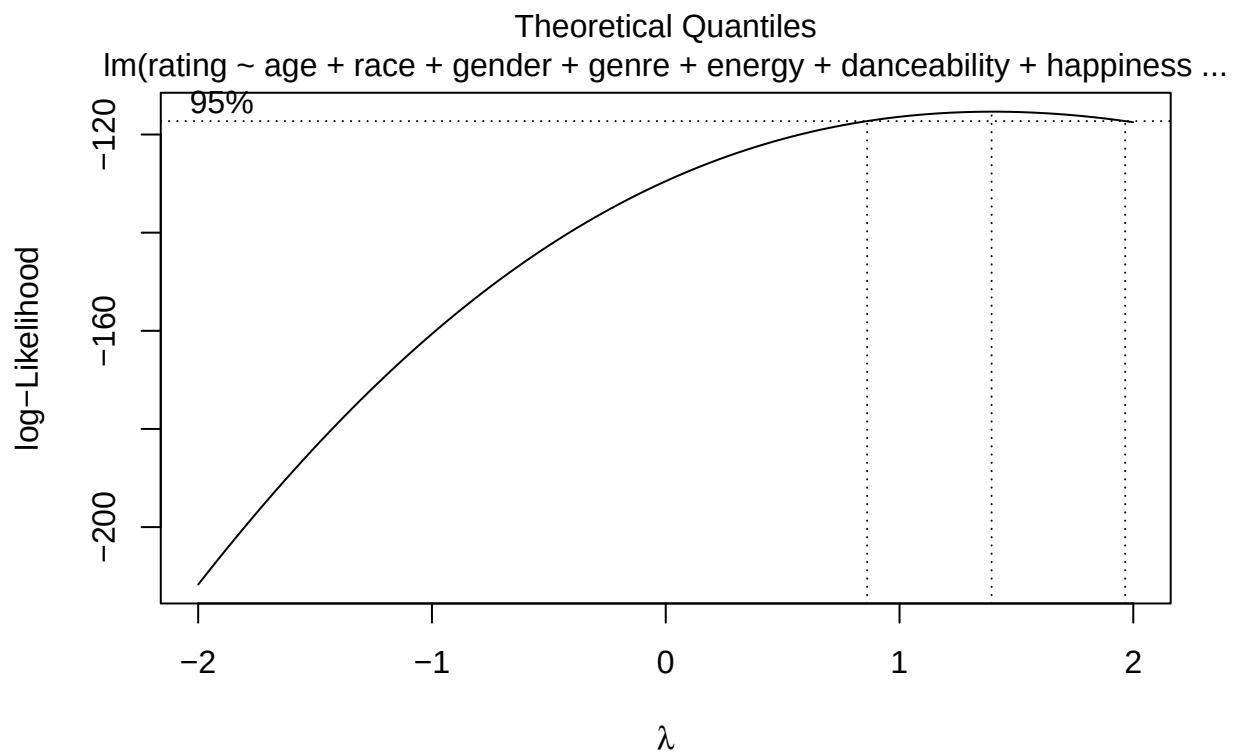
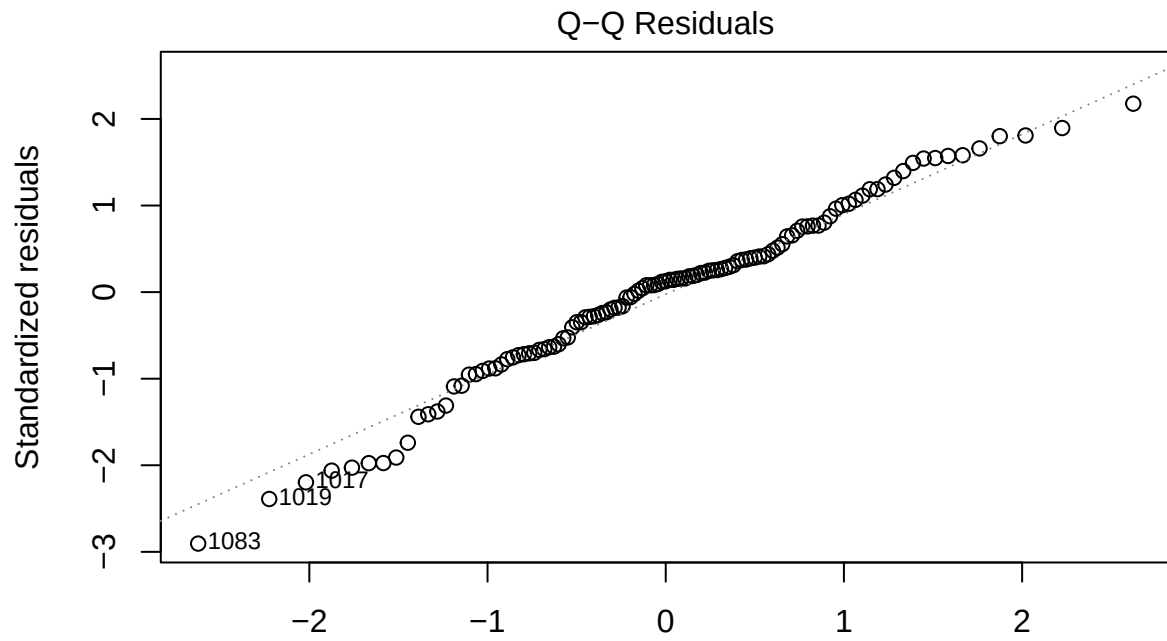
```

## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   age:loudness_db + race:genre + race:danceability + race:happiness +
##   race:bpm + race:length_sec + gender:energy + gender:danceability +
##   gender:happiness + gender:bpm
##
##
##      Df Sum of Sq  RSS   AIC
## - gender:energy      1    1.7232 205.17 105.86
## - age:loudness_db     1    2.6363 206.09 106.43
## <none>                  203.45 106.77
## - race:genre          1    3.5983 207.05 107.03
## - race:happiness      1    3.9824 207.43 107.27
## - race:length_sec     1    4.7272 208.18 107.73
## - age:danceability    1    4.8824 208.33 107.83
## - gender:happiness    1    5.6743 209.12 108.32
## - gender:bpm          1    6.7822 210.23 109.00
## - gender:danceability 1    8.2410 211.69 109.89
## - race:danceability   1   13.9885 217.44 113.35
## - race:bpm            1   16.7094 220.16 114.95
##
## Step: AIC=105.86
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   age:loudness_db + race:genre + race:danceability + race:happiness +
##   race:bpm + race:length_sec + gender:danceability + gender:happiness +
##   gender:bpm
##
##      Df Sum of Sq  RSS   AIC
## - age:loudness_db     1    2.9405 208.11 105.70
## <none>                  205.17 105.86
## - race:genre          1    3.2193 208.39 105.87
## - race:happiness      1    3.7242 208.90 106.18
## - gender:happiness    1    4.0686 209.24 106.39
## - energy              1    4.9055 210.08 106.91
## - age:danceability    1    5.3977 210.57 107.21
## - gender:bpm          1    5.4537 210.62 107.25
## - race:length_sec     1    5.6004 210.77 107.33
## - gender:danceability 1    7.3115 212.48 108.38
## - race:danceability   1   13.1129 218.28 111.85
## - race:bpm            1   16.2288 221.40 113.68
##
## Step: AIC=105.7
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   race:genre + race:danceability + race:happiness + race:bpm +
##   race:length_sec + gender:danceability + gender:happiness +
##   gender:bpm
##
##      Df Sum of Sq  RSS   AIC
## - loudness_db         1    0.5011 208.61 104.01
## <none>                  208.11 105.70
## - race:happiness      1    3.6051 211.72 105.91
## - race:genre          1    3.6755 211.79 105.95
## - gender:happiness    1    3.8465 211.96 106.06

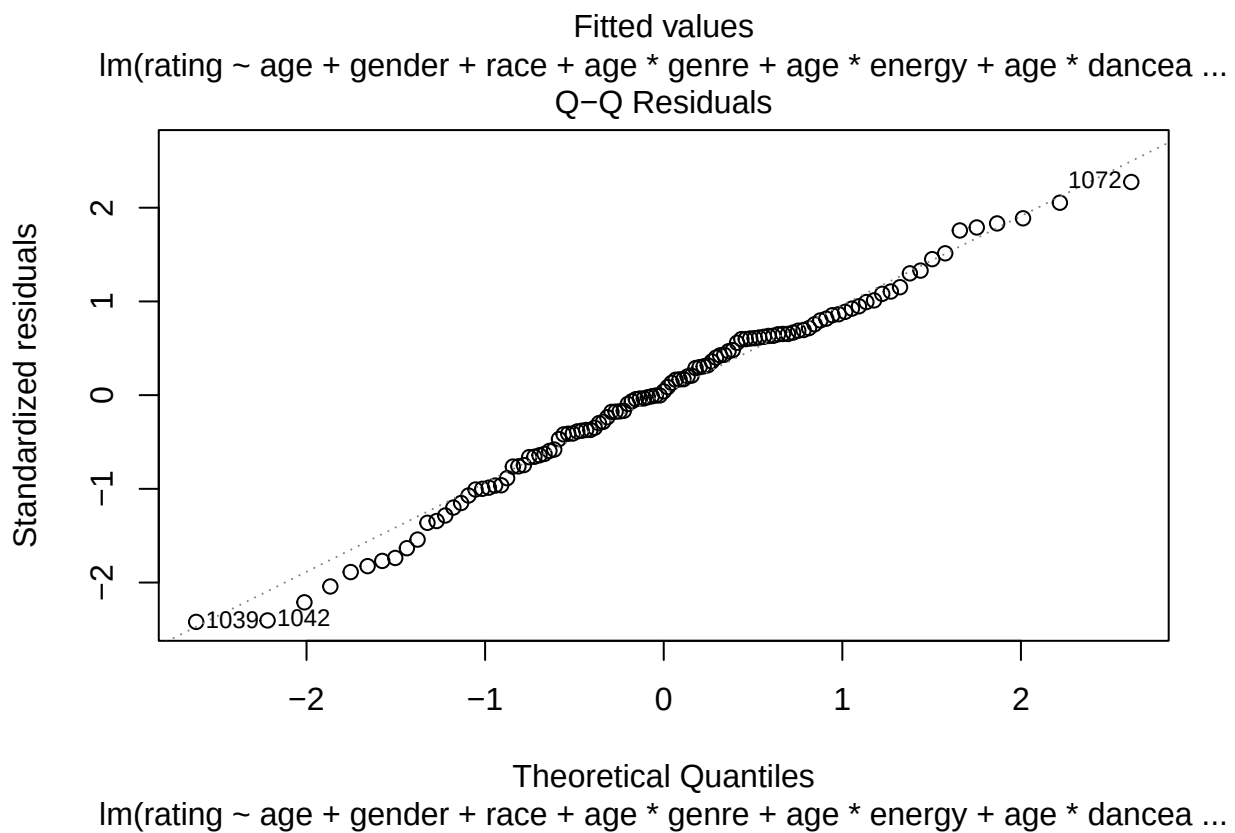
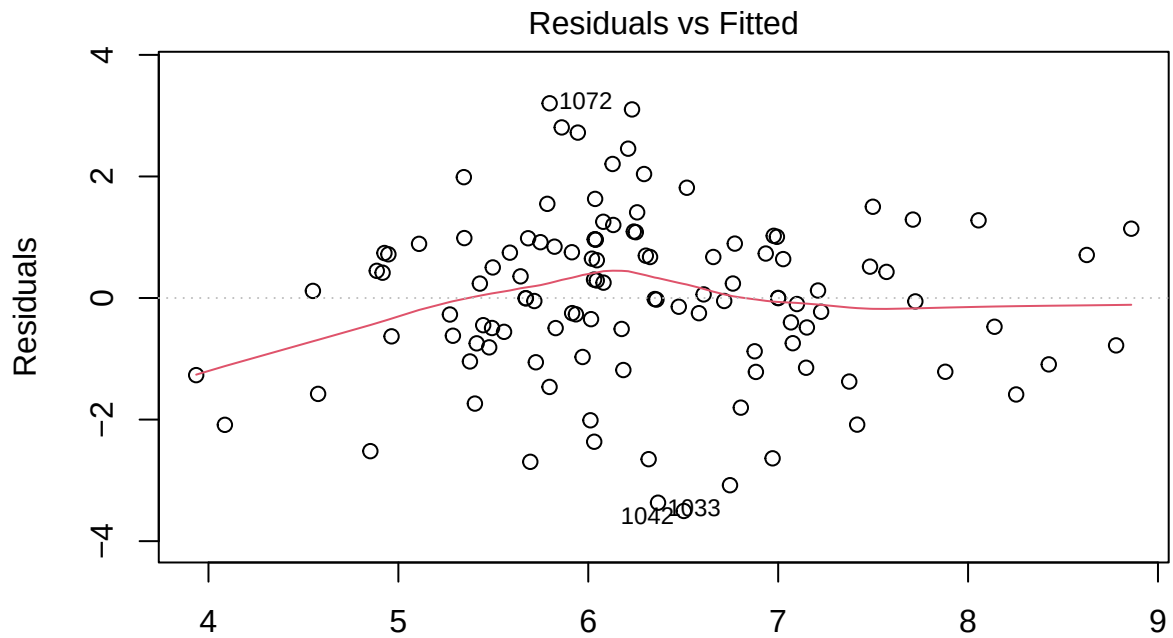
```

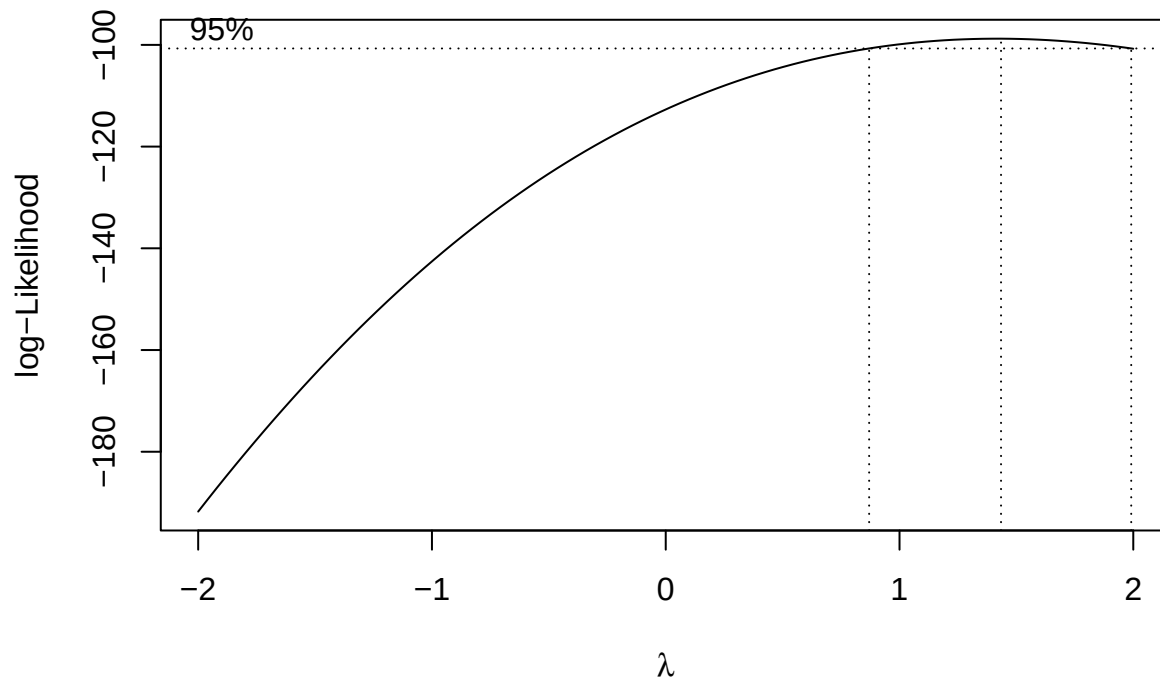
```
## - race:length_sec      1    5.2207 213.33 106.89
## - energy               1    5.2381 213.35 106.90
## - gender:bpm           1    5.4444 213.56 107.03
## - age:danceability     1    5.7563 213.87 107.22
## - gender:danceability  1    6.3411 214.45 107.57
## - race:danceability    1   13.7771 221.89 111.97
## - race:bpm             1   15.7977 223.91 113.14
##
## Step: AIC=104.01
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + length_sec + age:danceability + race:genre +
## race:danceability + race:happiness + race:bpm + race:length_sec +
## gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq  RSS   AIC
## <none>                208.61 104.01
## - race:happiness      1    3.3876 212.00 104.08
## - race:genre          1    3.6228 212.24 104.23
## - gender:happiness     1    4.1468 212.76 104.55
## - energy              1    5.3544 213.97 105.28
## - age:danceability    1    5.6004 214.21 105.42
## - race:length_sec     1    5.6508 214.26 105.45
## - gender:danceability  1    5.9037 214.52 105.61
## - gender:bpm          1    6.1967 214.81 105.78
## - race:danceability   1   13.7692 222.38 110.25
## - race:bpm            1   15.5847 224.20 111.30
## [1] "2000s"
```





```
## Warning: not plotting observations with leverage one:
## 41, 104
```





```
## Start: AIC=135.23
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##           Df Sum of Sq  RSS   AIC
## - race:length_sec      1    0.0490 206.39 133.26
## - age:loudness_db      1    0.1161 206.46 133.29
## - gender:danceability  1    0.3076 206.65 133.40
## - race:energy          1    0.3309 206.67 133.41
## - age:danceability     1    0.3807 206.72 133.44
## - gender:loudness_db   1    0.5786 206.92 133.55
## - race:bpm            1    0.6584 207.00 133.60
## - gender:energy        1    0.6699 207.01 133.60
## - race:genre           1    0.9574 207.30 133.76
## - race:happiness       1    1.0044 207.35 133.79
## - race:loudness_db     1    1.3673 207.71 133.99
## - gender:happiness     1    1.3906 207.73 134.00
## - age:happiness        1    1.4815 207.82 134.05
## - gender:length_sec    1    1.6502 207.99 134.15
## - race:danceability    1    1.7143 208.06 134.18
## - gender:bpm           1    1.9349 208.28 134.30
## - age:bpm             1    3.3788 209.72 135.10
## <none>                  206.34 135.23
## - age:energy           1    3.6401 209.98 135.24
## - gender:genre         1    4.1682 210.51 135.53
## - age:length_sec       1    4.6850 211.03 135.81
## - age:genre            1   14.3419 220.68 140.96
```

```

##
## Step: AIC=133.26
## rating ~ age + gender + race + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:energy + age:danceability + age:happiness + age:bpm +
##     age:loudness_db + age:length_sec + race:genre + race:energy +
##     race:danceability + race:happiness + race:bpm + race:loudness_db +
##     gender:genre + gender:energy + gender:danceability + gender:happiness +
##     gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq  RSS   AIC
## - age:loudness_db      1    0.1139 206.50 131.32
## - race:energy           1    0.2968 206.69 131.42
## - gender:danceability   1    0.3235 206.71 131.44
## - age:danceability      1    0.3640 206.75 131.46
## - gender:loudness_db    1    0.5707 206.96 131.57
## - race:bpm             1    0.6213 207.01 131.60
## - gender:energy         1    0.7372 207.13 131.67
## - race:genre           1    0.9750 207.37 131.80
## - race:happiness        1    0.9902 207.38 131.81
## - gender:happiness      1    1.3678 207.76 132.02
## - race:loudness_db      1    1.4274 207.82 132.05
## - age:happiness         1    1.5048 207.90 132.09
## - race:danceability     1    1.6843 208.07 132.19
## - gender:bpm           1    1.9496 208.34 132.34
## - gender:length_sec     1    2.0186 208.41 132.38
## - age:bpm              1    3.4745 209.87 133.18
## <none>                  1    206.39 133.26
## - age:energy           1    3.8029 210.19 133.36
## - gender:genre          1    4.1221 210.51 133.53
## - age:length_sec        1    5.2419 211.63 134.14
## - age:genre            1   14.3108 220.70 138.97
##
## Step: AIC=131.32
## rating ~ age + gender + race + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:energy + age:danceability + age:happiness + age:bpm +
##     age:length_sec + race:genre + race:energy + race:danceability +
##     race:happiness + race:bpm + race:loudness_db + gender:genre +
##     gender:energy + gender:danceability + gender:happiness +
##     gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq  RSS   AIC
## - gender:danceability   1    0.2932 206.80 129.48
## - gender:loudness_db    1    0.4647 206.97 129.58
## - race:energy           1    0.4876 206.99 129.59
## - age:danceability      1    0.5684 207.07 129.64
## - race:bpm             1    0.6983 207.20 129.71
## - gender:energy         1    0.9055 207.41 129.82
## - race:happiness        1    0.9794 207.48 129.86
## - race:genre           1    0.9916 207.50 129.87
## - gender:happiness      1    1.4148 207.92 130.10
## - age:happiness         1    1.4310 207.94 130.11
## - race:danceability     1    1.9473 208.45 130.40

```

```

## - gender:length_sec      1      1.9658 208.47 130.41
## - gender:bpm             1      1.9889 208.49 130.42
## - race:loudness_db       1      2.1307 208.64 130.50
## - age:bpm                1      3.3972 209.90 131.20
## <none>                    1      206.50 131.32
## - gender:genre           1      4.0628 210.57 131.56
## - age:energy              1      5.2417 211.75 132.20
## - age:length_sec         1      5.2885 211.79 132.23
## - age:genre              1     14.9934 221.50 137.38
##
## Step:  AIC=129.48
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db  1      0.3875 207.19 127.70
## - age:danceability    1      0.4720 207.27 127.75
## - race:energy          1      0.5310 207.33 127.78
## - race:bpm            1      0.6802 207.48 127.86
## - race:happiness       1      0.8607 207.66 127.96
## - race:genre           1      1.0030 207.80 128.04
## - gender:energy        1      1.0516 207.85 128.07
## - age:happiness        1      1.3840 208.18 128.25
## - race:danceability    1      1.7644 208.56 128.46
## - gender:happiness     1      1.9037 208.70 128.54
## - gender:bpm          1      1.9718 208.77 128.57
## - race:loudness_db     1      2.0605 208.86 128.62
## - gender:length_sec    1      2.1253 208.92 128.66
## - age:bpm              1      3.4085 210.21 129.36
## <none>                  1      206.80 129.48
## - gender:genre         1      4.5263 211.32 129.97
## - age:energy            1      5.0691 211.87 130.27
## - age:length_sec       1      5.9118 212.71 130.72
## - age:genre            1     16.0110 222.81 136.06
##
## Step:  AIC=127.7
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:danceability    1      0.5231 207.71 125.99
## - race:energy          1      0.5411 207.73 126.00
## - race:bpm            1      0.6923 207.88 126.08
## - race:happiness       1      1.0097 208.19 126.26
## - race:genre           1      1.2165 208.40 126.37

```

```

## - age:happiness      1      1.3071 208.49 126.42
## - gender:happiness   1      1.7375 208.92 126.66
## - race:danceability  1      1.7634 208.95 126.67
## - race:loudness_db   1      1.8873 209.07 126.74
## - gender:bpm         1      1.9759 209.16 126.79
## - gender:length_sec  1      2.5102 209.70 127.08
## - gender:energy      1      2.8893 210.07 127.29
## <none>                207.19 127.70
## - age:bpm            1      3.6769 210.86 127.72
## - gender:genre       1      4.2695 211.46 128.04
## - age:energy         1      5.1219 212.31 128.51
## - age:length_sec     1      6.9332 214.12 129.48
## - age:genre          1     15.9194 223.10 134.21
##
## Step:  AIC=125.99
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##         race:energy + race:danceability + race:happiness + race:bpm +
##         race:loudness_db + gender:genre + gender:energy + gender:happiness +
##         gender:bpm + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - race:energy      1    0.5083 208.22 124.27
## - race:bpm         1    0.6760 208.38 124.36
## - race:happiness    1    1.0497 208.76 124.57
## - race:genre        1    1.2306 208.94 124.67
## - race:danceability 1    1.3353 209.04 124.72
## - race:loudness_db  1    1.7469 209.46 124.95
## - age:happiness     1    1.8816 209.59 125.03
## - gender:bpm        1    1.8978 209.61 125.03
## - gender:happiness  1    1.9545 209.66 125.07
## - gender:length_sec 1    2.6662 210.38 125.45
## - gender:energy     1    3.0383 210.75 125.66
## - age:bpm           1    3.1538 210.86 125.72
## <none>              207.71 125.99
## - gender:genre      1    4.5753 212.28 126.49
## - age:energy         1    5.0486 212.76 126.75
## - age:length_sec     1    7.5679 215.28 128.10
## - age:genre          1   15.5192 223.23 132.28
##
## Step:  AIC=124.27
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##         race:danceability + race:happiness + race:bpm + race:loudness_db +
##         gender:genre + gender:energy + gender:happiness + gender:bpm +
##         gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - race:bpm         1    0.4440 208.66 122.51
## - race:happiness    1    0.6932 208.91 122.65
## - race:genre        1    0.9483 209.16 122.79
## - race:danceability 1    1.0990 209.32 122.88

```

```

## - race:loudness_db      1      1.2422 209.46 122.95
## - gender:bpm           1      1.7001 209.92 123.20
## - gender:happiness     1      2.0808 210.30 123.41
## - age:happiness        1      2.2454 210.46 123.50
## - gender:length_sec    1      2.3083 210.53 123.54
## - age:bpm              1      3.1638 211.38 124.00
## <none>                  208.22 124.27
## - gender:energy        1      3.7204 211.94 124.31
## - gender:genre         1      4.6078 212.82 124.79
## - age:energy           1      6.7450 214.96 125.94
## - age:length_sec       1      7.0736 215.29 126.11
## - age:genre            1     16.7728 224.99 131.18
##
## Step:  AIC=122.51
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:danceability + race:happiness + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.8452 209.51 120.98
## - race:danceability    1      1.0044 209.66 121.07
## - race:genre           1      1.1279 209.79 121.13
## - race:loudness_db     1      1.1556 209.82 121.15
## - gender:bpm           1      1.3033 209.96 121.23
## - gender:length_sec    1      1.9962 210.66 121.61
## - age:happiness        1      2.0236 210.69 121.62
## - gender:happiness     1      2.3165 210.98 121.78
## <none>                  208.66 122.51
## - gender:energy        1      3.9952 212.66 122.70
## - age:bpm              1      4.2849 212.95 122.85
## - gender:genre         1      4.8190 213.48 123.14
## - age:energy           1      6.3976 215.06 123.99
## - age:length_sec       1      7.1158 215.78 124.37
## - age:genre            1     16.3379 225.00 129.18
##
## Step:  AIC=120.98
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:danceability + race:loudness_db + gender:genre + gender:energy +
##      gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:genre           1      0.6079 210.11 119.31
## - race:danceability    1      0.6560 210.16 119.34
## - gender:bpm           1      1.4355 210.94 119.77
## - gender:length_sec    1      2.6481 212.15 120.42
## - race:loudness_db     1      2.7137 212.22 120.46
## - gender:happiness     1      3.4964 213.00 120.88
## - age:happiness        1      3.6142 213.12 120.95
## <none>                  209.51 120.98
## - gender:energy        1      4.0828 213.59 121.20

```

```

## - age:bpm          1    4.2957 213.80 121.31
## - gender:genre     1    5.9723 215.48 122.21
## - age:length_sec   1    6.8513 216.36 122.68
## - age:energy       1    7.6563 217.16 123.11
## - age:genre        1   18.8752 228.38 128.90
##
## Step:  AIC=119.31
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:danceability +
##      race:loudness_db + gender:genre + gender:energy + gender:happiness +
##      gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1    0.7035 210.82 117.70
## - gender:bpm        1    1.3836 211.50 118.07
## - gender:length_sec  1    2.8436 212.96 118.86
## <none>                210.11 119.31
## - race:loudness_db   1    3.7396 213.85 119.34
## - gender:happiness   1    3.8031 213.92 119.38
## - age:happiness      1    3.9144 214.03 119.44
## - age:bpm           1    4.2441 214.36 119.61
## - gender:energy      1    4.7148 214.83 119.86
## - age:length_sec     1    7.7796 217.89 121.49
## - age:energy         1    8.4517 218.57 121.85
## - gender:genre       1   10.0155 220.13 122.67
## - age:genre          1   23.5322 233.65 129.52
##
## Step:  AIC=117.7
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:loudness_db +
##      gender:genre + gender:energy + gender:happiness + gender:bpm +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - danceability       1    0.0373 210.85 115.72
## - gender:bpm         1    1.2836 212.10 116.39
## - gender:length_sec  1    2.6326 213.45 117.12
## - race:loudness_db   1    3.4155 214.23 117.55
## - age:happiness      1    3.4385 214.26 117.56
## <none>                210.82 117.70
## - gender:happiness   1    3.7042 214.52 117.70
## - gender:energy      1    4.3291 215.15 118.03
## - age:bpm           1    4.5001 215.32 118.13
## - age:length_sec     1    7.1707 217.99 119.54
## - age:energy         1    8.2803 219.10 120.13
## - gender:genre       1    9.3317 220.15 120.68
## - age:genre          1   22.8801 233.70 127.55
##
## Step:  AIC=115.72
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +

```



```

##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##      Df Sum of Sq    RSS    AIC
## - gender:bpm      1      1.2652 212.12 114.41
## - gender:length_sec 1      2.6203 213.47 115.14
## - age:happiness    1      3.4164 214.27 115.57
## - race:loudness_db 1      3.4459 214.30 115.58
## <none>                                210.85 115.72
## - gender:happiness 1      3.7546 214.61 115.75
## - gender:energy     1      4.3489 215.20 116.06
## - age:bpm           1      4.5016 215.36 116.15
## - age:length_sec    1      7.8418 218.70 117.92
## - age:energy        1      8.2812 219.14 118.15
## - gender:genre      1      9.4647 220.32 118.77
## - age:genre         1     22.8864 233.74 125.57
##
## Step:  AIC=114.41
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:length_sec
##
##      Df Sum of Sq    RSS    AIC
## - gender:length_sec 1      2.3130 214.43 113.65
## - age:happiness     1      2.5749 214.69 113.79
## - race:loudness_db  1      2.9498 215.07 113.99
## <none>                                212.12 114.41
## - gender:happiness  1      3.8730 215.99 114.49
## - age:bpm           1      4.2575 216.38 114.69
## - gender:energy     1      5.3010 217.42 115.24
## - age:energy        1      7.3038 219.42 116.30
## - age:length_sec    1      7.6141 219.73 116.46
## - gender:genre      1      8.7050 220.82 117.03
## - age:genre         1     21.8868 234.01 123.70
##
## Step:  AIC=113.65
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness
##
##      Df Sum of Sq    RSS    AIC
## - age:happiness     1      2.2305 216.66 112.84
## - race:loudness_db  1      2.9652 217.40 113.23
## <none>                                214.43 113.65
## - gender:happiness  1      4.1606 218.59 113.86
## - age:bpm           1      4.8673 219.30 114.23
## - gender:energy     1      5.7007 220.13 114.67
## - age:energy        1      6.8720 221.31 115.28
## - age:length_sec    1      8.3442 222.78 116.04
## - gender:genre      1      8.8745 223.31 116.32
## - age:genre         1     20.3945 234.83 122.10
##
## Step:  AIC=112.84

```

```

## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:bpm +
##   age:length_sec + race:loudness_db + gender:genre + gender:energy +
##   gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - race:loudness_db  1     2.8189 219.48 112.33
## <none>                                216.66 112.84
## - age:bpm           1     4.6675 221.33 113.29
## - age:energy        1     4.7498 221.41 113.34
## - gender:happiness  1     5.4999 222.16 113.72
## - gender:energy     1     6.3739 223.04 114.18
## - age:length_sec    1     8.8338 225.50 115.44
## - gender:genre      1    10.8930 227.56 116.48
## - age:genre         1    18.5262 235.19 120.28
##
## Step:  AIC=112.33
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:bpm +
##   age:length_sec + gender:genre + gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - loudness_db      1     0.1716 219.65 110.42
## - race              1     0.2294 219.71 110.45
## - age:energy        1     2.7840 222.27 111.78
## <none>                                219.48 112.33
## - age:bpm          1     4.9981 224.48 112.92
## - gender:happiness  1     5.4417 224.92 113.15
## - gender:energy     1     5.5351 225.02 113.19
## - age:length_sec    1     9.3725 228.85 115.14
## - gender:genre      1     9.5943 229.08 115.25
## - age:genre         1    17.0670 236.55 118.94
##
## Step:  AIC=110.42
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   length_sec + age:genre + age:energy + age:bpm + age:length_sec +
##   gender:genre + gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - race              1     0.2188 219.87 108.53
## - age:energy        1     2.7121 222.37 109.83
## <none>                                219.65 110.42
## - age:bpm          1     4.8897 224.54 110.95
## - gender:happiness  1     5.3659 225.02 111.19
## - gender:energy     1     5.4898 225.14 111.26
## - age:length_sec    1     9.2176 228.87 113.15
## - gender:genre      1     9.7357 229.39 113.41
## - age:genre         1    18.2207 237.88 117.58
##
## Step:  AIC=108.53
## rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:energy + age:bpm + age:length_sec + gender:genre +
##   gender:energy + gender:happiness
##

```

```

##           Df Sum of Sq   RSS   AIC
## - age:energy      1    2.9154 222.79 108.05
## <none>                219.87 108.53
## - age:bpm         1    4.7691 224.64 109.00
## - gender:happiness 1    5.2167 225.09 109.23
## - gender:energy    1    5.3407 225.21 109.29
## - age:length_sec   1    9.1286 229.00 111.21
## - gender:genre     1    9.6088 229.48 111.45
## - age:genre        1   18.5139 238.39 115.83
##
## Step:  AIC=108.05
## rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##       age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##       gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## <none>                222.79 108.05
## - gender:happiness    1    4.8910 227.68 108.55
## - age:bpm             1    5.0446 227.83 108.62
## - gender:energy       1    8.0804 230.87 110.14
## - age:length_sec      1    9.8753 232.66 111.04
## - gender:genre        1   10.8644 233.65 111.52
## - age:genre           1   15.6786 238.47 113.87
## [1] "2010s"
## Skipping 2020s due to single-level factor: genre
# Loop through each decade in the model_results list
for (decade in names(model_results)) {

  # Check if the 'optimal' model exists for this decade
  if (!is.null(model_results[[decade]]$optimal)) {

    cat("\n", "=====", "\n")
    cat("  SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
    cat("=====", "\n")

    # Print the summary
    print(summary(model_results[[decade]]$optimal))

  } else {
    cat("\nNo optimal model found for:", decade, "\n")
  }
}

##
## =====
##   SUMMARY FOR OPTIMAL MODEL: 1960s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)

```

```

##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.235892   2.001360   3.116  0.00214 **
## age             -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male   -1.820687   1.293523  -1.408  0.16101
## raceWhite        1.604179   1.795663   0.893  0.37287
## genreJazz        1.473039   1.332354   1.106  0.27040
## genreOther       1.978211   0.999147   1.980  0.04926 *
## genrePop         0.988531   0.889417   1.111  0.26788
## genreRock        0.260014   0.629236   0.413  0.67994
## energy          -0.005765   0.015336  -0.376  0.70743
## danceability    -0.013030   0.022465  -0.580  0.56264
## happiness        0.004845   0.009013   0.538  0.59154
## length_sec       0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz -1.353373   2.222521  -0.609  0.54334
## raceWhite:genreOther -3.249204   1.187123  -2.737  0.00683 **
## raceWhite:genrePop  -1.786329   0.784348  -2.277  0.02395 *
## raceWhite:genreRock      NA         NA         NA         NA
## raceWhite:energy      0.024399   0.017347   1.407  0.16132
## raceWhite:danceability -0.046982   0.022464  -2.091  0.03791 *
## genderNot Male:danceability 0.076584   0.023552   3.252  0.00137 **
## genderNot Male:happiness -0.030694   0.016106  -1.906  0.05830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314, Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF, p-value: 4.551e-08
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1970s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:happiness + age:bpm + age:length_sec + race:energy +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1773 -1.0450  0.0378  1.0913  4.6561
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.7599806   6.0727470   0.784  0.43397
## age              0.0742298   0.1986635   0.374  0.70902
## genderNot Male   -3.3233372   1.3575270  -2.448  0.01513 *

```

```

## raceWhite          -1.5914109  0.7655701  -2.079  0.03878 *
## genreJazz          2.9396820  2.5394662   1.158  0.24826
## genreOther        -5.4187148  4.2740093  -1.268  0.20618
## genrePop          -3.9847287  4.0475549  -0.984  0.32594
## genreRock         -6.6280029  4.3994636  -1.507  0.13334
## energy            -0.0217547  0.0131035  -1.660  0.09827 .
## happiness          0.0681171  0.0256047   2.660  0.00837 **
## bpm              -0.0531808  0.0330717  -1.608  0.10923
## loudness_db        0.0702293  0.0514647   1.365  0.17374
## length_sec         0.0505438  0.0166865   3.029  0.00274 **
## age:genreJazz             NA             NA             NA             NA
## age:genreOther          0.2154805  0.1350105   1.596  0.11189
## age:genrePop            0.1145362  0.1272669   0.900  0.36910
## age:genreRock           0.2268970  0.1398953   1.622  0.10623
## age:happiness          -0.0020628  0.0008623  -2.392  0.01758 *
## age:bpm                0.0017767  0.0011320   1.570  0.11794
## age:length_sec         -0.0015432  0.0005939  -2.599  0.00998 **
## raceWhite:energy        0.0207272  0.0131130   1.581  0.11537
## genderNot Male:length_sec 0.0151038  0.0060578   2.493  0.01338 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.772 on 224 degrees of freedom
## Multiple R-squared:  0.2958, Adjusted R-squared:  0.233
## F-statistic: 4.705 on 20 and 224 DF,  p-value: 2.095e-09
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1980s
## =====
##
## Call:
## lm(formula = rating ~ race + genre + bpm + length_sec + race:genre +
##     race:bpm + race:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4534 -0.8562  0.0485  0.9861  3.9808
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.289636    1.681818   1.361  0.1748
## raceWhite      2.987554    1.951337   1.531  0.1272
## genrePop      -0.726606    0.413305  -1.758  0.0801 .
## genreRock     -0.153197    0.492855  -0.311  0.7562
## bpm            0.011539    0.007813   1.477  0.1412
## length_sec     0.011023    0.005228   2.108  0.0361 *
## raceWhite:genrePop  1.329615    0.537348   2.474  0.0141 *
## raceWhite:genreRock  0.819251    0.591083   1.386  0.1672
## raceWhite:bpm    -0.016139    0.009210  -1.752  0.0811 .
## raceWhite:length_sec -0.009072    0.006014  -1.509  0.1329
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## Residual standard error: 1.492 on 218 degrees of freedom
## Multiple R-squared:  0.06512,    Adjusted R-squared:  0.02652
## F-statistic: 1.687 on 9 and 218 DF,  p-value: 0.09338
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1990s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + loudness_db + length_sec + age:energy +
##     age:danceability + age:loudness_db + race:danceability +
##     race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7557 -0.9125  0.0000  0.7949  3.6288
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.151452   4.862418   2.088 0.039069 *
## age            -0.188701   0.136443  -1.383 0.169390
## genderNot Male    4.498917   3.029352   1.485 0.140299
## raceWhite        5.519565   3.041628   1.815 0.072227 .
## genreOther      -0.174664   1.634218  -0.107 0.915074
## genrePop         0.268361   1.665154   0.161 0.872253
## genreRock        0.835016   1.632311   0.512 0.609961
## energy          -0.102502   0.032920  -3.114 0.002341 **
## danceability     0.119070   0.050887   2.340 0.021045 *
## happiness        0.017499   0.007628   2.294 0.023637 *
## loudness_db      1.147865   0.316756   3.624 0.000437 ***
## length_sec       0.011877   0.007043   1.686 0.094503 .
## age:energy       0.002489   0.001100   2.262 0.025630 *
## age:danceability -0.003376   0.001633  -2.067 0.040979 *
## age:loudness_db  -0.028004   0.010459  -2.678 0.008522 **
## raceWhite:danceability -0.035957 0.024752  -1.453 0.149090
## raceWhite:length_sec -0.014350 0.008659  -1.657 0.100228
## genderNot Male:genreOther  1.606214 0.980833   1.638 0.104286
## genderNot Male:genrePop    0.660151 1.026215   0.643 0.521341
## genderNot Male:genreRock    NA         NA         NA         NA
## genderNot Male:danceability -0.037786 0.023108  -1.635 0.104790
## genderNot Male:loudness_db -0.199855 0.119747  -1.669 0.097891 .
## genderNot Male:length_sec  -0.015603 0.008158  -1.913 0.058336 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.429 on 113 degrees of freedom
## Multiple R-squared:  0.374,    Adjusted R-squared:  0.2577
## F-statistic: 3.215 on 21 and 113 DF,  p-value: 3.432e-05
##
##
## =====

```

```

## SUMMARY FOR OPTIMAL MODEL: 2000s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      14.254799   5.689019   2.506 0.013714 *
## age              -0.262392   0.164223  -1.598 0.113013
## genderNot Male    -0.551642   2.208098  -0.250 0.803196
## raceWhite        -13.479191   3.724145  -3.619 0.000451 ***
## genreOther         0.362481   1.668695   0.217 0.828443
## genrePop          0.218854   1.608297   0.136 0.892013
## genreRock         0.090807   1.604975   0.057 0.954986
## energy            0.018882   0.011341   1.665 0.098826 .
## danceability      -0.083852   0.065397  -1.282 0.202519
## happiness         0.001354   0.011393   0.119 0.905583
## bpm              -0.014549   0.008120  -1.792 0.075985 .
## length_sec        -0.009048   0.005130  -1.764 0.080619 .
## age:danceability   0.003737   0.002195   1.703 0.091492 .
## raceWhite:genreOther -1.186109   0.866090  -1.369 0.173685
## raceWhite:genrePop      NA         NA         NA      NA
## raceWhite:genreRock     NA         NA         NA      NA
## raceWhite:danceability   0.084078   0.031491   2.670 0.008759 **
## raceWhite:happiness     -0.025028   0.018899  -1.324 0.188199
## raceWhite:bpm           0.035295   0.012426   2.840 0.005385 **
## raceWhite:length_sec     0.019703   0.011520   1.710 0.090064 .
## genderNot Male:danceability -0.041970   0.024007  -1.748 0.083262 .
## genderNot Male:happiness  0.021226   0.014487   1.465 0.145771
## genderNot Male:bpm       0.019019   0.010619   1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198
## F-statistic: 2.803 on 20 and 108 DF,  p-value: 0.0003367
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2010s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + genre + energy + happiness +
##     bpm + length_sec + age:genre + age:bpm + age:length_sec +

```

```
##      gender:genre + gender:energy + gender:happiness, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6153 -0.7497  0.0326  0.8805  3.4548
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.7479129   5.9555613   -0.126  0.90032
## age            0.2078850   0.2138097    0.972  0.33328
## genderNot Male -1.4345612   1.2782808   -1.122  0.26447
## genrePop       4.0067252   1.8431272    2.174  0.03210 *
## genreRock     -1.3841166   6.8130047   -0.203  0.83943
## energy        -0.0346294   0.0140445   -2.466  0.01540 *
## happiness      0.0176167   0.0096788    1.820  0.07176 .
## bpm           -0.0504318   0.0327145   -1.542  0.12637
## length_sec     0.0611063   0.0235217    2.598  0.01081 *
## age:genrePop   -0.1641477   0.0621883   -2.640  0.00965 **
## age:genreRock  0.0665065   0.2087839    0.319  0.75074
## age:bpm        0.0017263   0.0011530    1.497  0.13752
## age:length_sec -0.0017601   0.0008402   -2.095  0.03874 *
## genderNot Male:genrePop 1.5747664   0.7167070    2.197  0.03034 *
## genderNot Male:genreRock      NA         NA         NA         NA
## genderNot Male:energy  0.0408389   0.0215520    1.895  0.06102 .
## genderNot Male:happiness -0.0271834   0.0184388   -1.474  0.14359
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.5 on 99 degrees of freedom
## Multiple R-squared:  0.2797, Adjusted R-squared:  0.1706
## F-statistic: 2.563 on 15 and 99 DF, p-value: 0.002865
```

Preliminary Exploration of Data through Visual Analysis

Plots of Individual Relationships

The plots show the relationship between the predictors and the response variable, rating.

```
library(ggplot2)
library(patchwork)

##
## Attaching package: 'patchwork'

## The following object is masked from 'package:MASS':
##
##      area

plot_age <- ggplot(cleaned_data, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

plot_energy <- ggplot(cleaned_data, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
```



```

geom_smooth(method = "lm", color = "red", se = F) +
theme_classic() +
ggtitle("Rating vs Energy")

plot_danceability <- ggplot(cleaned_data, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Danceability")

plot_happiness <- ggplot(cleaned_data, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")

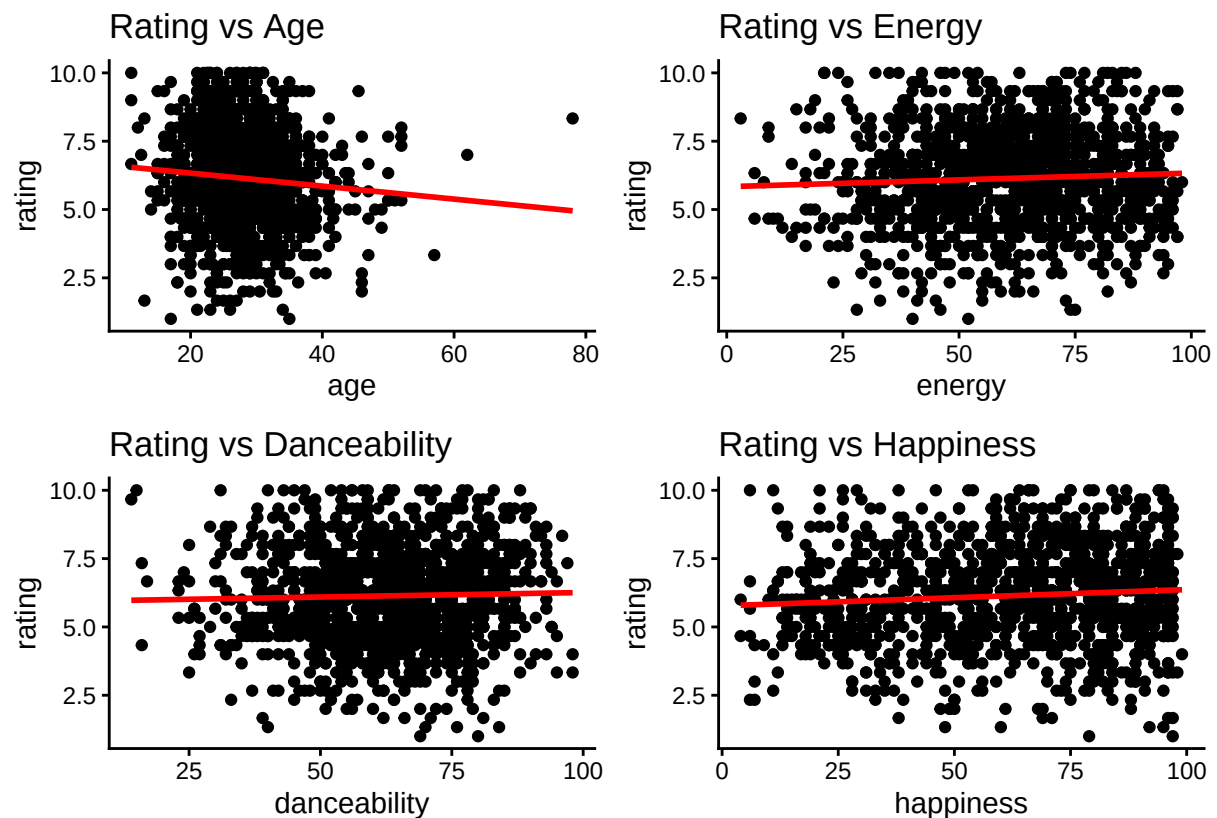
plot_age + plot_energy + plot_danceability + plot_happiness

```

```

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



```

plot_bpm <- ggplot(cleaned_data, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +

```

```

ggtitle("Rating vs BPM")

plot_loudness_db <- ggplot(cleaned_data, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Loudness(db)")

plot_length_sec <- ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length(sec)")

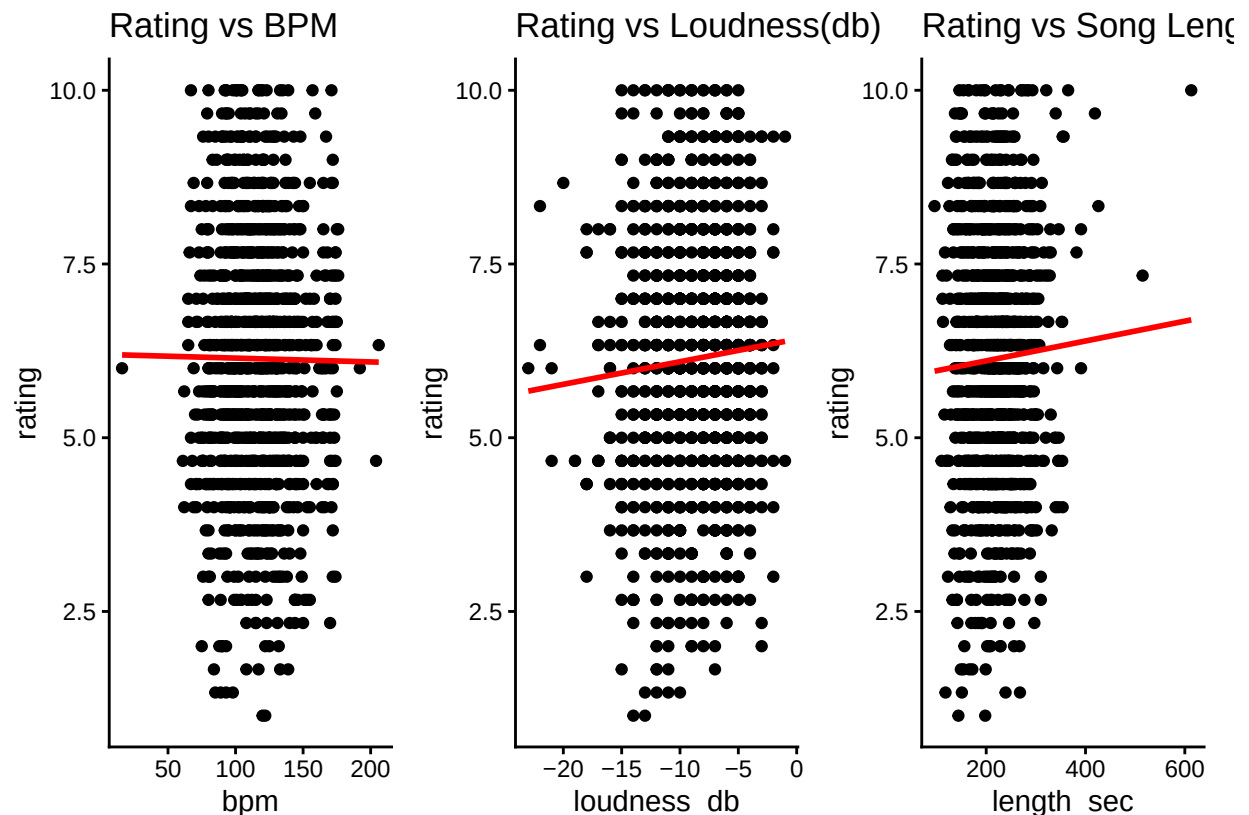
plot_bpm + plot_loudness_db + plot_length_sec

```

```

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



```

plot_race <- ggplot(cleaned_data, aes(x = race, y = rating, fill = race)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Race") +
  guides(fill = "none")

plot_gender <- ggplot(cleaned_data, aes(x = gender, y = rating, fill = gender)) +
  geom_boxplot() +

```

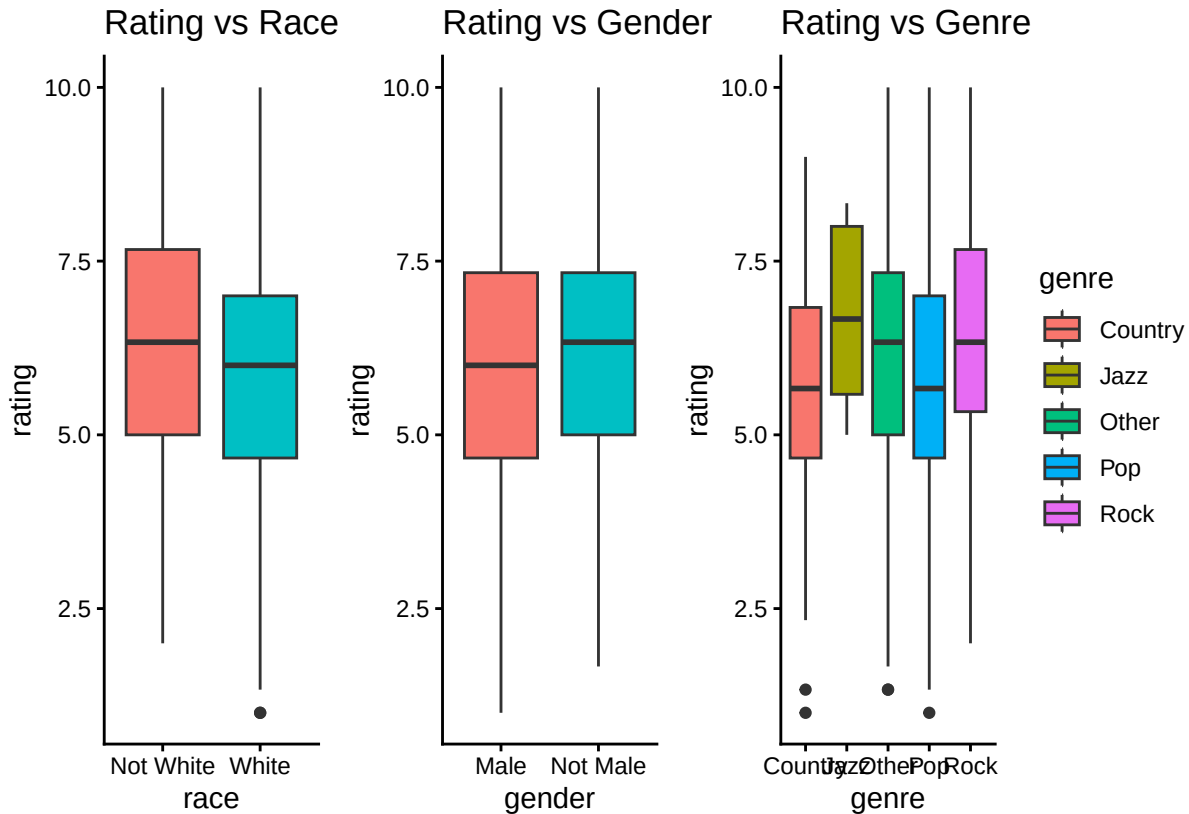
```

theme_classic() +
ggtitle("Rating vs Gender") +
guides(fill = "none")

plot_genre <- ggplot(cleaned_data, aes(x = genre, y = rating, fill = genre)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Genre")

plot_race + plot_gender + plot_genre

```

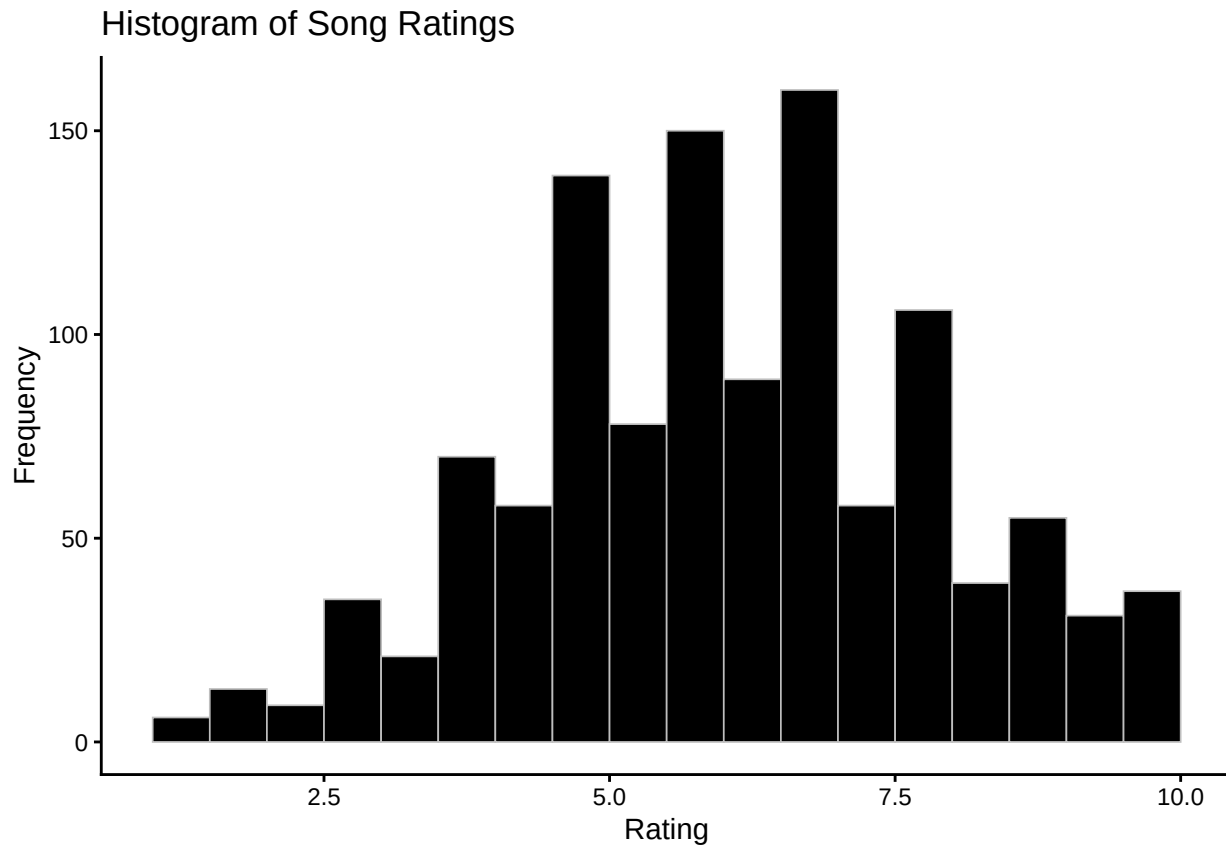


As expected, banding is observed in the scatterplots because the response variable, rating, is discrete. Since race, gender, and genre are categorical predictors, the relationship with rating is displayed by boxplots instead.

```

ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()

```



However, the histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. While the ratings are discrete (integer-valued) and there are many categorical predictors, the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

Plots of Grouped Relationships

Data visualizations for grouped relationships help us examine whether relationships between predictors and ratings differ across categories. These plots justify the inclusion of categorical predictors in the regression models and help identify potential interaction effects.

```
model_60s <- model_results[["1960s"]]$optimal
model_70s <- model_results[["1970s"]]$optimal
model_80s <- model_results[["1980s"]]$optimal
model_90s <- model_results[["1990s"]]$optimal
model_2000s <- model_results[["2000s"]]$optimal
model_2010s <- model_results[["2010s"]]$optimal

vars_by_decade <- lapply(model_results, function(x) {
  attr(terms(x$optimal), "term.labels")
})

all_vars <- unique(unlist(vars_by_decade))

presence_matrix <- sapply(vars_by_decade, function(vars) {
  all_vars %in% vars
})
```

```
rownames(presence_matrix) <- all_vars
presence_matrix
```

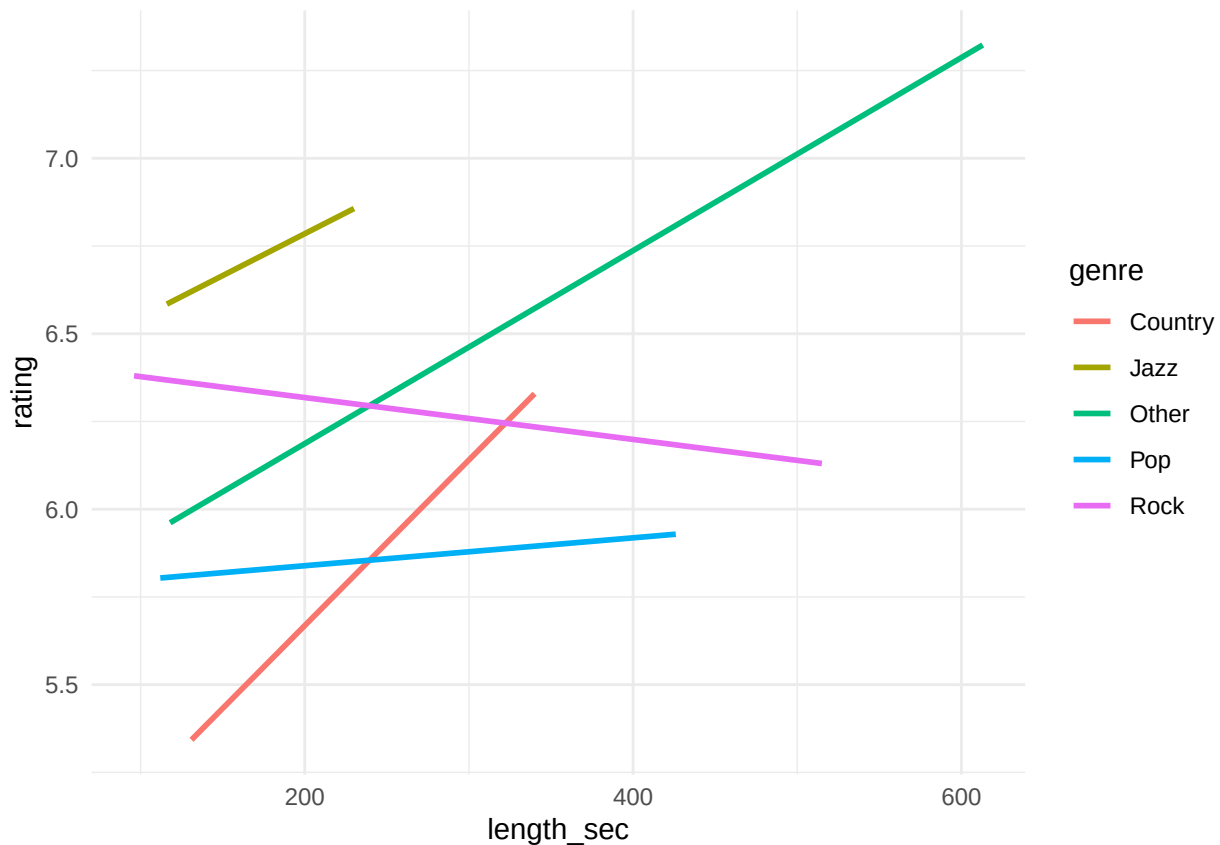
```
##           1960s 1970s 1980s 1990s 2000s 2010s
## age          TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## gender        TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## race          TRUE  TRUE  TRUE  TRUE  TRUE FALSE
## genre         TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## energy        TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## danceability  TRUE FALSE FALSE  TRUE  TRUE FALSE
## happiness     TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## length_sec    TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## race:genre    TRUE FALSE  TRUE FALSE  TRUE FALSE
## race:energy   TRUE  TRUE FALSE FALSE FALSE FALSE
## race:danceability TRUE FALSE FALSE  TRUE  TRUE FALSE
## gender:danceability TRUE FALSE FALSE  TRUE  TRUE FALSE
## gender:happiness TRUE FALSE FALSE FALSE  TRUE  TRUE
## bpm          FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## loudness_db   FALSE  TRUE FALSE  TRUE FALSE FALSE
## age:genre     FALSE  TRUE FALSE FALSE FALSE  TRUE
## age:happiness FALSE  TRUE FALSE FALSE FALSE FALSE
## age:bpm       FALSE  TRUE FALSE FALSE FALSE  TRUE
## age:length_sec FALSE  TRUE FALSE FALSE FALSE  TRUE
## gender:length_sec FALSE  TRUE FALSE  TRUE FALSE FALSE
## race:bpm      FALSE FALSE  TRUE FALSE  TRUE FALSE
## race:length_sec FALSE FALSE  TRUE  TRUE  TRUE FALSE
## age:energy    FALSE FALSE FALSE  TRUE FALSE FALSE
## age:danceability FALSE FALSE FALSE  TRUE  TRUE FALSE
## age:loudness_db FALSE FALSE FALSE  TRUE FALSE FALSE
## gender:genre   FALSE FALSE FALSE  TRUE FALSE  TRUE
## gender:loudness_db FALSE FALSE FALSE  TRUE FALSE FALSE
## race:happiness FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:bpm     FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:energy  FALSE FALSE FALSE FALSE FALSE  TRUE
```

```
rownames(presence_matrix)[rowSums(presence_matrix) == ncol(presence_matrix)]
```

```
## [1] "genre"      "length_sec"
```

```
ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_smooth(aes(color = genre), method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The fitted regression lines indicate that the relationship between song length and rating varies by genre. In particular, certain genres (e.g., Other and Country) exhibit steeper positive slopes, while others (e.g., Rock and Pop) show weaker or slightly negative trends, suggesting a potential interaction between length and genre.

model_60s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Coefficients:
##              (Intercept)              age
##                6.235892             -0.036813
##      genderNot Male      raceWhite
##        -1.820687             1.604179
##      genreJazz      genreOther
##         1.473039             1.978211
##      genrePop      genreRock
##         0.988531             0.260014
##          energy      danceability
##        -0.005765            -0.013030
##        happiness      length_sec
##         0.004845             0.008965
##      raceWhite:genreJazz      raceWhite:genreOther
```

```
##           -1.353373           -3.249204
##           raceWhite:genrePop       raceWhite:genreRock
##           -1.786329                NA
##           raceWhite:energy       raceWhite:danceability
##           0.024399                -0.046982
## genderNot Male:danceability       genderNot Male:happiness
##           0.076584                -0.030694
```

model_70s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:happiness + age:bpm + age:length_sec + race:energy +
##     gender:length_sec, data = subpop)
##
## Coefficients:
##           (Intercept)                age
##           4.759981                0.074230
##           genderNot Male                raceWhite
##           -3.323337                -1.591411
##           genreJazz                genreOther
##           2.939682                -5.418715
##           genrePop                genreRock
##           -3.984729                -6.628003
##           energy                happiness
##           -0.021755                0.068117
##           bpm                loudness_db
##           -0.053181                0.070229
##           length_sec                age:genreJazz
##           0.050544                NA
##           age:genreOther                age:genrePop
##           0.215480                0.114536
##           age:genreRock                age:happiness
##           0.226897                -0.002063
##           age:bpm                age:length_sec
##           0.001777                -0.001543
##           raceWhite:energy       genderNot Male:length_sec
##           0.020727                0.015104
```

model_80s

```
##
## Call:
## lm(formula = rating ~ race + genre + bpm + length_sec + race:genre +
##     race:bpm + race:length_sec, data = subpop)
##
## Coefficients:
##           (Intercept)                raceWhite                genrePop
##           2.289636                2.987554                -0.726606
##           genreRock                bpm                length_sec
##           -0.153197                0.011539                0.011023
##           raceWhite:genrePop       raceWhite:genreRock       raceWhite:bpm
##           1.329615                0.819251                -0.016139
##           raceWhite:length_sec
```

```
## -0.009072
```

```
model_90s
```

```
##
```

```
## Call:
```

```
## lm(formula = rating ~ age + gender + race + genre + energy +  
##     danceability + happiness + loudness_db + length_sec + age:energy +  
##     age:danceability + age:loudness_db + race:danceability +  
##     race:length_sec + gender:genre + gender:danceability + gender:loudness_db +  
##     gender:length_sec, data = subpop)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept) age  
## 10.151452 -0.188701  
## genderNot Male raceWhite  
## 4.498917 5.519565  
## genreOther genrePop  
## -0.174664 0.268361  
## genreRock energy  
## 0.835016 -0.102502  
## danceability happiness  
## 0.119070 0.017499  
## loudness_db length_sec  
## 1.147865 0.011877  
## age:energy age:danceability  
## 0.002489 -0.003376  
## age:loudness_db raceWhite:danceability  
## -0.028004 -0.035957  
## raceWhite:length_sec genderNot Male:genreOther  
## -0.014350 1.606214  
## genderNot Male:genrePop genderNot Male:genreRock  
## 0.660151 NA  
## genderNot Male:danceability genderNot Male:loudness_db  
## -0.037786 -0.199855  
## genderNot Male:length_sec  
## -0.015603
```

```
model_2000s
```

```
##
```

```
## Call:
```

```
## lm(formula = rating ~ age + gender + race + genre + energy +  
##     danceability + happiness + bpm + length_sec + age:danceability +  
##     race:genre + race:danceability + race:happiness + race:bpm +  
##     race:length_sec + gender:danceability + gender:happiness +  
##     gender:bpm, data = subpop)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept) age  
## 14.254799 -0.262392  
## genderNot Male raceWhite  
## -0.551642 -13.479191  
## genreOther genrePop  
## 0.362481 0.218854  
## genreRock energy
```



```
##          0.090807          0.018882
##          danceability      happiness
##          -0.083852          0.001354
##          bpm                length_sec
##          -0.014549          -0.009048
##          age:danceability    raceWhite:genreOther
##          0.003737            -1.186109
##          raceWhite:genrePop   raceWhite:genreRock
##          NA                  NA
##          raceWhite:danceability raceWhite:happiness
##          0.084078            -0.025028
##          raceWhite:bpm        raceWhite:length_sec
##          0.035295            0.019703
## genderNot Male:danceability  genderNot Male:happiness
##          -0.041970            0.021226
##          genderNot Male:bpm
##          0.019019

model_2010s

##
## Call:
## lm(formula = rating ~ age + gender + genre + energy + happiness +
##     bpm + length_sec + age:genre + age:bpm + age:length_sec +
##     gender:genre + gender:energy + gender:happiness, data = subpop)
##
## Coefficients:
##          (Intercept)          age          genderNot Male
##          -0.747913          0.207885          -1.434561
##          genrePop          genreRock          energy
##          4.006725          -1.384117          -0.034629
##          happiness          bpm          length_sec
##          0.017617          -0.050432          0.061106
##          age:genrePop          age:genreRock          age:bpm
##          -0.164148          0.066506          0.001726
##          age:length_sec  genderNot Male:genrePop  genderNot Male:genreRock
##          -0.001760          1.574766          NA
##          genderNot Male:energy  genderNot Male:happiness
##          0.040839          -0.027183

anova(model_results[["1960s"]], first, model_60s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##     happiness + length_sec + race:genre + race:energy + race:danceability +
##     gender:danceability + gender:happiness
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     183 673.25
## 2     178 575.01  5    98.245 6.0826 3.166e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_60s, model_results[["1960s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + length_sec + race:genre + race:energy + race:danceability +
##   gender:danceability + gender:happiness
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      178 575.01
## 2      155 535.98 23    39.027 0.4907 0.9759
```

```
anova(model_results[["1970s"]])$first, model_70s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##   age:length_sec + race:energy + gender:length_sec
##   Res.Df    RSS Df Sum of Sq    F  Pr(>F)
## 1      231 764.63
## 2      224 703.08  7    61.547 2.8012 0.008172 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_70s, model_results[["1970s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##   age:length_sec + race:energy + gender:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      224 703.08
## 2      204 666.81 20    36.266 0.5547 0.9388
```

```
anova(model_results[["1980s"]])$first, model_80s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
```

```
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
##      race:length_sec
##   Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1      216 494.88
## 2      218 485.57 -2      9.3015
```

```
anova(model_80s, model_results[["1980s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
##      race:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      218 485.57
## 2      192 459.91 26      25.667 0.4121 0.9953
```

```
anova(model_results[["1990s"]])$first, model_90s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      age:loudness_db + race:danceability + race:length_sec + gender:genre +
##      gender:danceability + gender:loudness_db + gender:length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      122 269.95
## 2      113 230.77  9      39.186 2.132 0.03227 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_90s, model_results[["1990s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      age:loudness_db + race:danceability + race:length_sec + gender:genre +
##      gender:danceability + gender:loudness_db + gender:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
```

```

## 1    113 230.77
## 2     98 217.50 15    13.262 0.3984 0.9766
anova(model_results[["2000s"]] $\$$ first, model_2000s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + length_sec + age:danceability + race:genre +
##   race:danceability + race:happiness + race:bpm + race:length_sec +
##   gender:danceability + gender:happiness + gender:bpm
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     116 270.33
## 2     108 208.61  8     61.719 3.994 0.0003515 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_2000s, model_results[["2000s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + length_sec + age:danceability + race:genre +
##   race:danceability + race:happiness + race:bpm + race:length_sec +
##   gender:danceability + gender:happiness + gender:bpm
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     108 208.61
## 2      93 194.20 15     14.415 0.4602 0.9544

anova(model_results[["2010s"]] $\$$ first, model_2010s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##   gender:happiness
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     103 274.94
## 2      99 222.79  4     52.151 5.7935 0.0003127 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_2010s, model_results[["2010s"]] $\$$ second)

## Analysis of Variance Table
##

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## Model 1: rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##   gender:happiness
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      99 222.79
## 2      81 206.34 18    16.446 0.3587 0.9916

```